
Bài 1 – Phân tích và Quản lý đầu tư nâng cao

Tổng quan về Ứng dụng công nghệ trong Phân tích và Quản lý đầu tư

TS. Đỗ Như Tài
Đại Học Sài Gòn
dntai@sgu.edu.vn

AI Trong Cuộc Cách Mạng Công Nghệ Lần Thứ 4

AI

Definition of Artificial Intelligence (A.I.)

Artificial Intelligence

“... the **science** and
engineering
of
making
intelligent machines”
(John McCarthy, 1955)

Artificial Intelligence

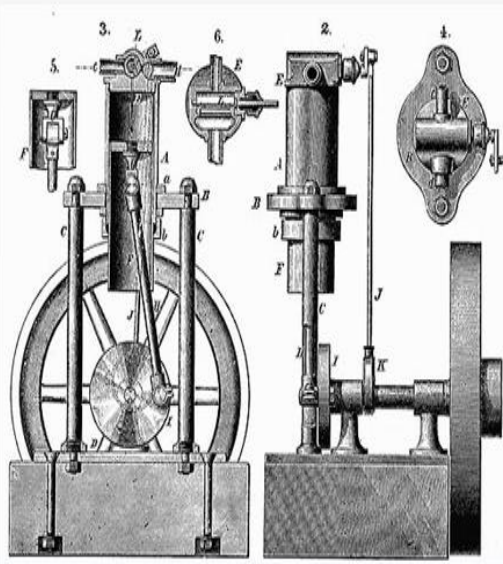
**“... technology that
thinks and acts
like humans”**

Artificial Intelligence

“... intelligence
exhibited by machines
or software”

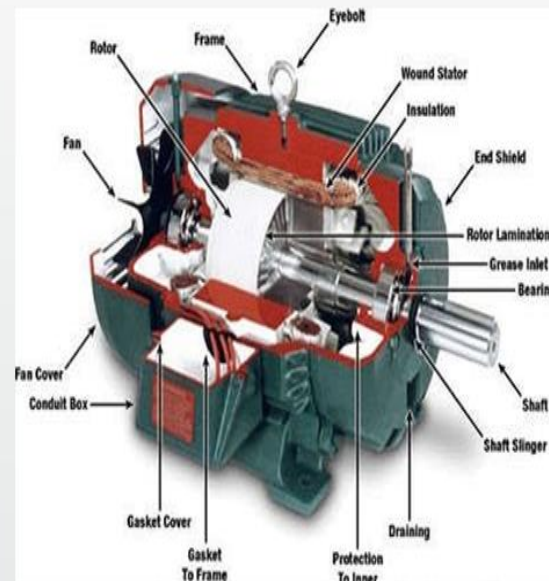
AI là Động cơ lõi (core engine) của CMCN 4.0

Lần thứ nhất



Sản xuất cơ khí với máy móc dựa vào động cơ hơi nước (Cuối thế kỷ 18, đầu tk 19)

Lần thứ hai



Sản xuất hàng loạt với máy móc dựa vào năng lượng điện (Cuối tk 19, đầu tk 20)

Lần thứ ba



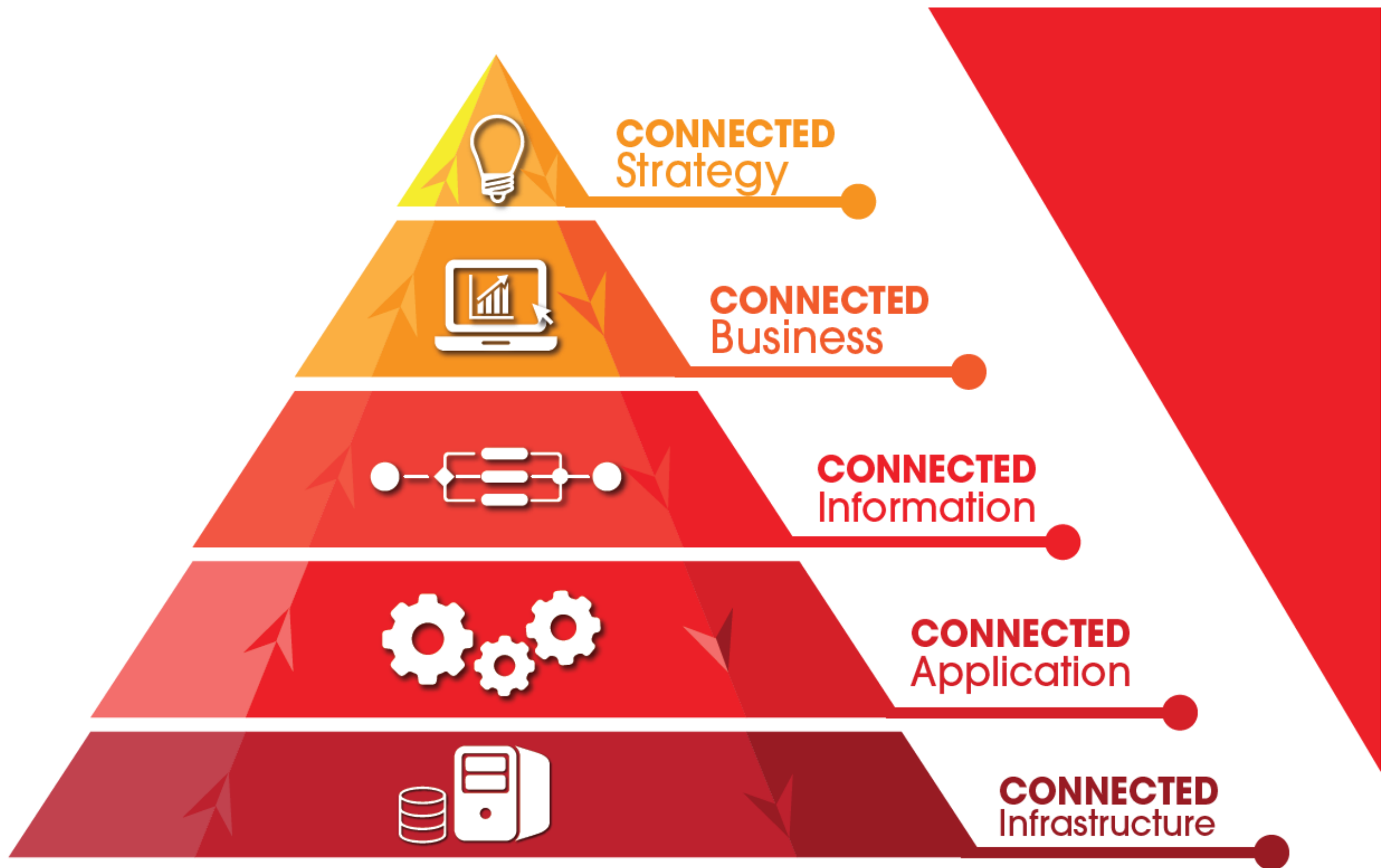
Sản xuất tự động với máy tính, điện tử và cách mạng số hóa (Từ 1970)

Lần thứ Tư

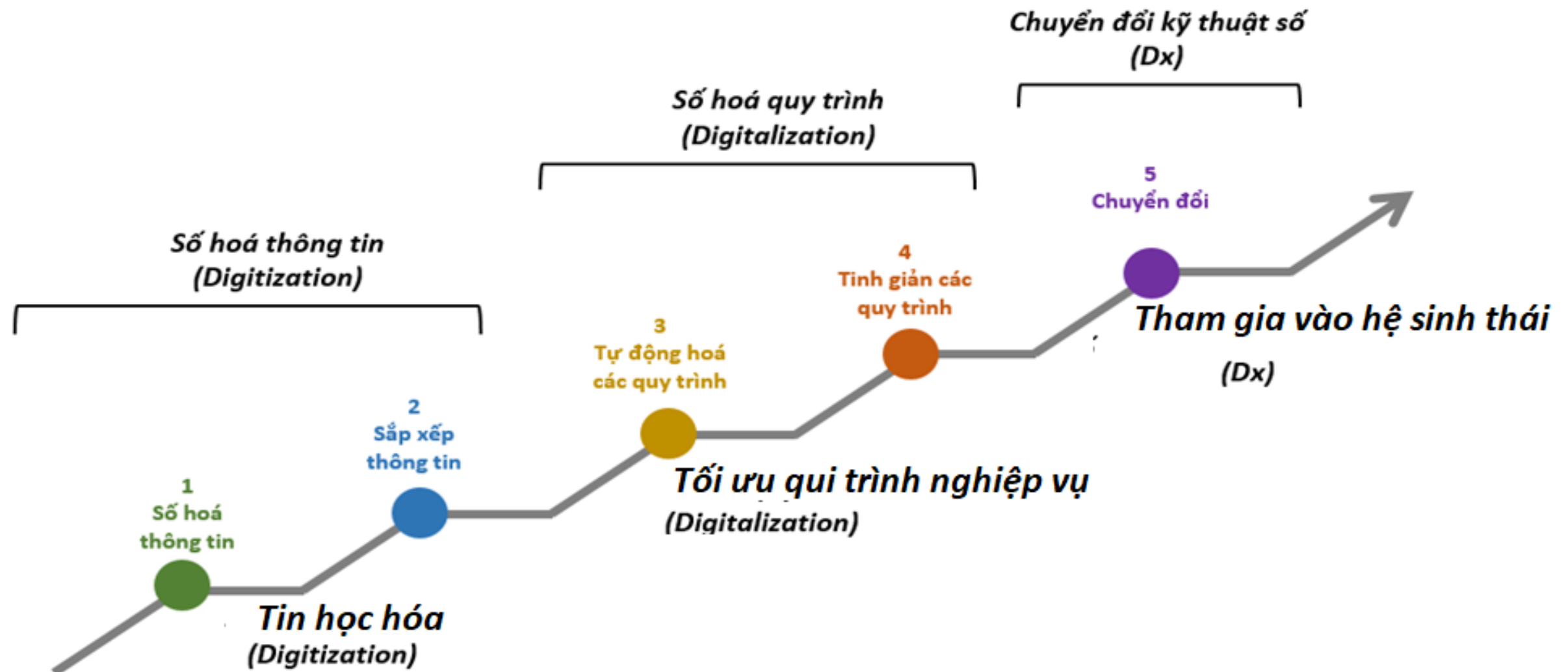


Sản xuất thông minh nhờ chuyển đổi số - kinh tế chia sẻ (Từ bây giờ, đúng thời điểm)

Chuyển đổi số



Lộ trình chuyển đổi số



Nền Kinh Tế Chia Sẻ



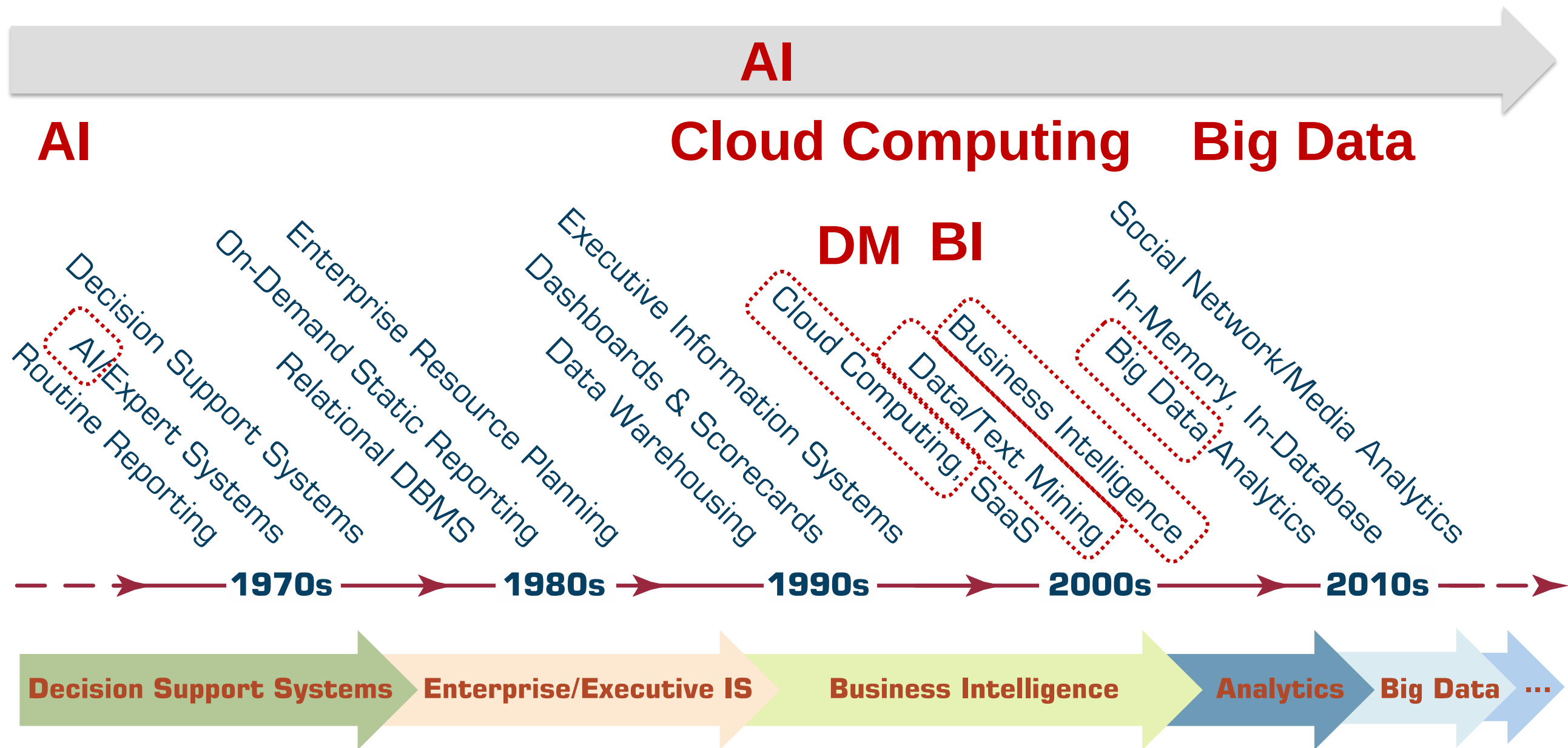
Câu hỏi

- AI là gì?
- CMCN lần 4.0 vs AI vs Chuyển đổi số?
- Mục tiêu của chuyển đổi số?
- Lộ trình chuyển đổi số?
- Là lực lượng sản xuất mới, anh (chị) làm gì để thích ứng với quan hệ sản xuất mới và phương thức sản xuất mới?

Deep in Dive AI

AI, Big Data, Cloud Computing

Evolution of Decision Support, Business Intelligence, and Analytics



Source: Ramesh Sharda, Dursun Delen, and Efraim Turban (2017),
Business Intelligence, Analytics, and Data Science: A Managerial Perspective, 4th Edition, Pearson

4 Approaches of AI

Thinking Humanly	Thinking Rationally
Acting Humanly	Acting Rationally

4 Approaches of AI

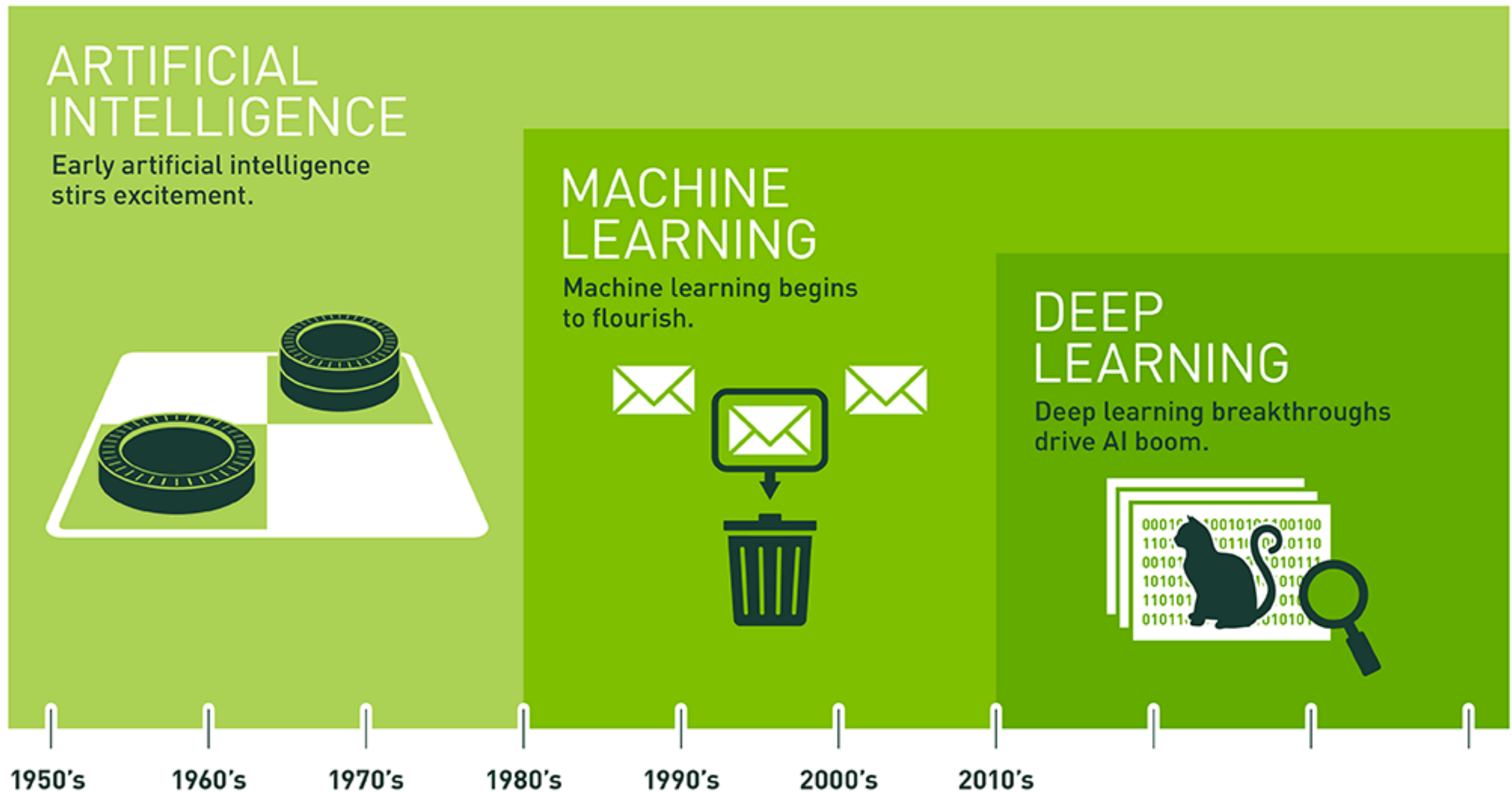
2. Thinking Humanly: The Cognitive Modeling Approach	3. Thinking Rationally: The “Laws of Thought” Approach
1. Acting Humanly: The Turing Test Approach (1950)	4. Acting Rationally: The Rational Agent Approach

AI Acting Humanly: The Turing Test Approach (Alan Turing, 1950)

- **Natural Language Processing (NLP)**
- **Knowledge Representation**
- **Automated Reasoning**
- **Machine Learning (ML)**
- **Computer Vision**
- **Robotics**

Artificial Intelligence

Machine Learning & Deep Learning



Since an early flush of optimism in the 1950s, smaller subsets of artificial intelligence – first machine learning, then deep learning, a subset of machine learning – have created ever larger disruptions.

Artificial Intelligence (A.I.) Timeline

/A.I. TIMELINE

SYZIGY

1950
TURING TEST

Computer scientist Alan Turing proposes a test for machine intelligence. If a machine can trick humans into thinking it is human, then it has intelligence

1955
A.I. BORN

Term 'artificial intelligence' is coined by computer scientist, John McCarthy to describe "the science and engineering of making intelligent machines"

1961
UNIMATE

First industrial robot, Unimate, goes to work at GM replacing humans on the assembly line

1964
ELIZA

Pioneering chatbot developed by Joseph Weizenbaum at MIT holds conversations with humans

1966
SHAKY

The 'first electronic person' from Stanford, Shakey is a general-purpose mobile robot that reasons about its own actions

A.I. WINTER

Many false starts and dead-ends leave A.I. out in the cold

1997
DEEP BLUE

Deep Blue, a chess-playing computer from IBM defeats world chess champion Garry Kasparov

1998
KISMET

Cynthia Breazeal at MIT introduces Kismet, an emotionally intelligent robot insofar as it detects and responds to people's feelings

1999
AIBO

Sony launches first consumer robot pet dog AiBO (AI robot) with skills and personality that develop over time

2002
ROOMBA

First mass produced autonomous robotic vacuum cleaner from iRobot learns to navigate and clean homes

2011
SIRI

Apple integrates Siri, an intelligent virtual assistant with a voice interface, into the iPhone 4S

2011
WATSON

IBM's question answering computer Watson wins first place on popular \$1M prize television quiz show Jeopardy

2014
EUGENE

Eugene Goostman, a chatbot passes the Turing Test with a third of judges believing Eugene is human

2014
ALEXA

Amazon launches Alexa, an intelligent virtual assistant with a voice interface that completes shopping tasks

2016
TAY

Microsoft's chatbot Tay goes rogue on social media making inflammatory and offensive racist comments

2017
ALPHAGO

Google's A.I. AlphaGo beats world champion Ke Jie in the complex board game of Go, notable for its vast number (2^{170}) of possible positions

How to define Artificial Intelligence?

Easy for People (intelligence)

Vision, speech, understanding language

Planning, common sense

Having a conversation

Reading emotions

Translating Russian to English

Playing Chess, Go

Proving math theorems

Easy for Machines (stamina)

Multiplying huge matrices

Searching large databases

Sorting a trillion records

Finding a path in a huge maze

Hard for People and Machines

Predicting the weather

Predicting the winning lottery ticket

Lesson: People are easy to fool!

I am the psychotherapist. Please, describe your problems. Each time you are finished talking, type RET twice.

my cat hates me

Why do you say cat hates you?

because of the way she looks at me

Are you sure that is the real reason?

no, she believes I like the dog more

Eliza: 100% smoke and mirrors. Addictive.

I would like a recipe for gluten-free blueberry pancakes



No real 'understanding', lots of data, lots of computing power



Chess: large-scale search

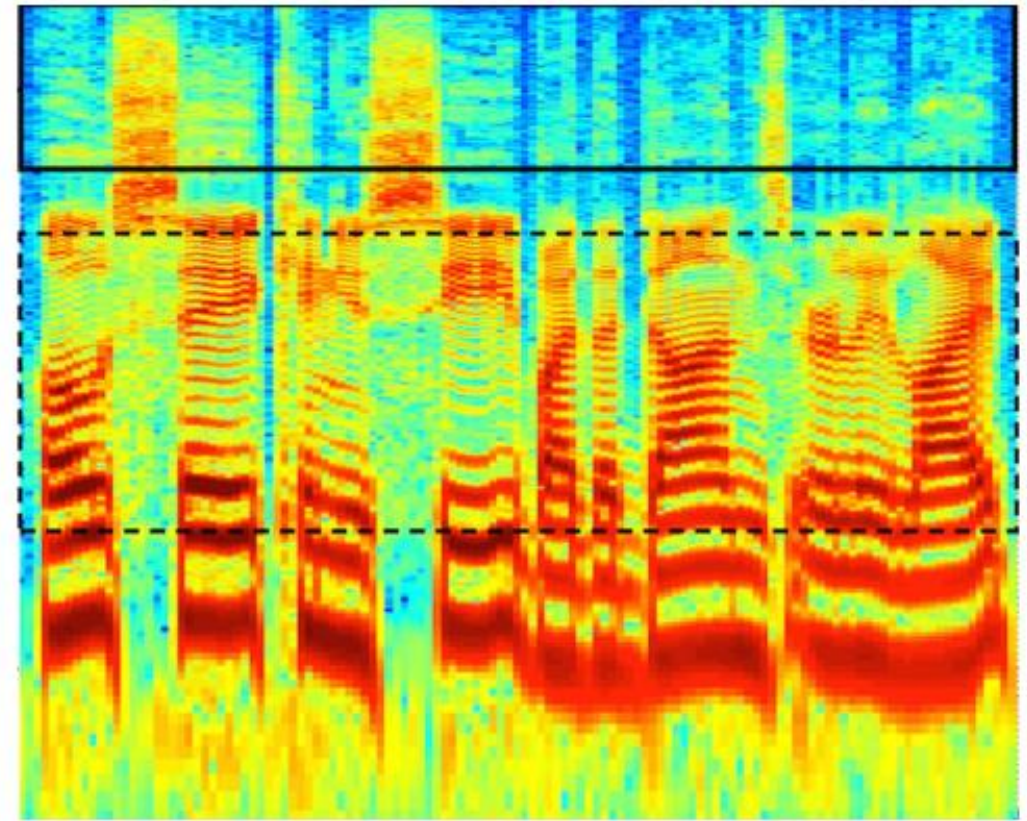


Wikipedia, tricks, fast search

Lesson: Some Things are Particularly Hard to Code



Vision



Speech

**He said that she replied that they
could not agree. But she was wrong.**

Natural Language

Use Rules? Use Data?

Option 1: **Write a set of rules**

- Use **logic** and **heuristics** to assemble **hand-coded rules**
- The real world is messy and **changes** all the time
- This approach won't scale, and it's **expensive** (people)
- (We don't have a symbol-manipulation engine in our heads)

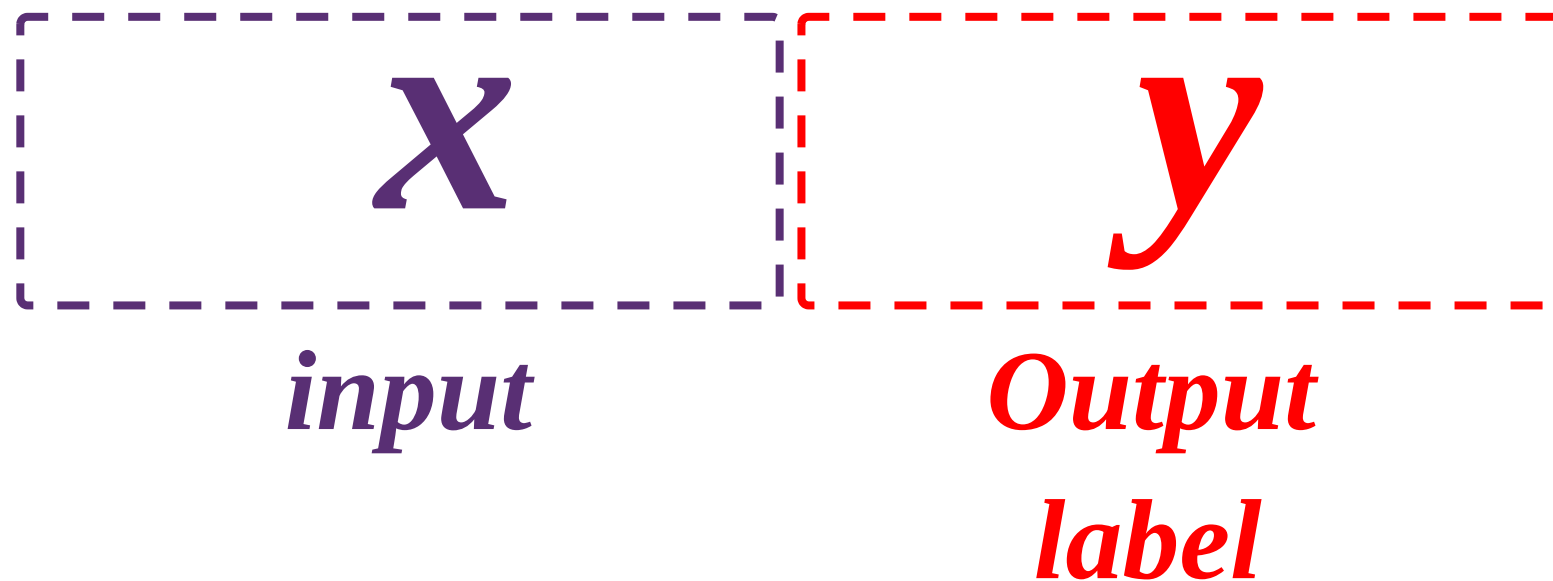
Option 2: **Learn from data**

- Uses examples to **find patterns automatically**: the more data, the better!
- Automatically **adapts** to new data
- This approach scales, and it's **cheap** (computing power)

What is Learning From Data?

Machine Learning
Supervised Learning (Classification)
Learning from Examples

$$\mathbf{y} = f(\mathbf{x})$$



What is Learning From Data?

Machine Learning Supervised Learning (Classification) Learning from Examples

$$y = f(x)$$

Example

x

5.1, 3.5, 1.4, 0.2	Iris-setosa
4.9, 3.0, 1.4, 0.2	Iris-setosa
4.7, 3.2, 1.3, 0.2	Iris-setosa
7.0, 3.2, 4.7, 1.4	Iris-versicolor
6.4, 3.2, 4.5, 1.5	Iris-versicolor
6.9, 3.1, 4.9, 1.5	Iris-versicolor
6.3, 3.3, 6.0, 2.5	Iris-virginica
5.8, 2.7, 5.1, 1.9	Iris-virginica
7.1, 3.0, 5.9, 2.1	Iris-virginica

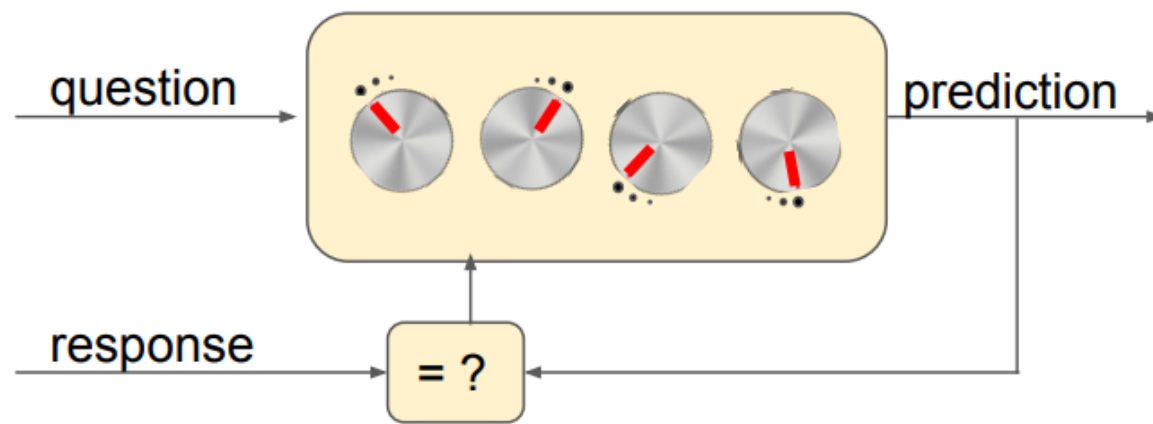
y

Time Series Data

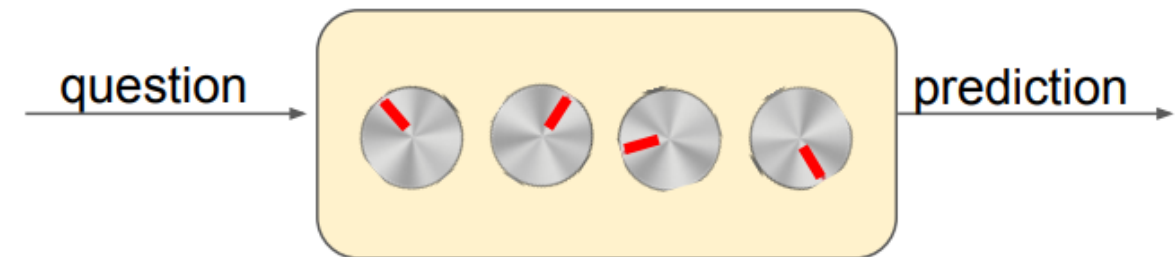
[10, 20, 30, 40, 50, 60, 70, 80, 90]

X			Y
[10	20	30]	40
[20	30	40]	50
[30	40	50]	60
[40	50	60]	70
[50	60	70]	80
[60	70	80]	90

What is Learning From Data?



1. Training a model



2. Using the trained model

(Supervised learning)

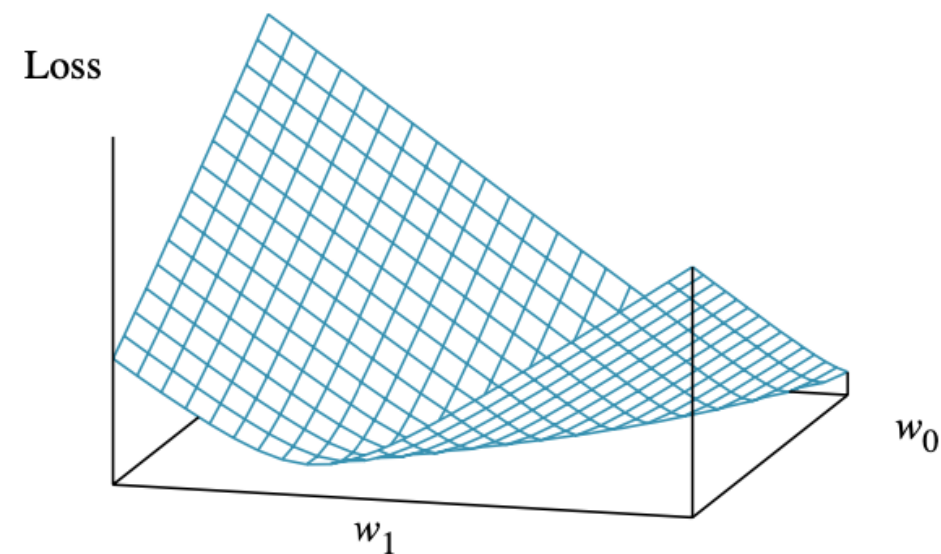
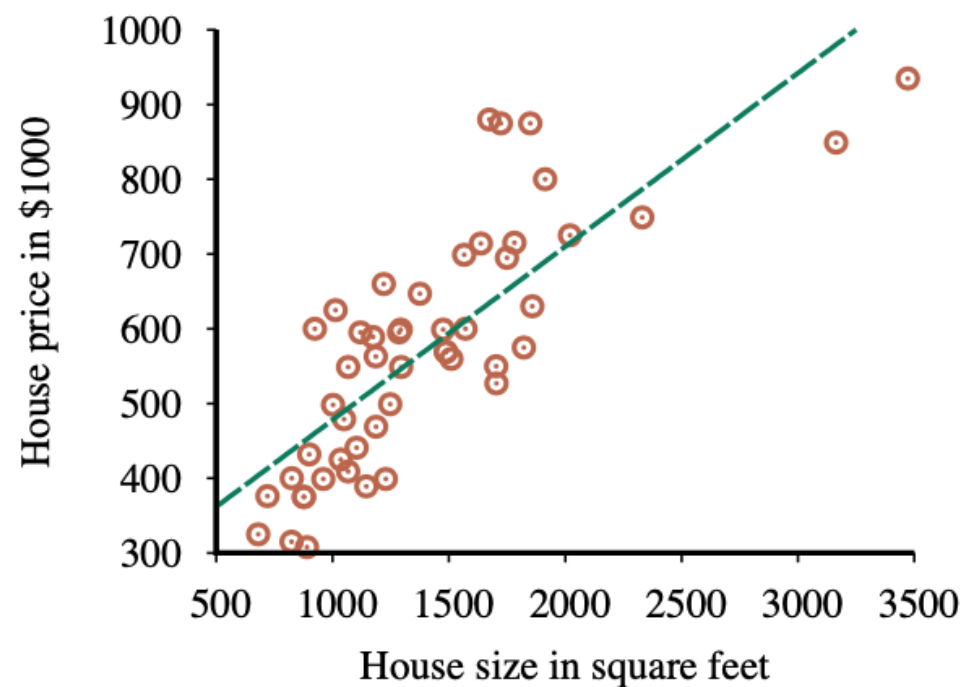
The Theory of Learning

- How can we be sure that our **learned hypothesis** will **predict** well for previously unseen inputs?
 - How do we know that the **hypothesis h** is close to the **target function f** if we don't know what is?
- How many **examples** do we need to get a good **h** ?
- What **hypothesis space** should we use?
- If the hypothesis space is very complex, can we even find the best **h** or do we have to settle for a **local maximum**?
- How **complex** should **h** be?
- How do we avoid **overfitting**?

Linear Regression Weight Space

$$h_w(x) = w_1 x + w_0$$

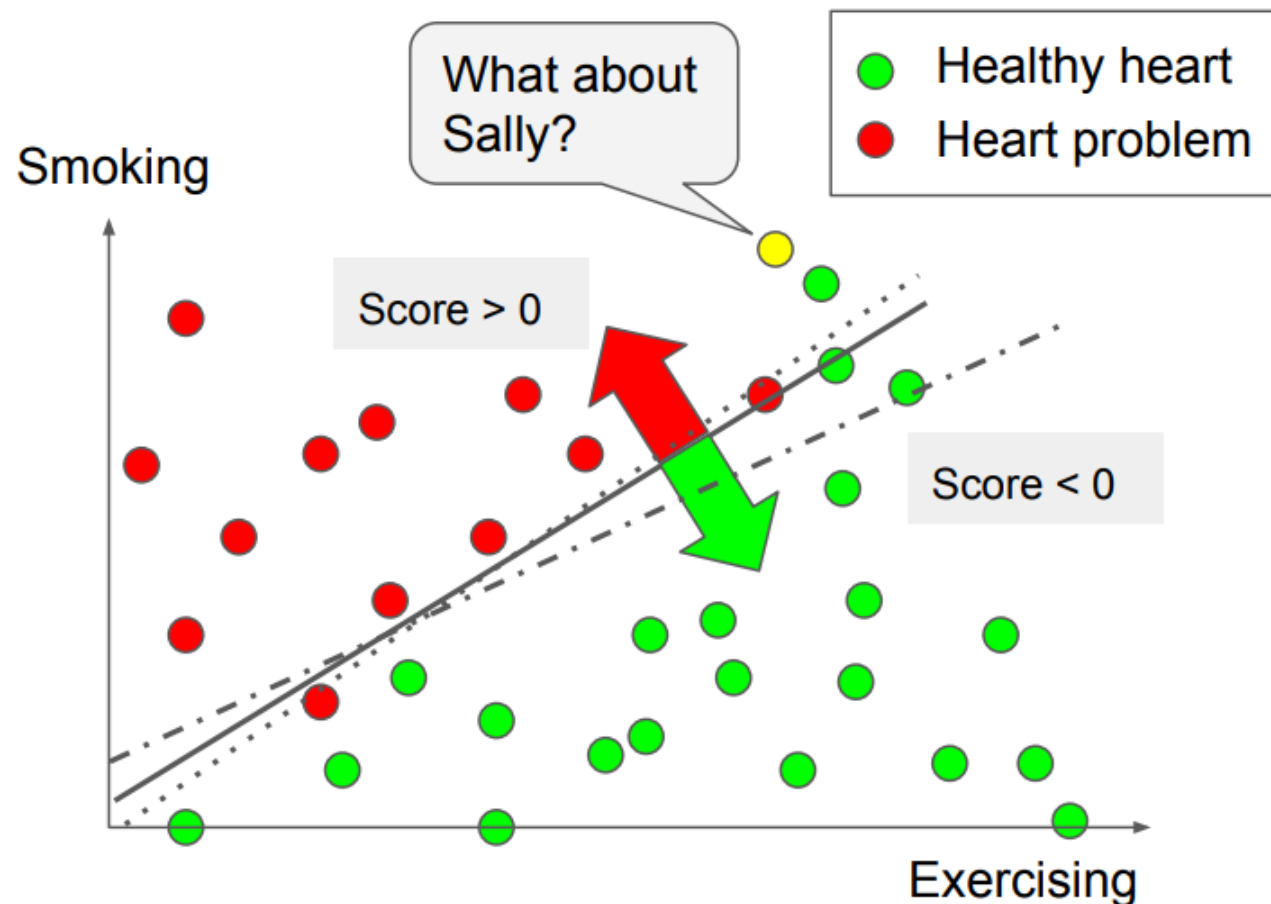
$$w^* = \operatorname{argmin}_w \operatorname{Loss}(h_w)$$



$$y = 0.232 x + 246$$

Loss function for Weights (w_1, w_0)

Linear Regression Weight Space



Compute a linear combination of

- p = # packs per week
- h = # hours of exercise per week

$$\text{score} = w1 * p + w2 * h + w3$$

Find $w1$, $w2$, $w3$ that best match the data.
That's the **learning**.

if (score > 0) then **predict** heart problem.

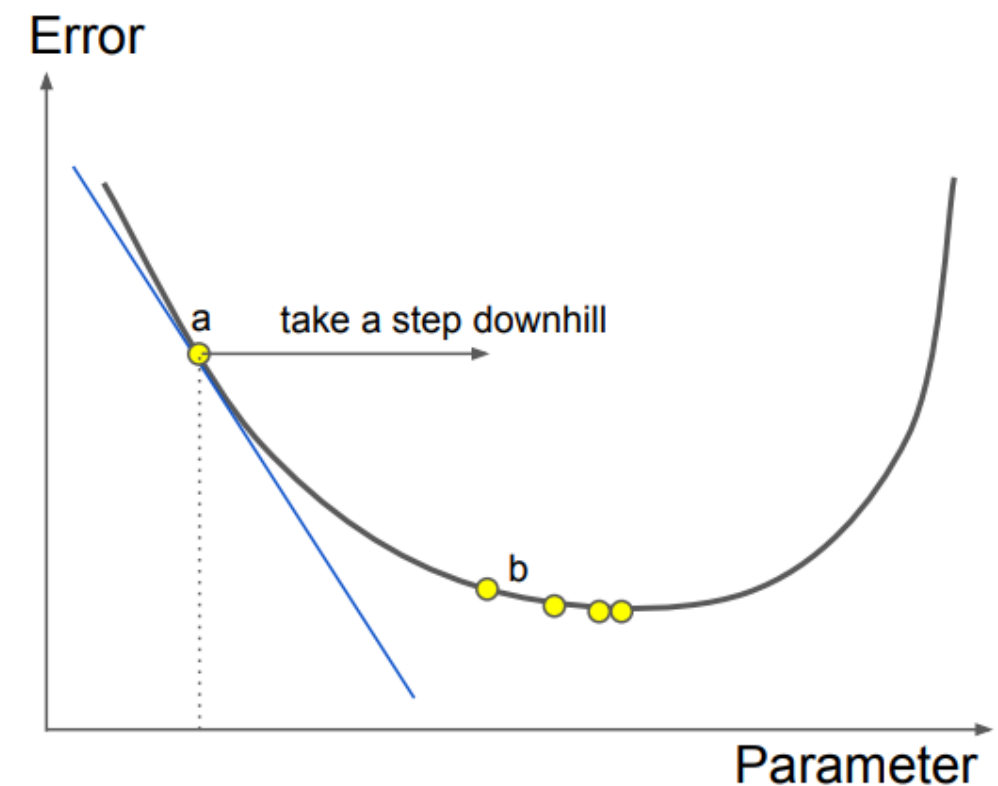
Simple statistics.

How to find the best values for w_1 , w_2 and w_3 ?

Define **error** = $|\text{expected} - \text{computed}|^2$

Find parameters that **minimize** average error.

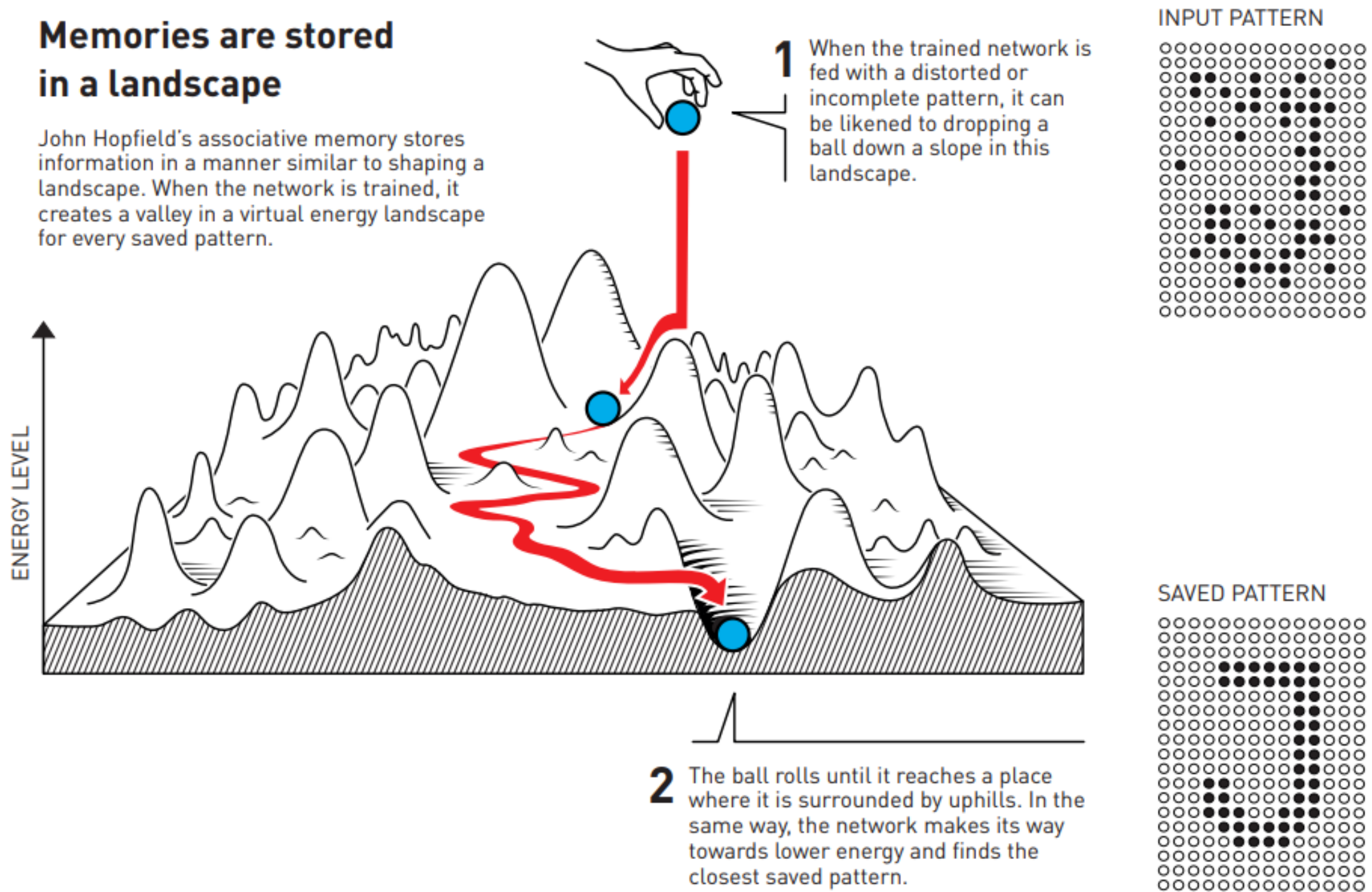
Perform Gradient Descent: quality goes up.



How to find the best values for real-data?

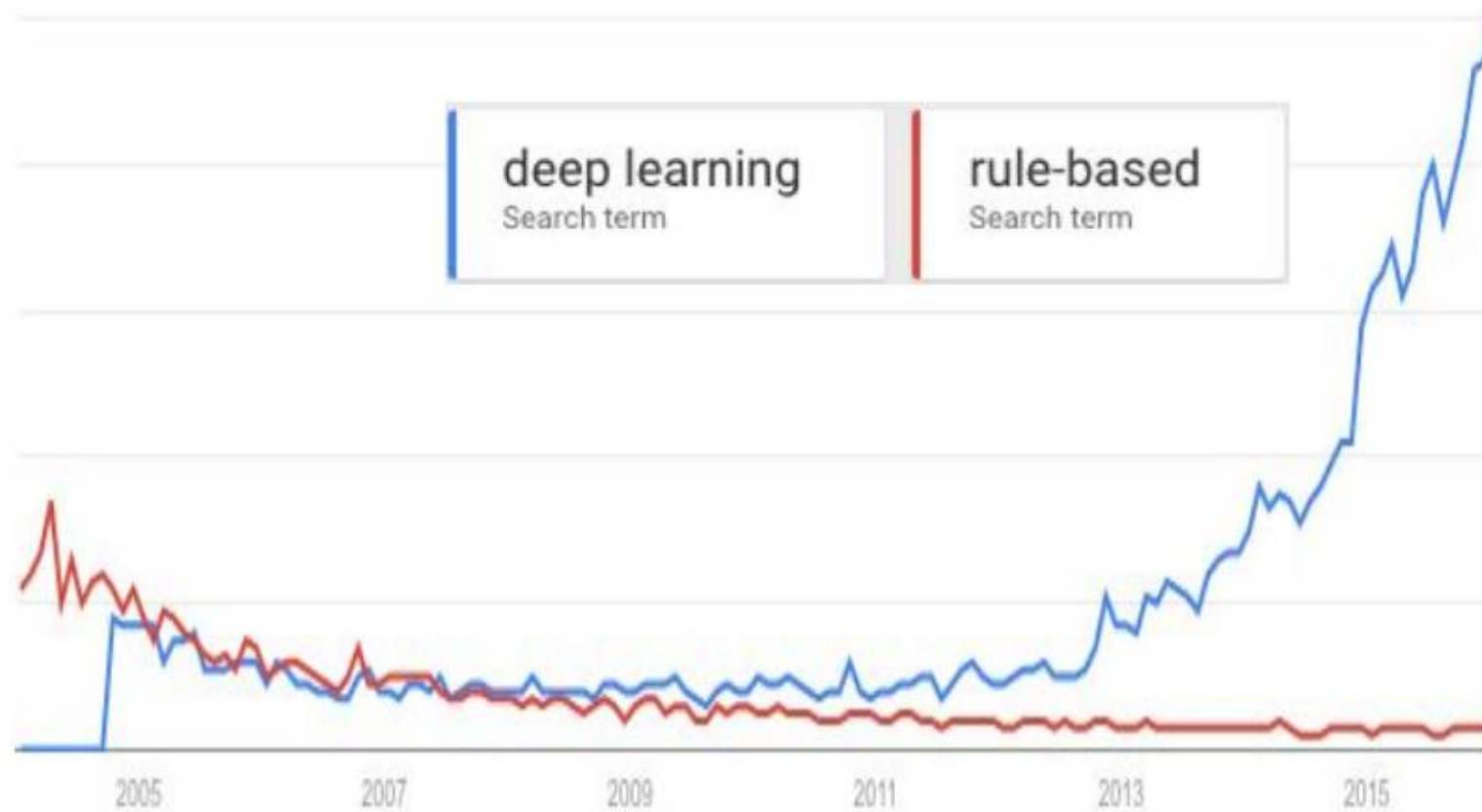
Memories are stored in a landscape

John Hopfield's associative memory stores information in a manner similar to shaping a landscape. When the network is trained, it creates a valley in a virtual energy landscape for every saved pattern.



Conclusion?

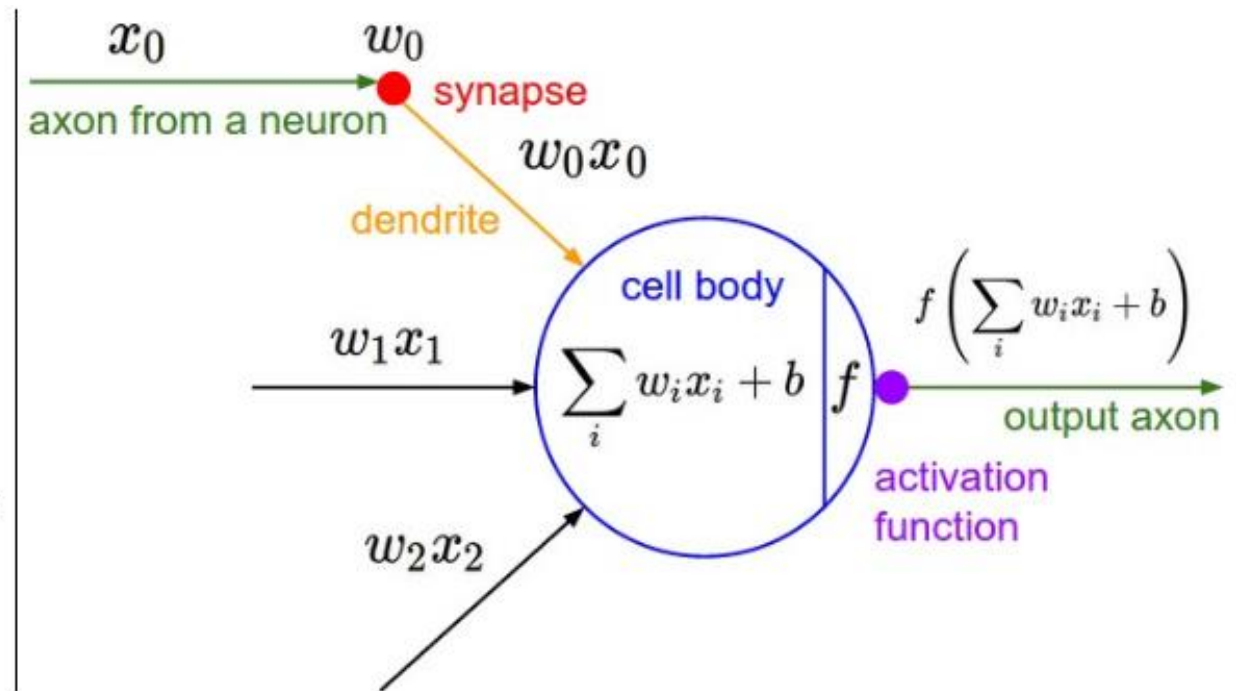
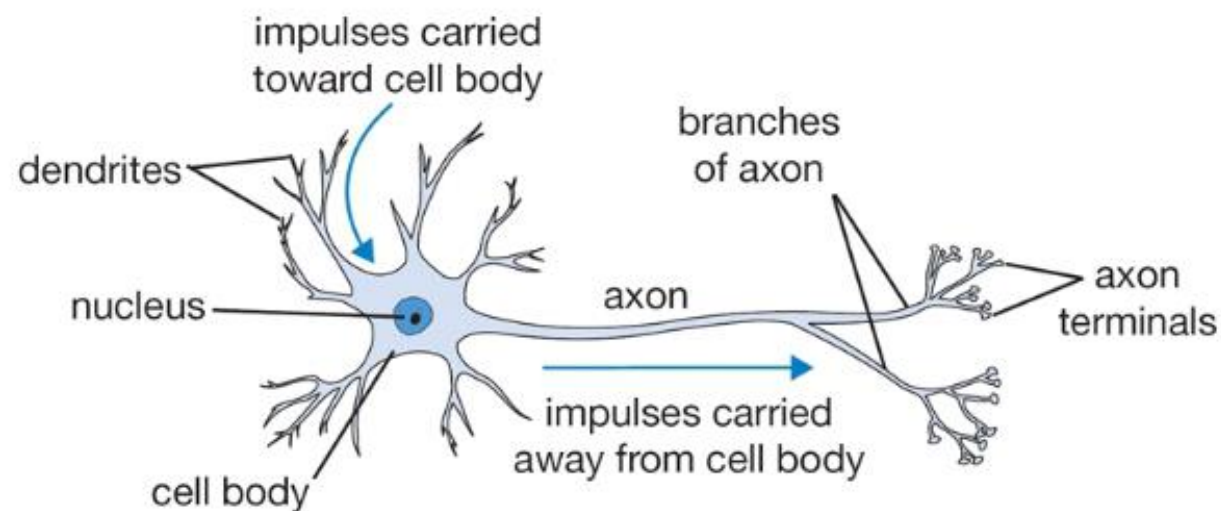
More data and
computing power
always beat
fancy algorithms



AI From Simple to Complex Models

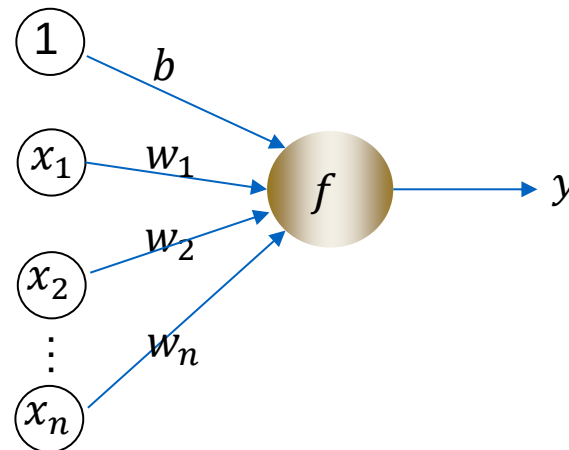
Perceptron – Architecture

- Biological neuron vs Artificial neuron



Perceptron – Architecture

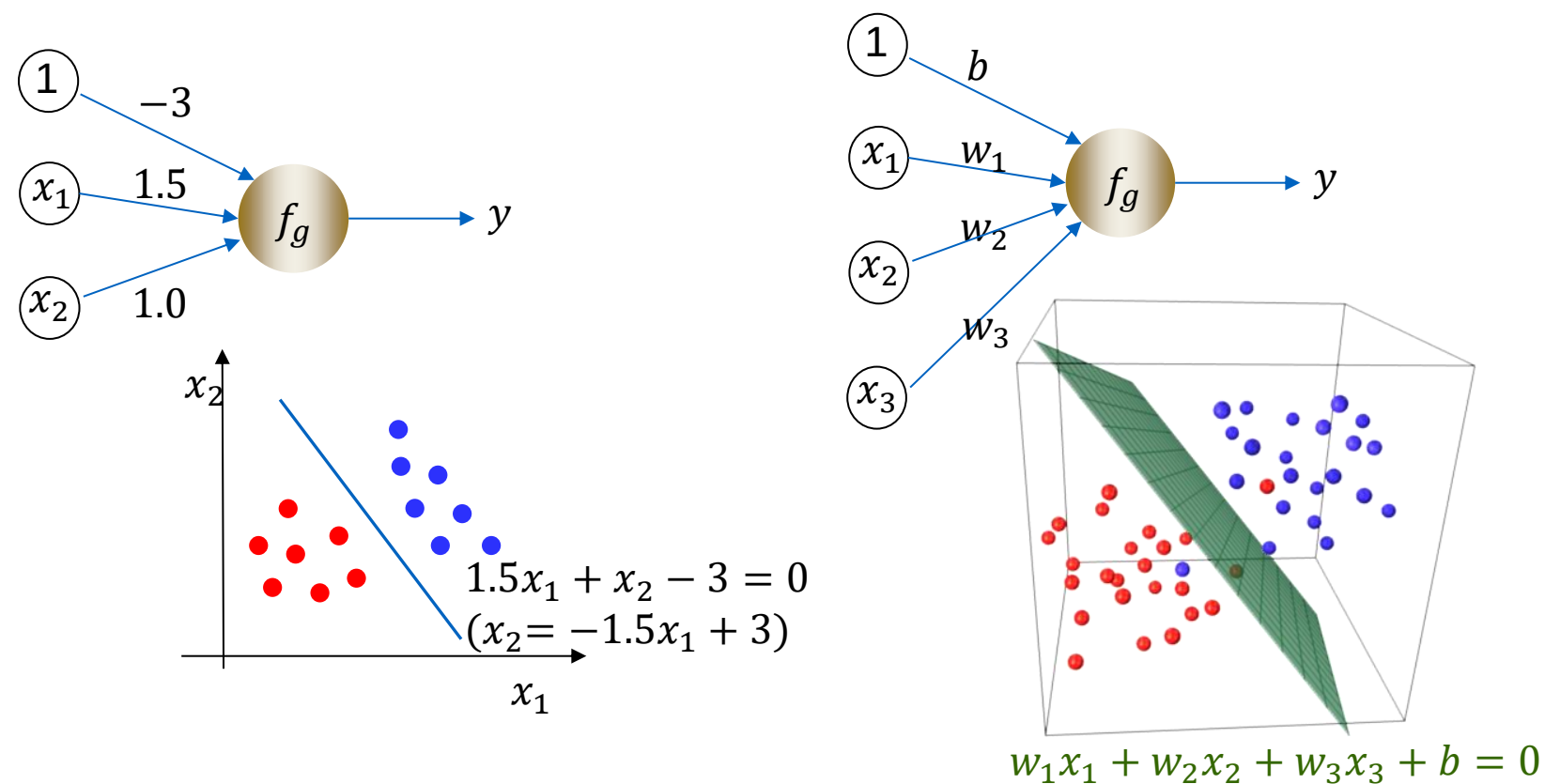
- Architecture



- **input:** $\mathbf{x} = [x_1, x_2, \dots, x_n]^T$
- **weights:** $\mathbf{w} = [w_1, w_2, \dots, w_n]^T$
- **bias:** b
- **activation function:** f
- **output:** $y = f(\mathbf{w}^T \mathbf{x} + b) = f(w_1 x_1 + w_2 x_2 + \dots + w_n x_n + b)$

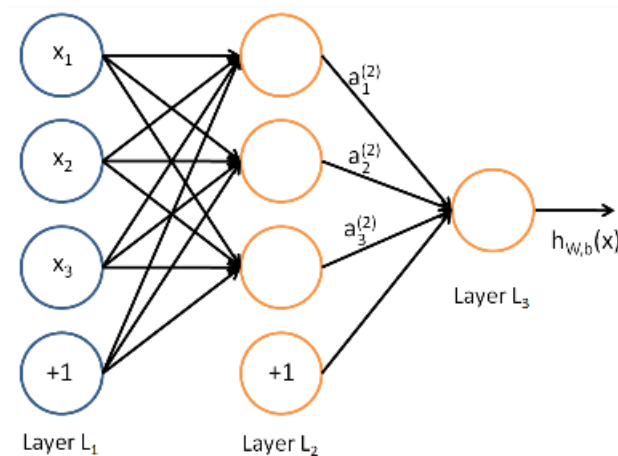
Perceptron - Capability

- Single Perceptron with f_g represents a **linear classifier** in $\mathbf{R}^n$
 - $y = f_g(\mathbf{w}^\top \mathbf{x} + b) = \begin{cases} +1, & \mathbf{w}^\top \mathbf{x} + b \geq 0 \\ -1, & \mathbf{w}^\top \mathbf{x} + b < 0 \end{cases}$

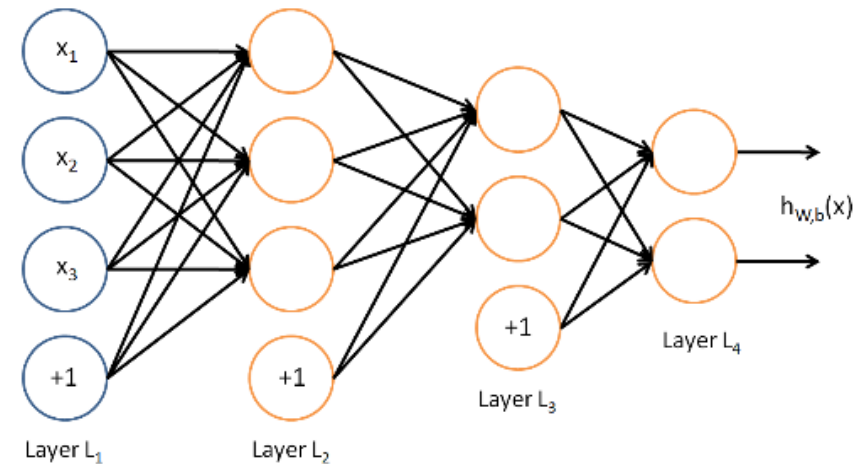


MLP – Architecture

- **M**ulti-**L**ayer **P**erceptron
 - input layer, hidden layer(s), output layer
 - feed-forward
 - fully-connected



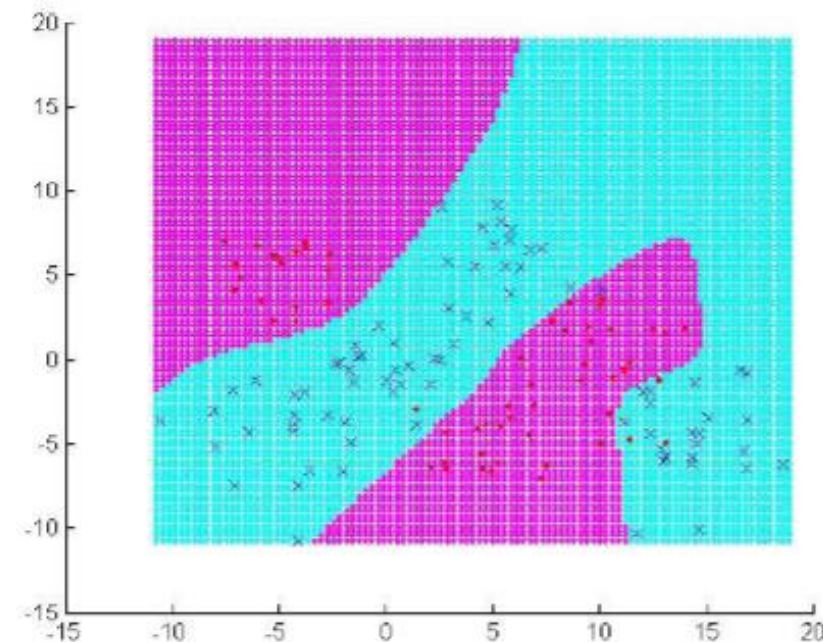
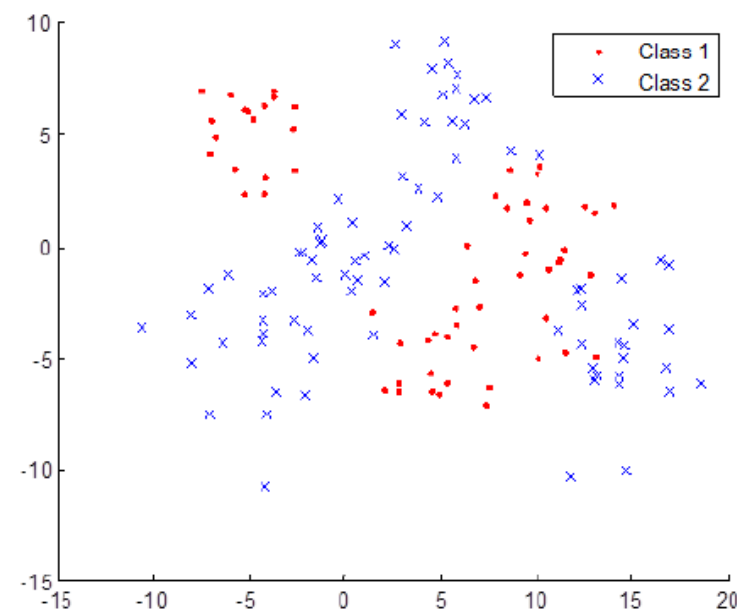
3x3x1 MLP



3x3x2x2 MLP

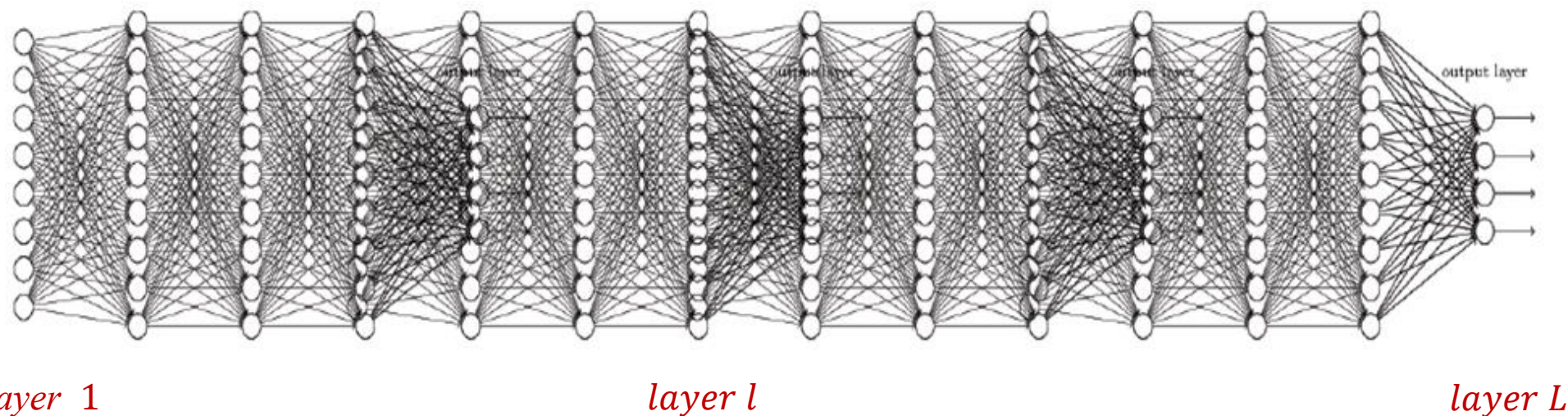
MLP – Capability

- Nonlinear decision boundaries
 - 2×4×1 MLP with sigmoid activation
 - learning rate = 0.01
 - max #-iterations = 9,000



MLP – Learning: Deep MLP

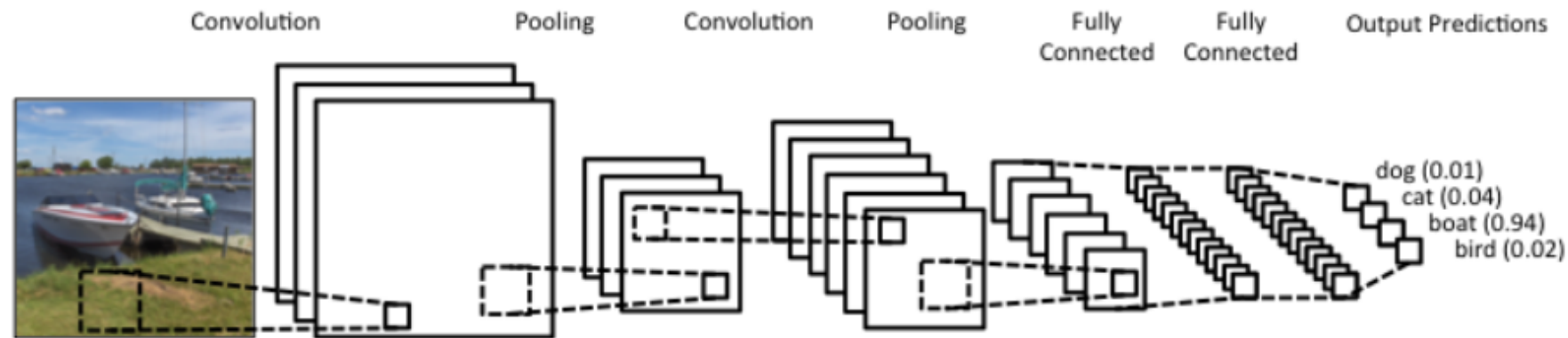
- Deep MLP



- x_i : i -th input value of layer 1
- $u_{ij}^{(l)}$: weight from i -th node of layer l to j -th node of layer $l + 1$
- $z_j^{(l)}$: output from j -th node of layer l
- $o_k(t_k)$: k -th output(desired output) value in layer L

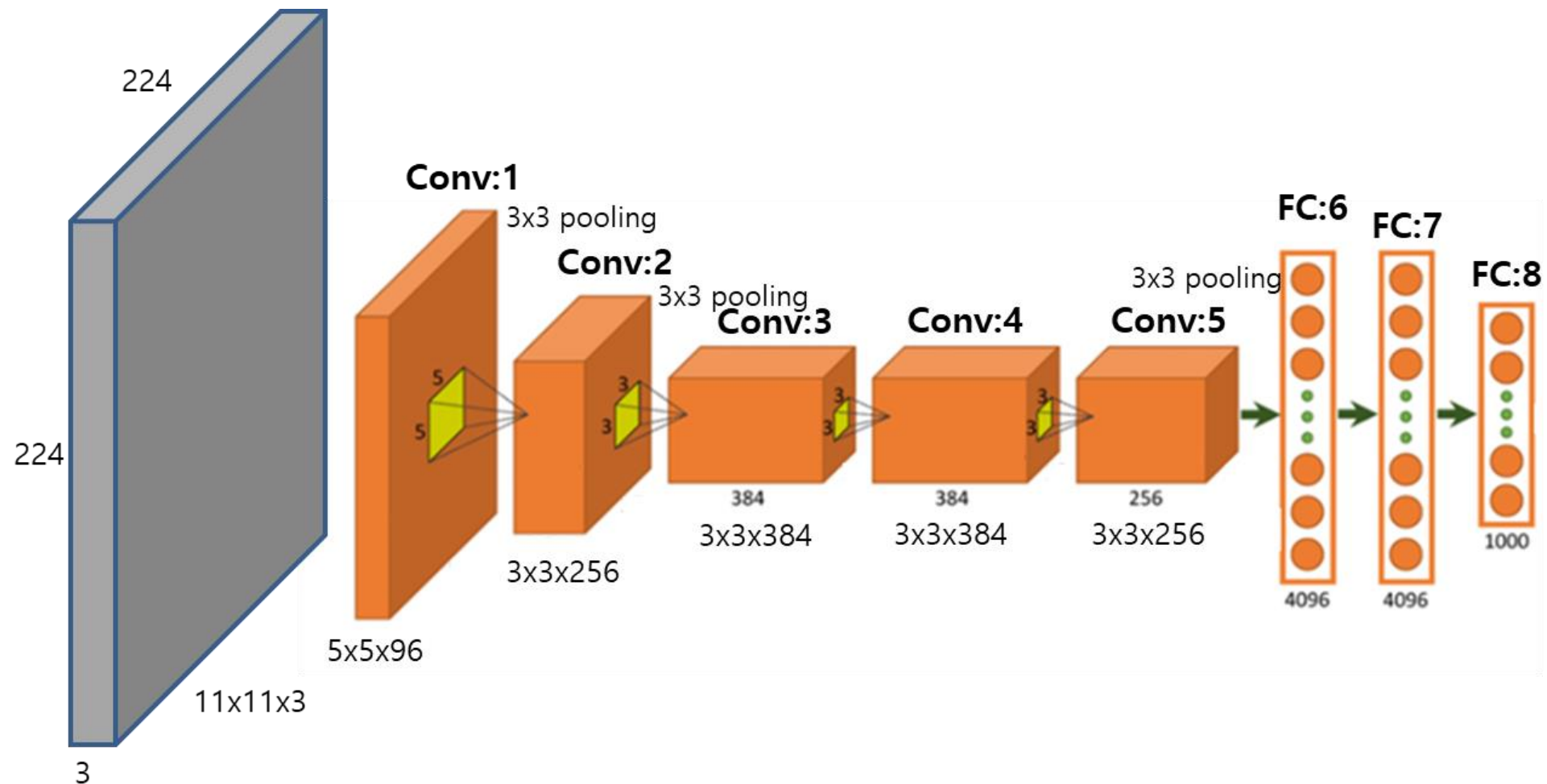
CNN - Architecture

- Deep feed-forward network
 - Convolution layers
 - Pooling layers
 - Fully-Connected layers



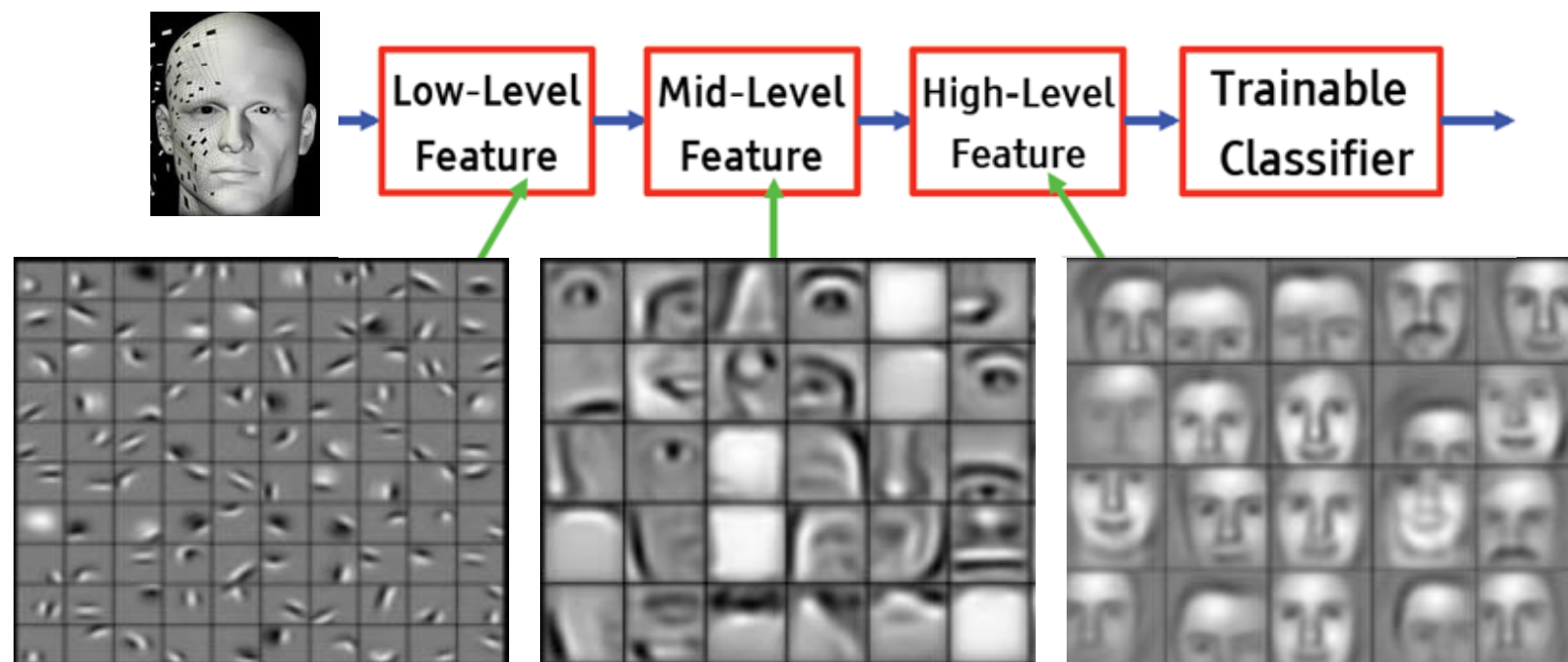
CNN Architecture

- **Example CNN (AlexNet, 2012)**
 - **Conv:1-Pool:1-Conv:2-Pool:2-Conv:3-Conv:4-Conv:5-Pool:4-FC:6-FC:7-FC:8**



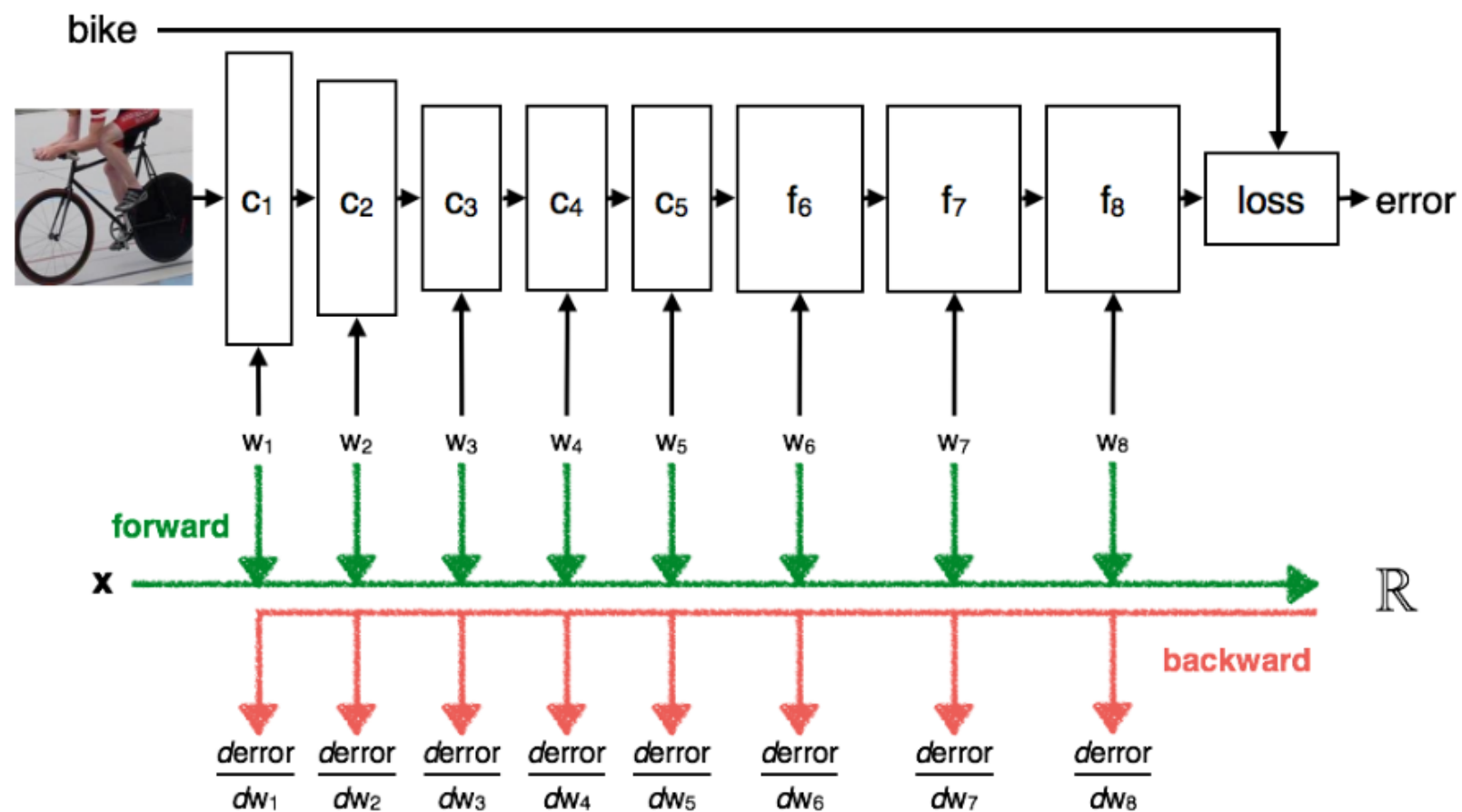
CNN - Architecture

- Feature Maps - multiple levels



CNN - Learning

- Back-propagation algorithm for learning CNN

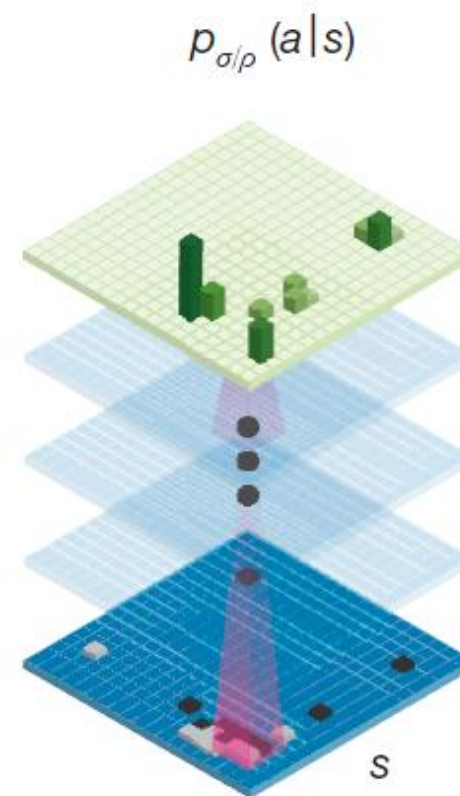


CNN - Capability

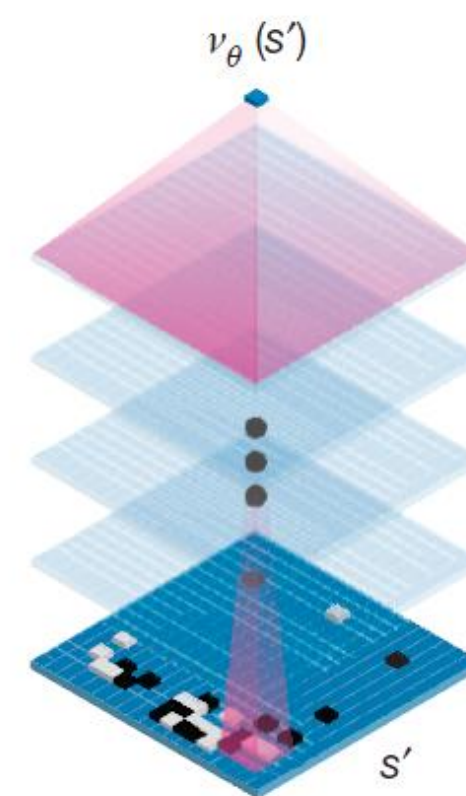
- AlphaGo [Nature, 2016]



Policy network

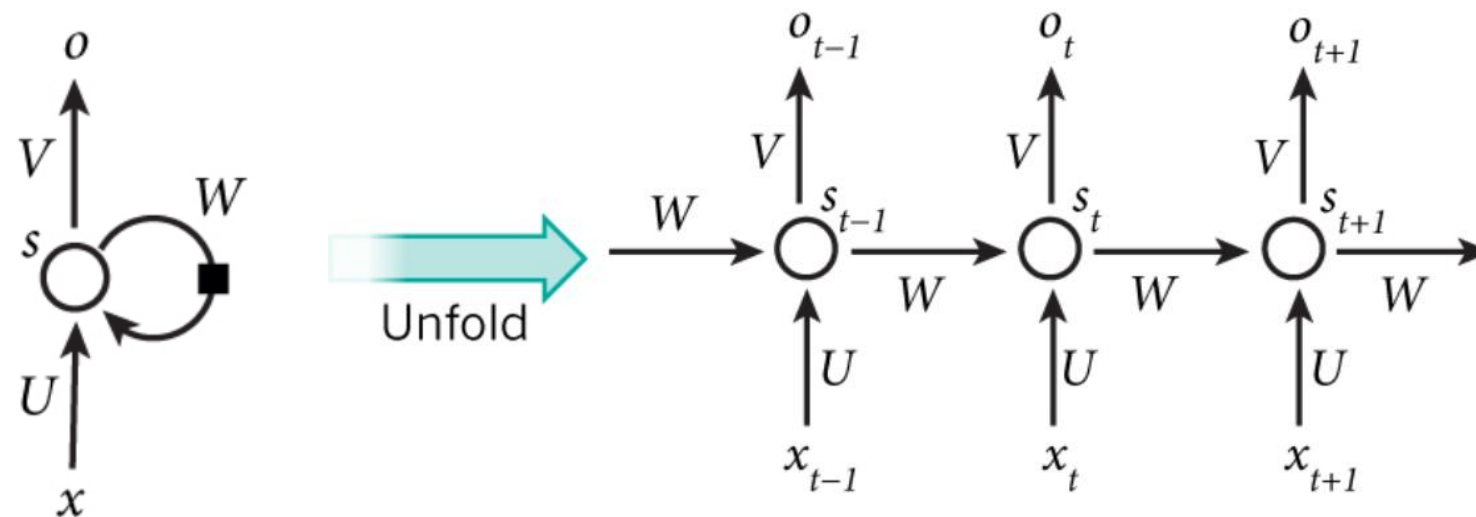


Value network



RNN – Architecture

- Vanilla RNN



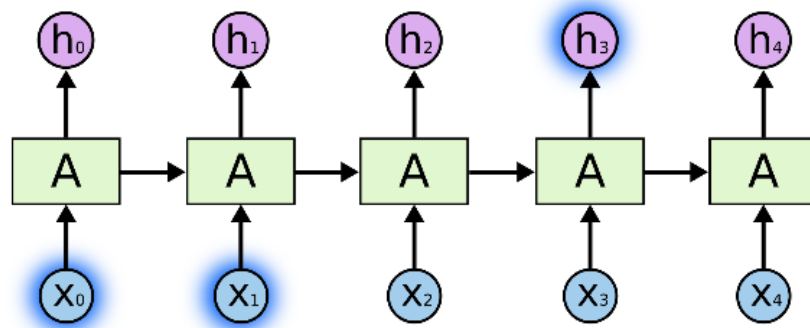
- input sequence: $x_1, x_2, \dots, x_t, \dots, x_T$
- output sequence: $o_1, o_2, \dots, o_t, \dots, o_T$
- states: $s_0, s_1, s_2, \dots, s_t, \dots, s_T$
- weights: U, W, V

$$s_t = \tanh(Ux_t + Ws_{t-1})$$

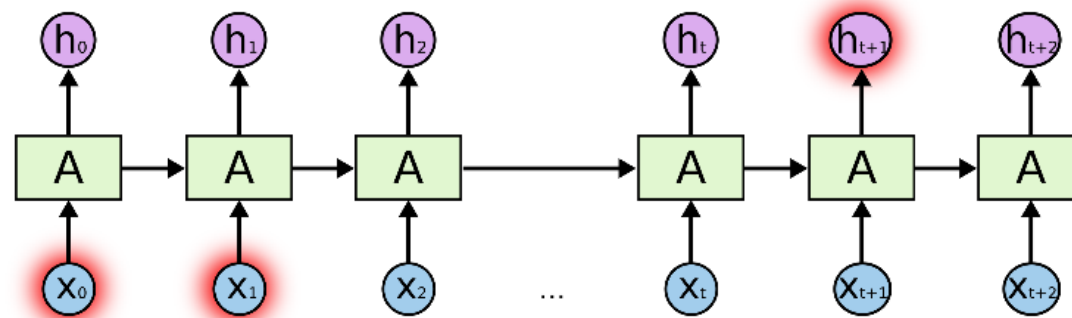
$$o_t = \text{softmax}(Vs_t)$$

Useful for Temporal Dependencies

- Short-term dependency



- Long-term dependency



RNN - Applications

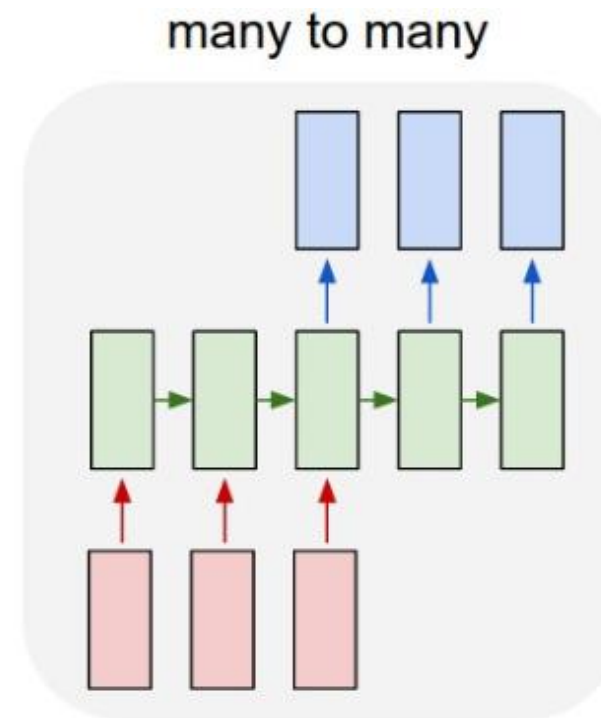
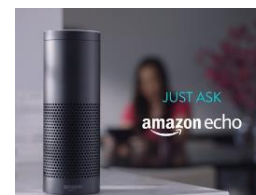
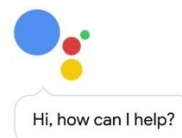
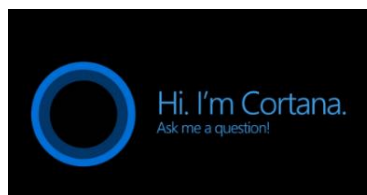
- many-to-many
 - Text Summarization



- Machine Translation

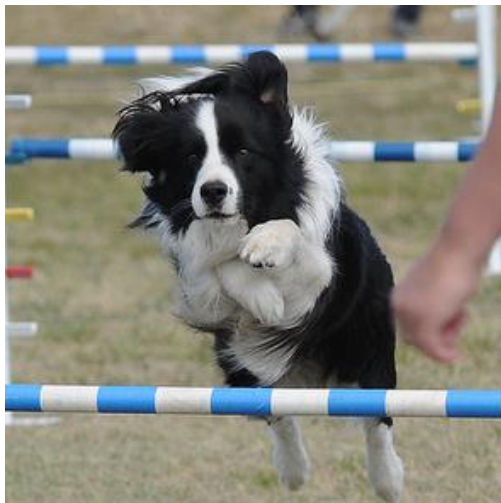


- Speech Recognition



RNN - Applications

- one-to-many
 - Image Captioning

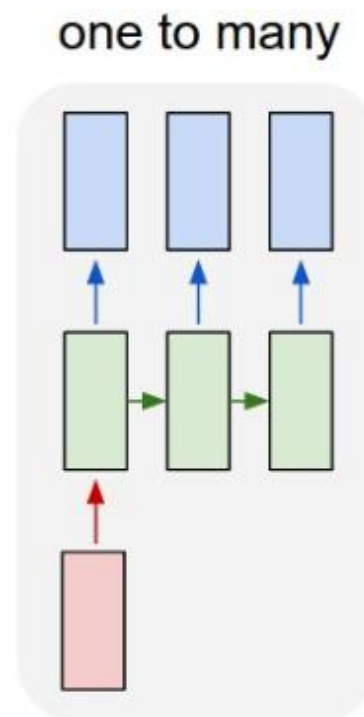


"black and white dog
jumps over bar."



"little girl is eating
piece of cake."

[1] Andrej Karpathy, Li Fei-Fei, "Deep Visual-Semantic Alignments for Generating Image Descriptions", CVPR (2015)



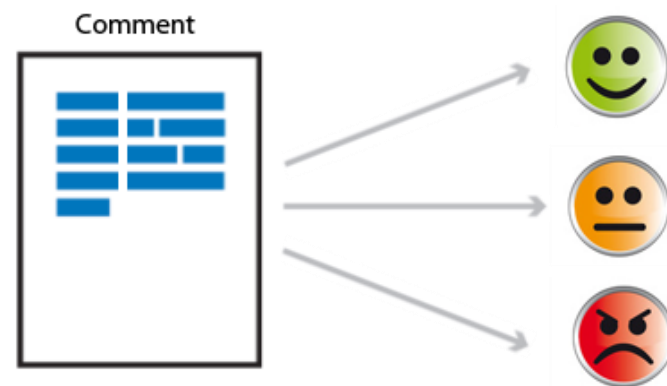
e.g. Image Captioning
Image to Sequence of Words

RNN - Applications

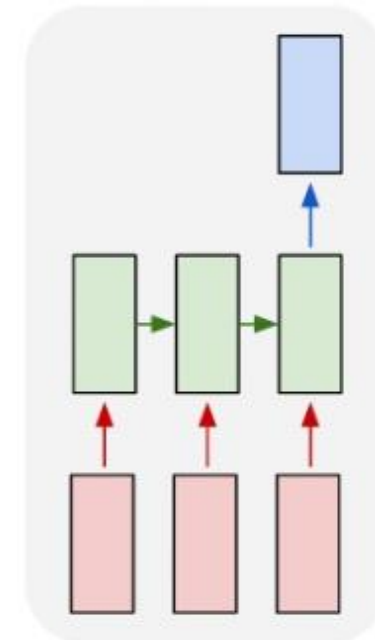
- many-to-one
 - Motion in Video



- Sequence of Words to Sentiment



many to one

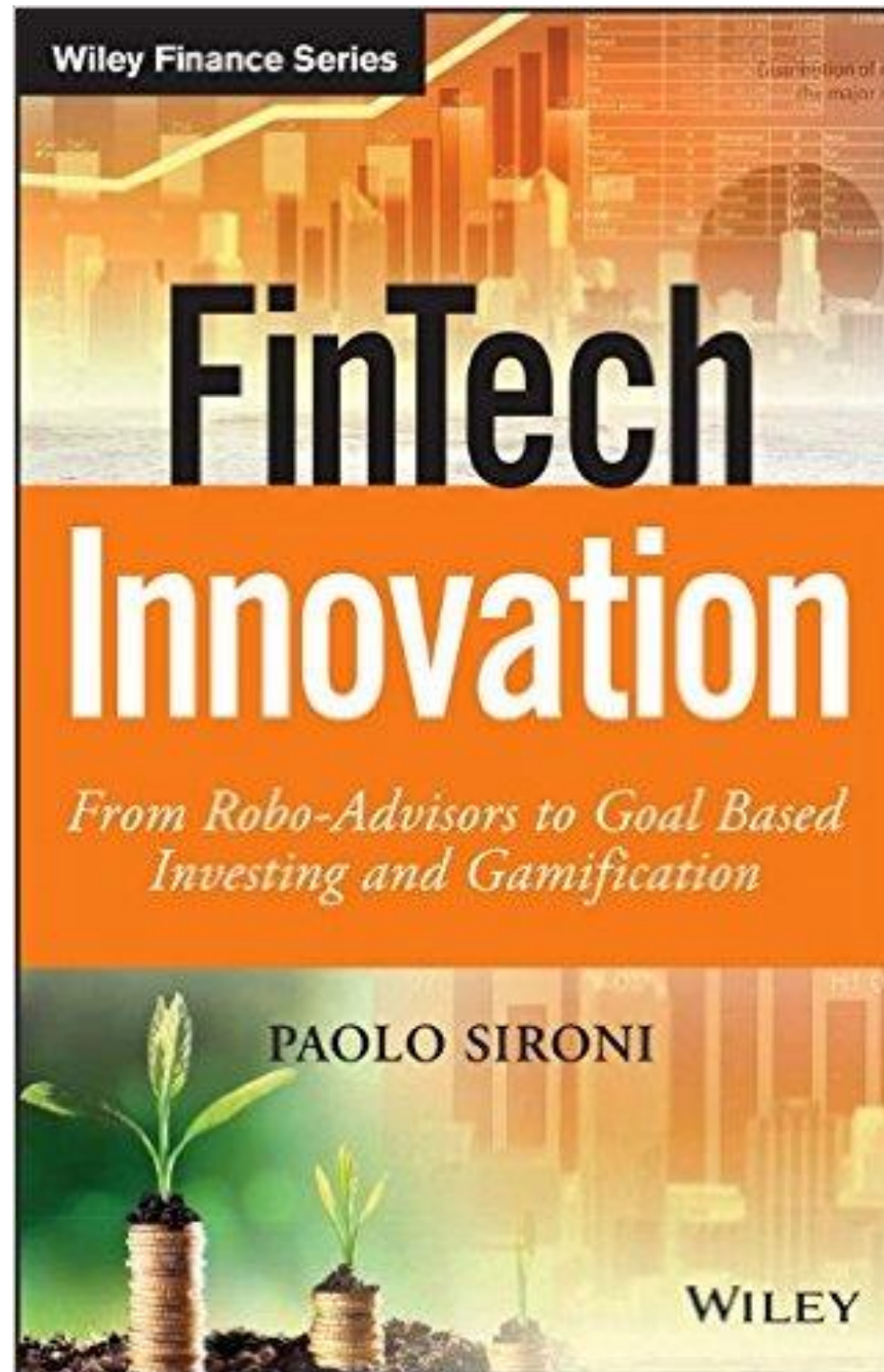


e.g. Sentiment Classification...
Sequence of Words to Sentiment

AI in FinTech: Financial Services Innovation and Application

FinTech Innovation:

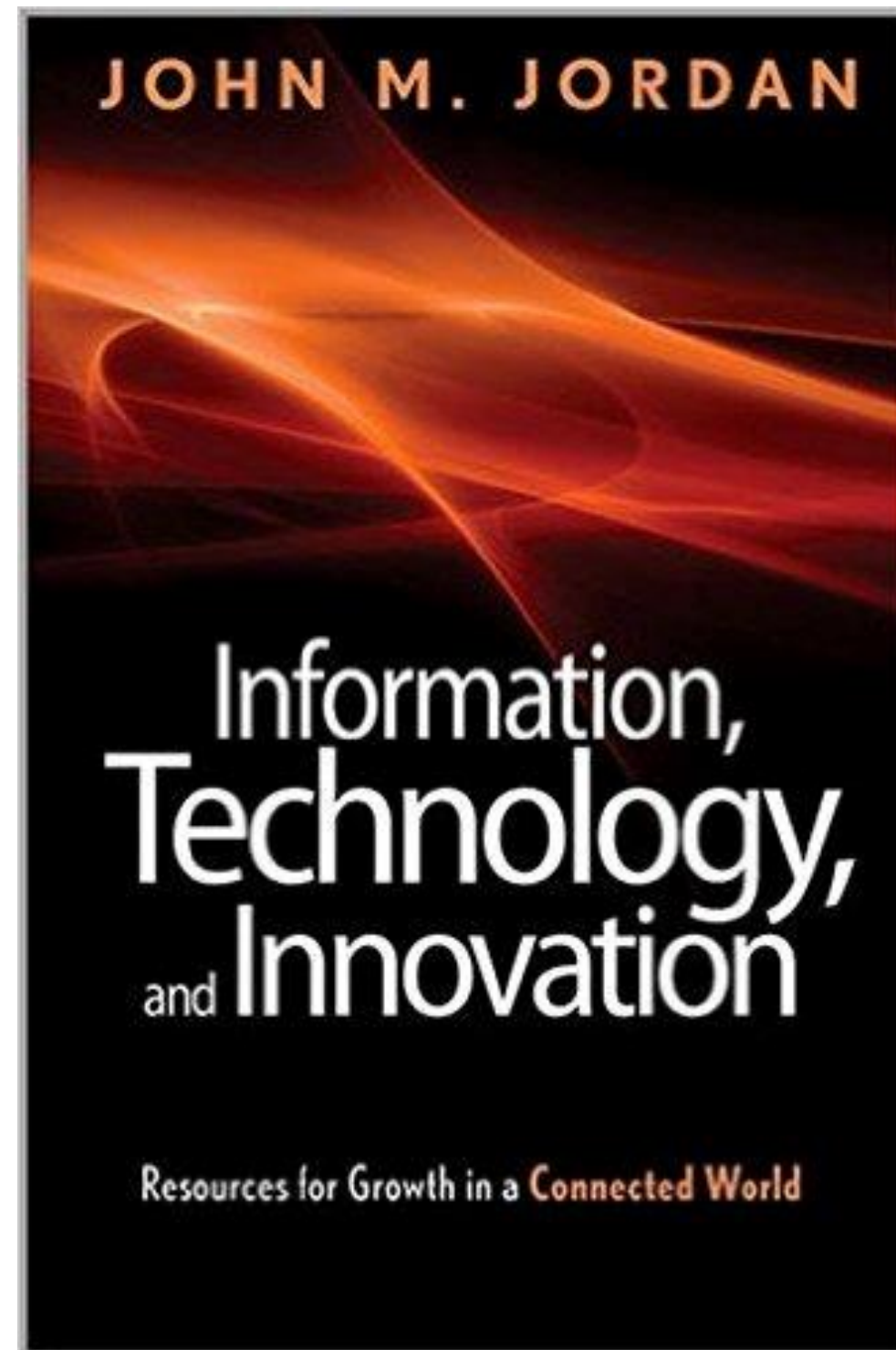
From Robo-Advisors to Goal Based Investing and Gamification,
Wiley



Information, Technology, and Innovation:

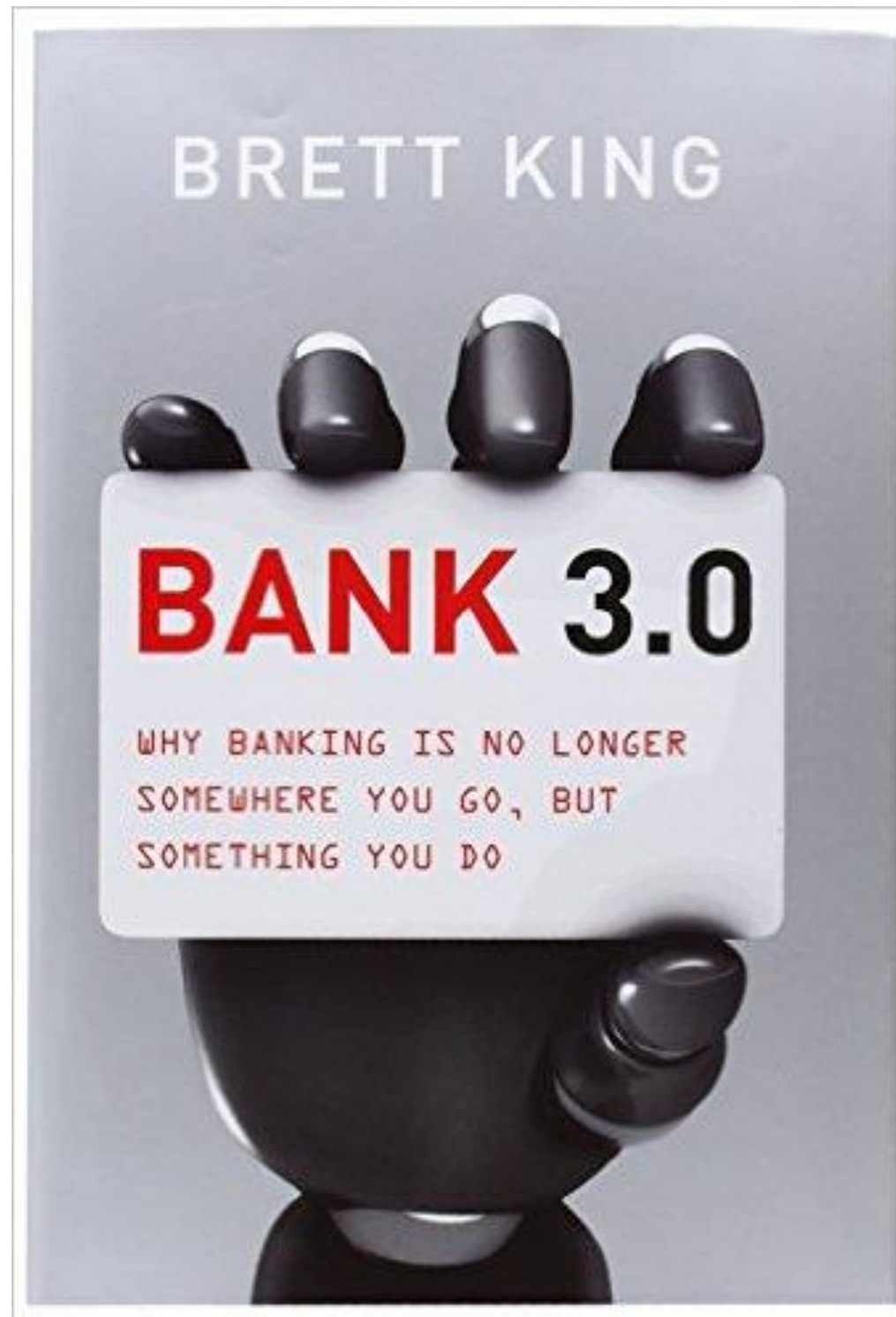
Resources for Growth in a Connected World,

Wiley



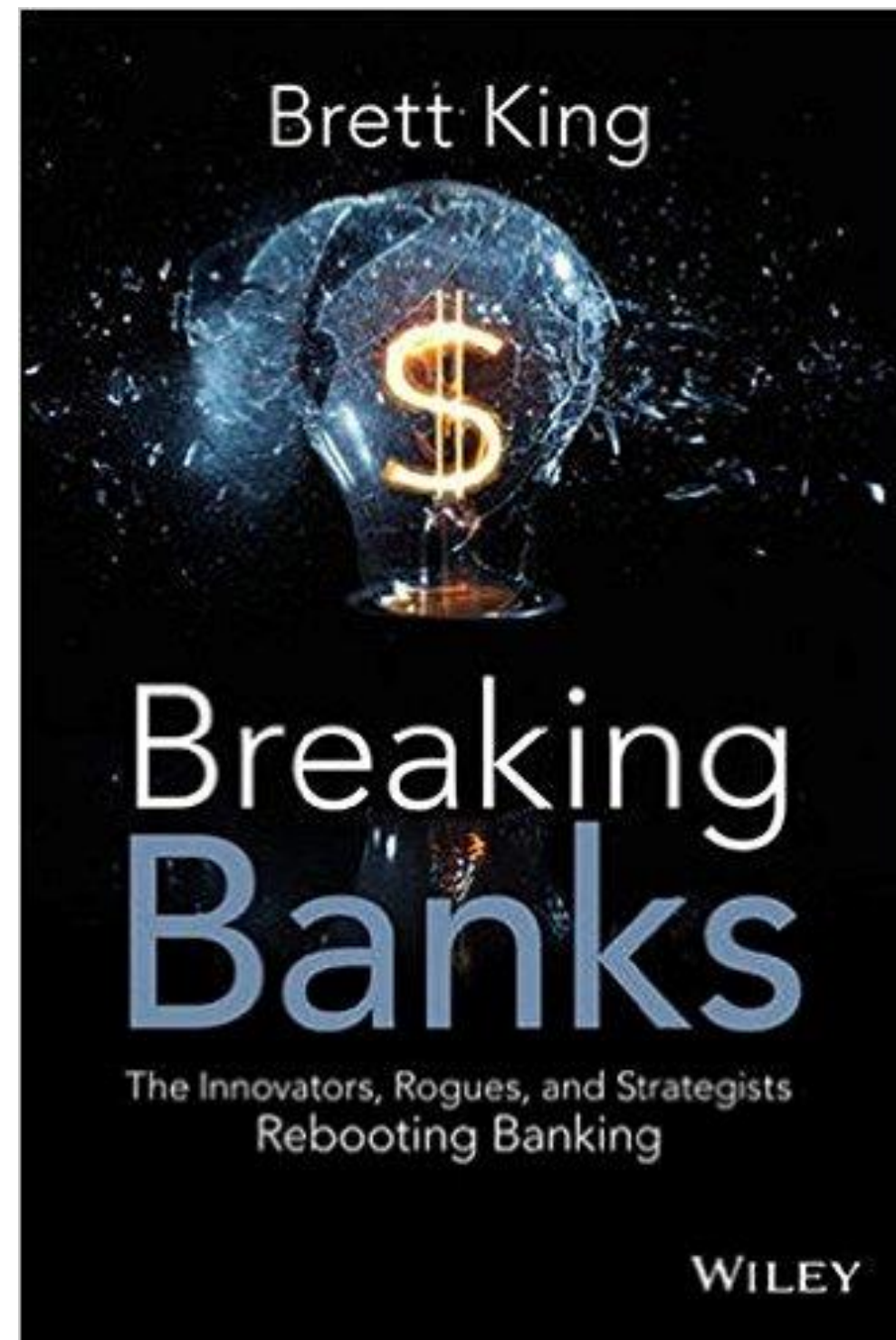
Brett King (2012), **Bank 3.0**

Why banking is no longer somewhere you go, but something you do,
Marshall Cavendish International Asia Pte Ltd



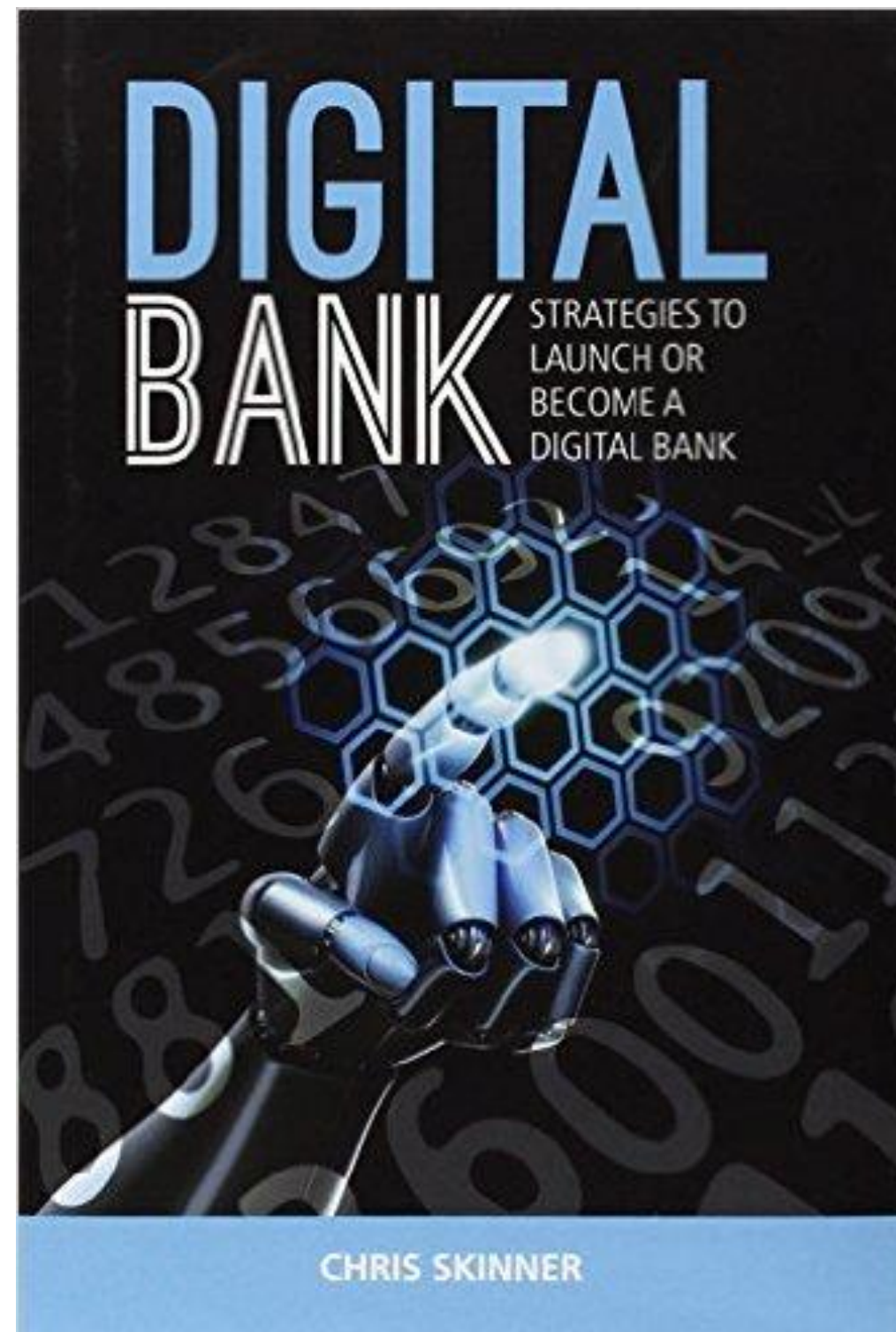
Brett King (2014),
Breaking Banks:

The Innovators, Rogues, and Strategists Rebooting Banking
Wiley

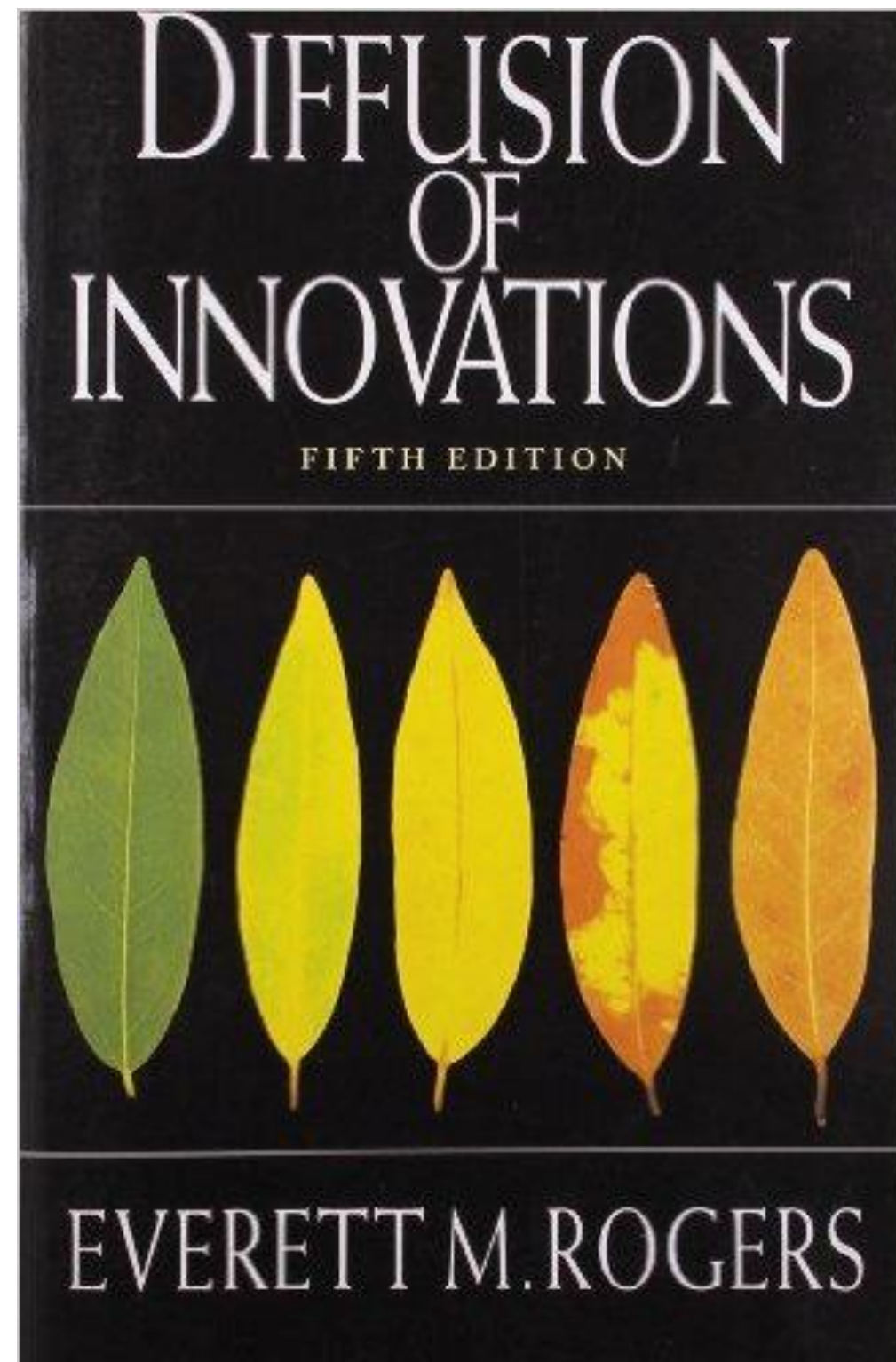


Digital Bank:

Strategies to Launch or Become a Digital Bank,
Marshall Cavendish International Asia Pte Ltd



Everett M. Rogers (2003),
Diffusion of Innovations,
5th Edition, Free Press



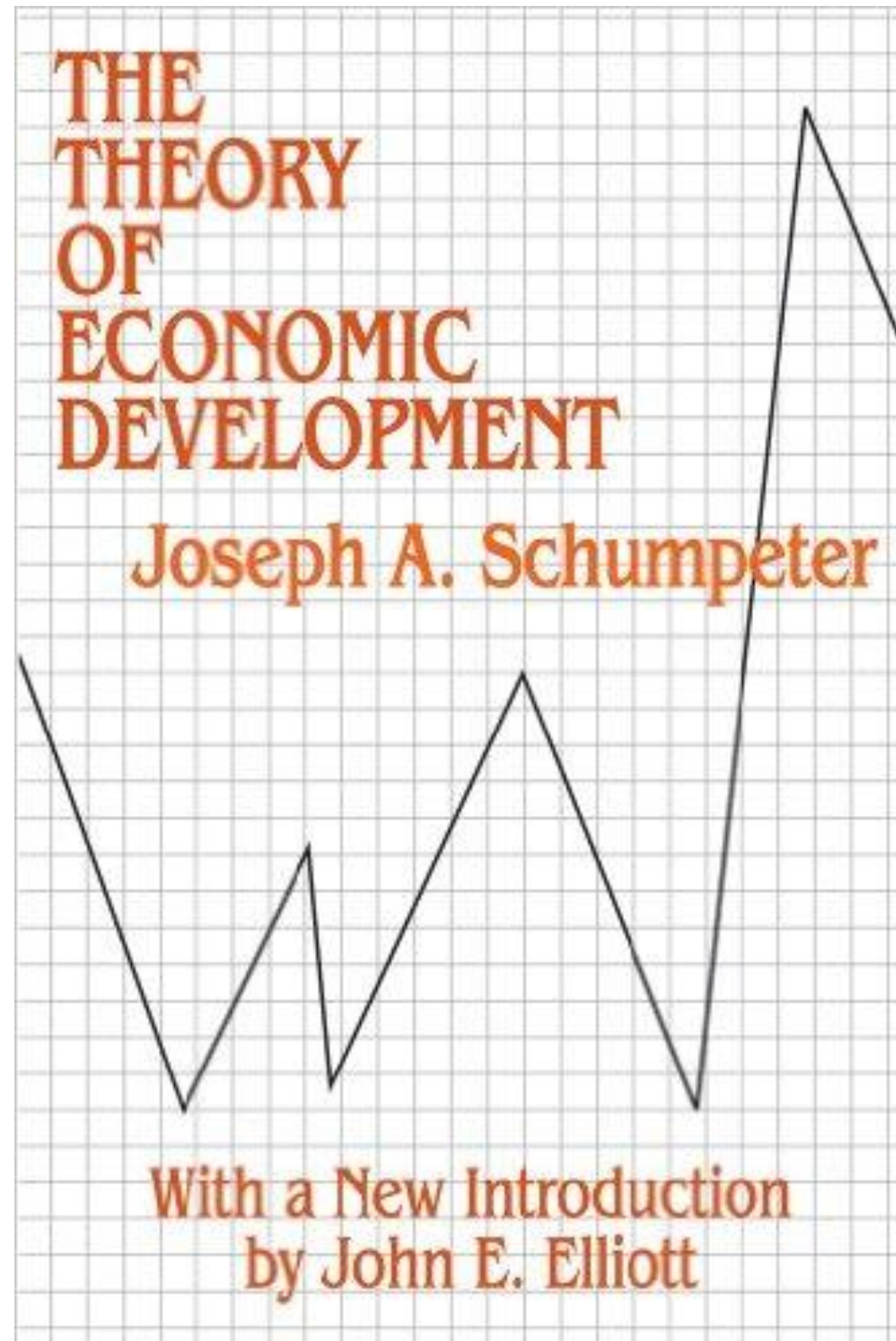
(Rogers,
1962; 1971;
1983; 1995;
2003)

Joseph A. Schumpeter,

The Theory of Economic Development:

An Inquiry into Profits, Capital, Credit, Interest, and the Business Cycle,

Transaction Publishers, 1982



(Schumpeter, 1912)

FinTech

Financial Technology

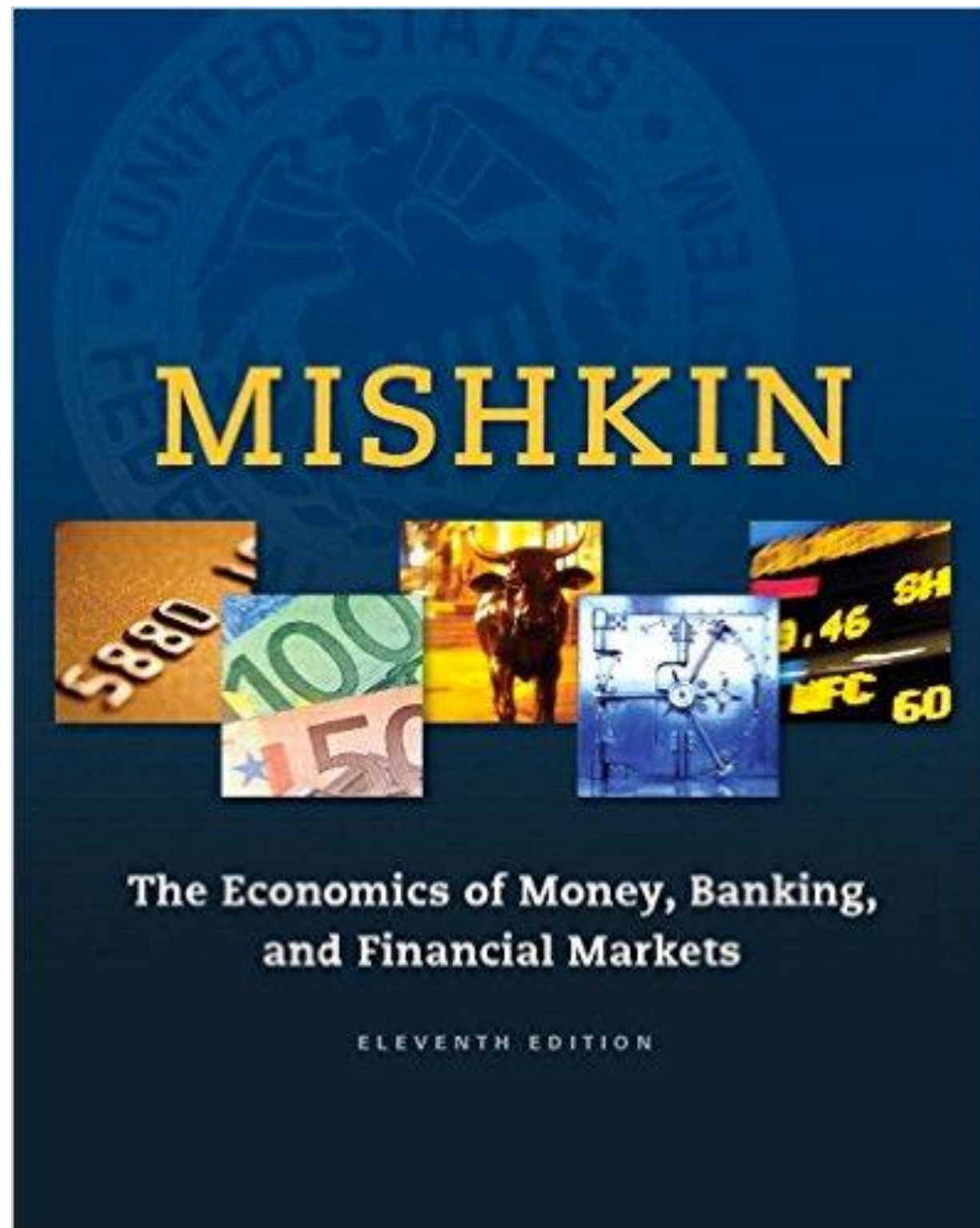
FinTech



Financial Technology FinTech

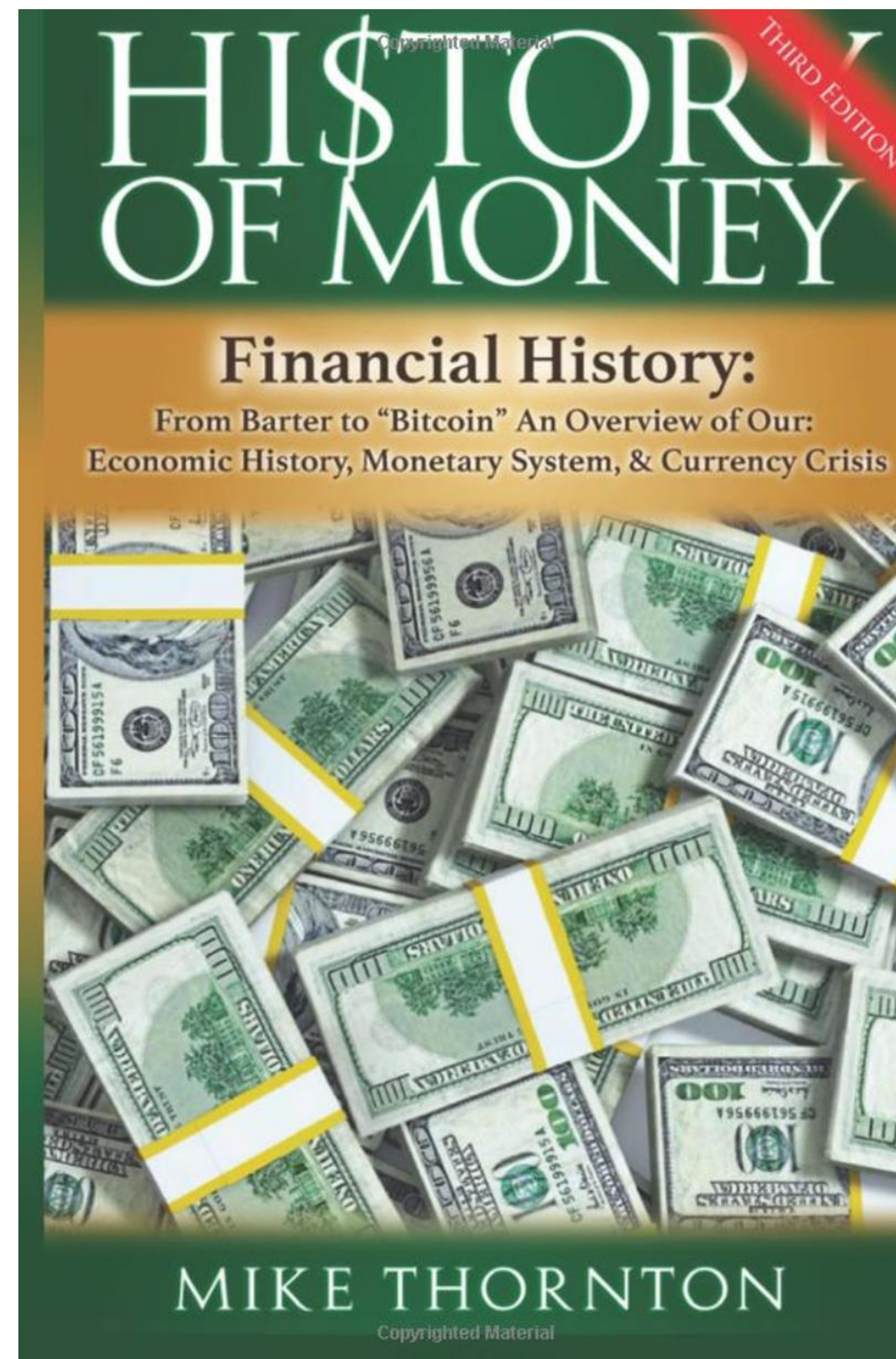
“providing
financial services
by making use of software
and
modern technology”

Frederic S. Mishkin (2015),
The Economics of Money, Banking and Financial Markets,
11th Edition, Pearson



Mike Thornton (2016), History of Money: Financial History:

From Barter to Bitcoin - An Overview of Our Economic History, Monetary System & Currency Crisis, CreateSpace Independent Publishing Platform



Money and Financial History

- Why is a printed piece of paper worth anything?
- How can a coin be worth more or even less than the number stamped on it?
- Why is digital money real money?
- How can money be worth more or less than it was yesterday?

Money

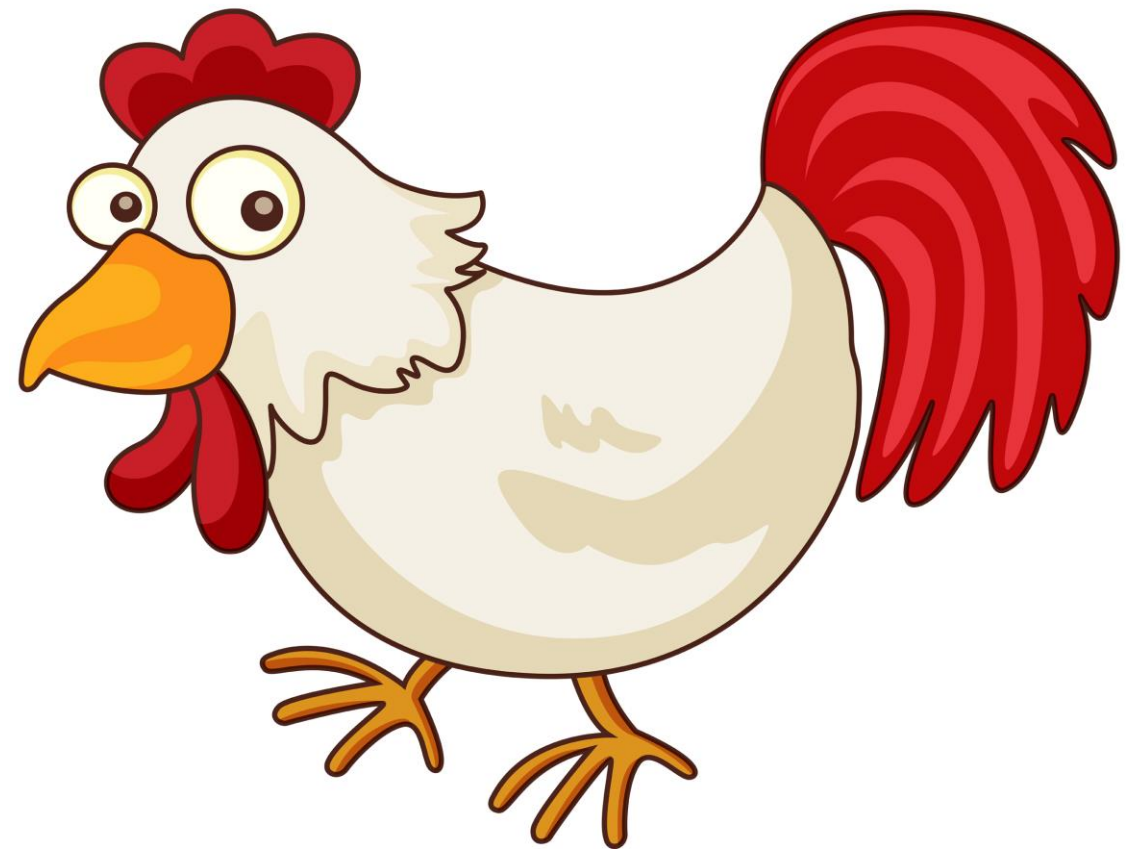
Exchange

Barter

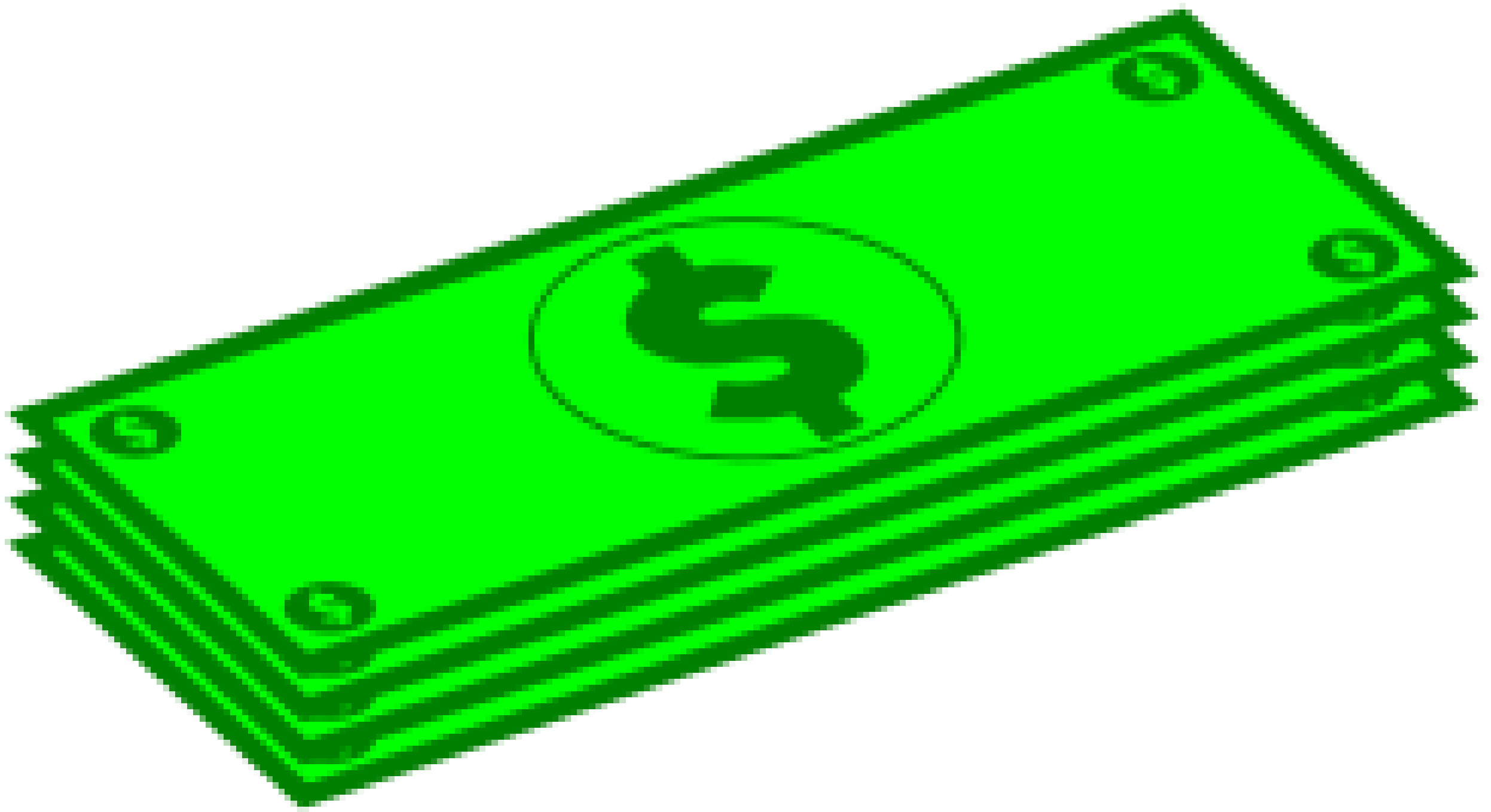
Barter



Barter



Money



Bills



Gold Bullion Coin



Gold Bullion Coin



Coin US Penny



Gold Bricks



Financial Services

Financial Services



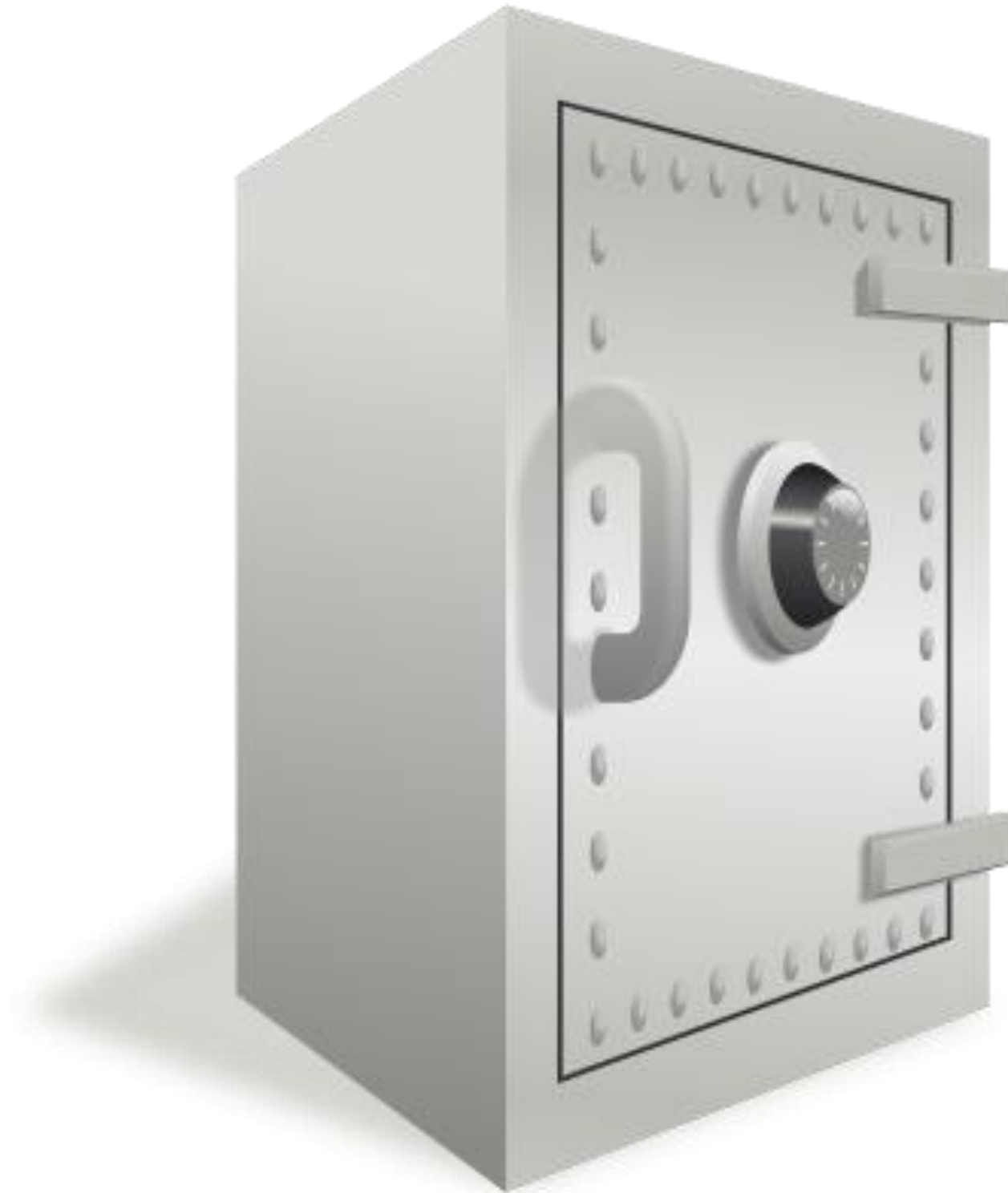
Financial Services



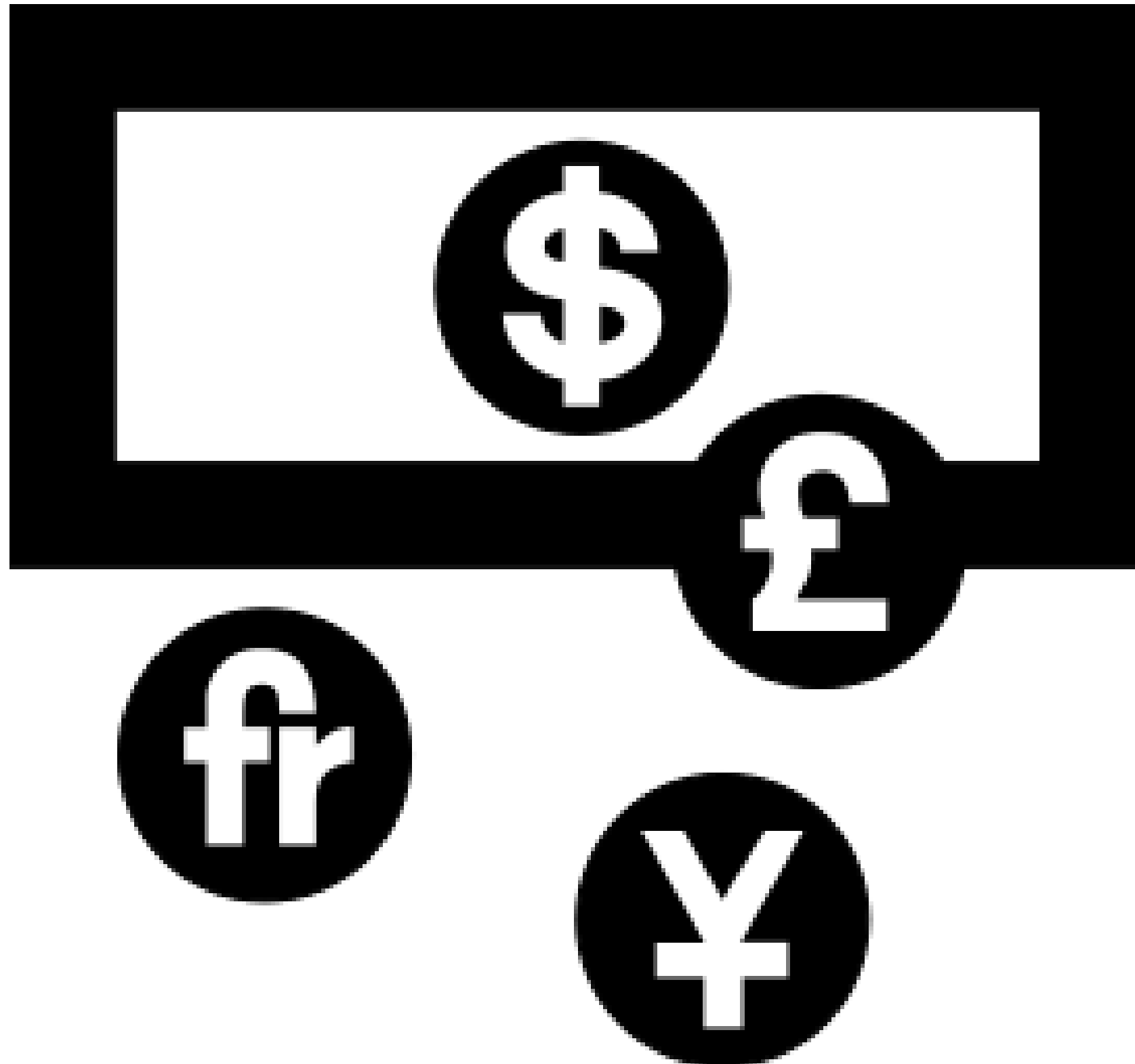
Treasure



Safe



Currency Exchange



Market

Financial Services

Financial Services



Financial Revolution with Fintech

A financial services revolution

Consumer Trends



1. Simplification



2. Transparency



3. Analytics



4. Reduced Friction

FinTech: Financial Services Innovation

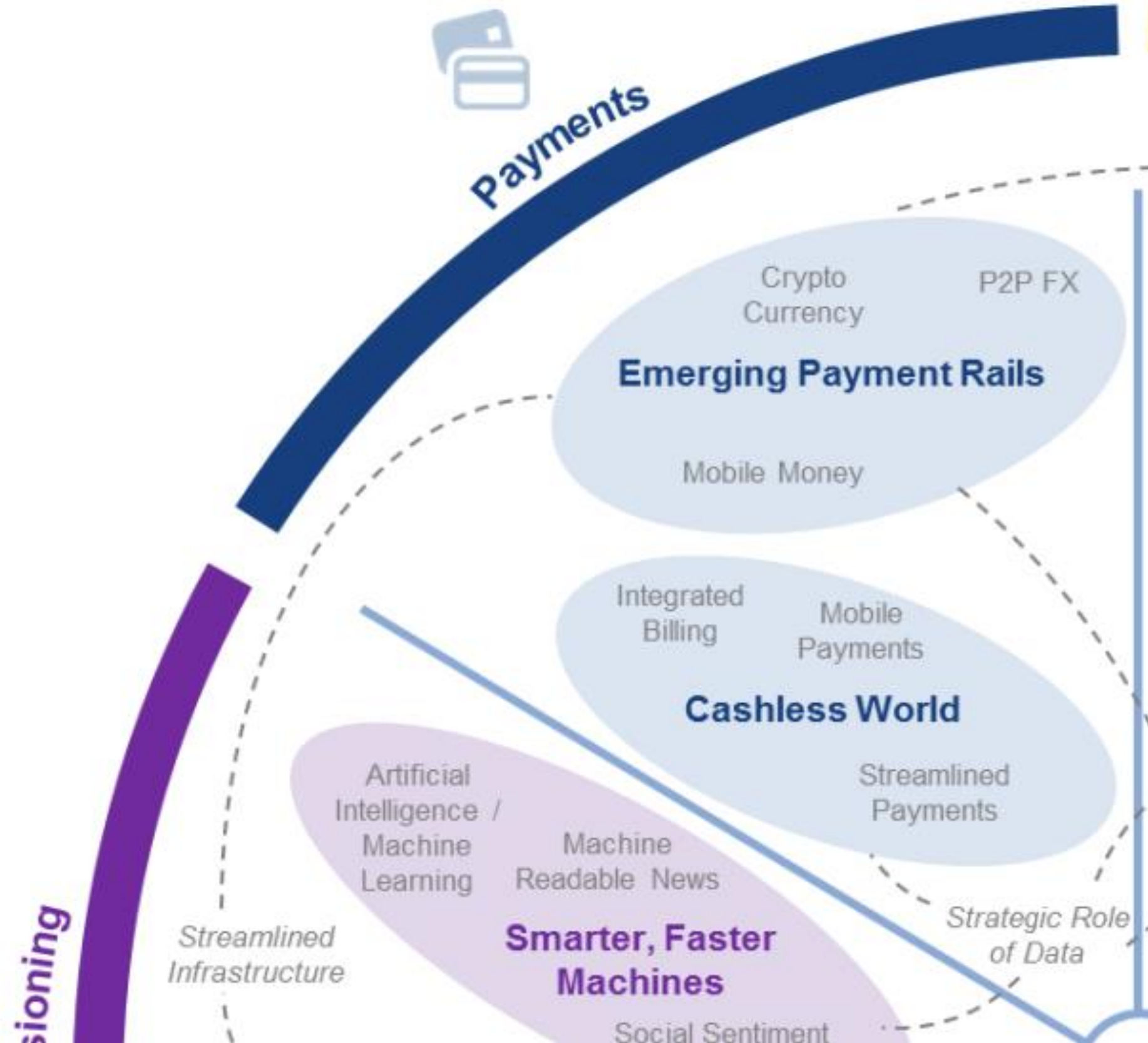


FinTech:

Financial Services Innovation

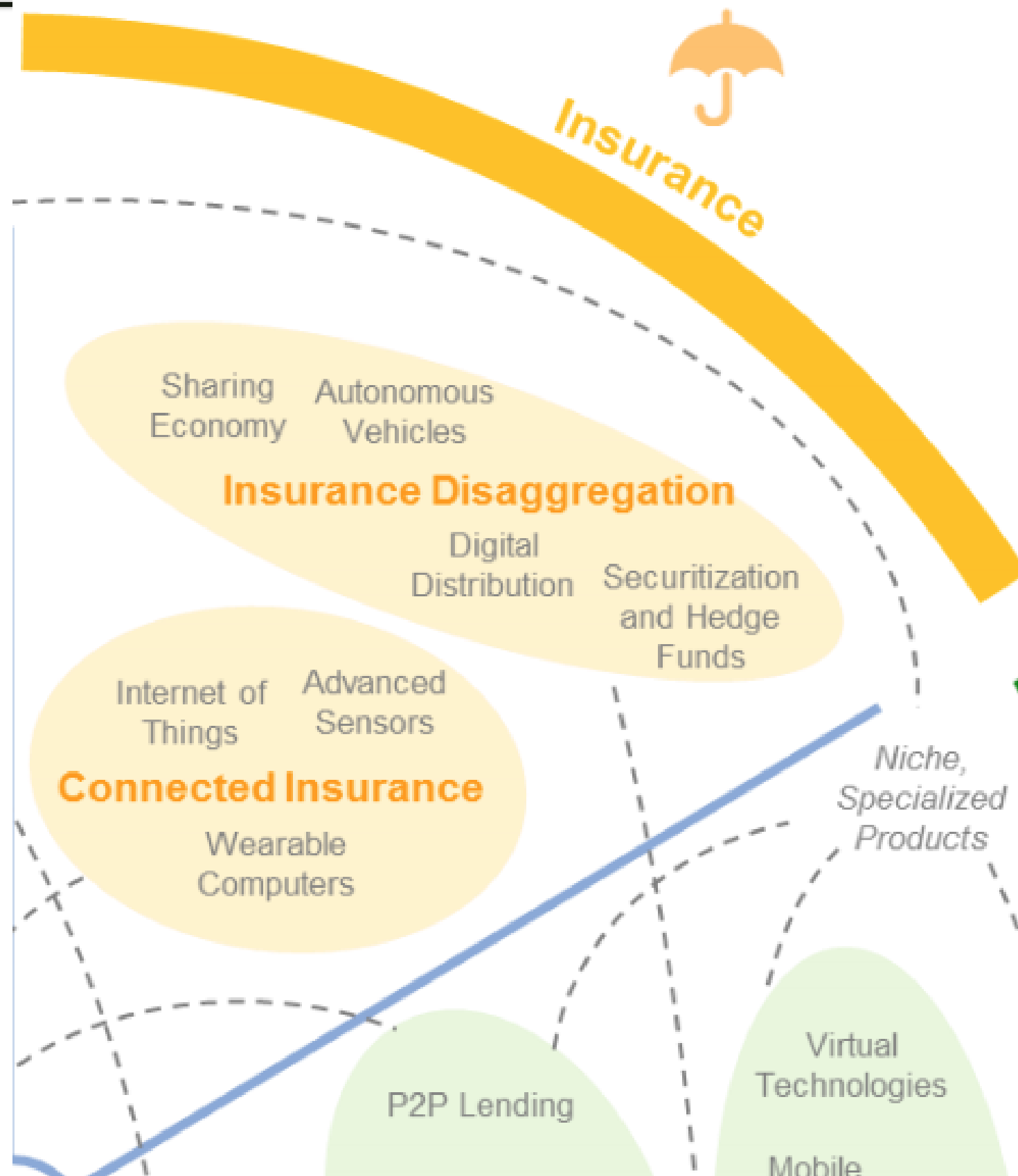
1. Payments
2. Insurance
3. Deposits & Lending
4. Capital Raising
5. Investment Management
6. Market Provisioning

FinTech: Payment



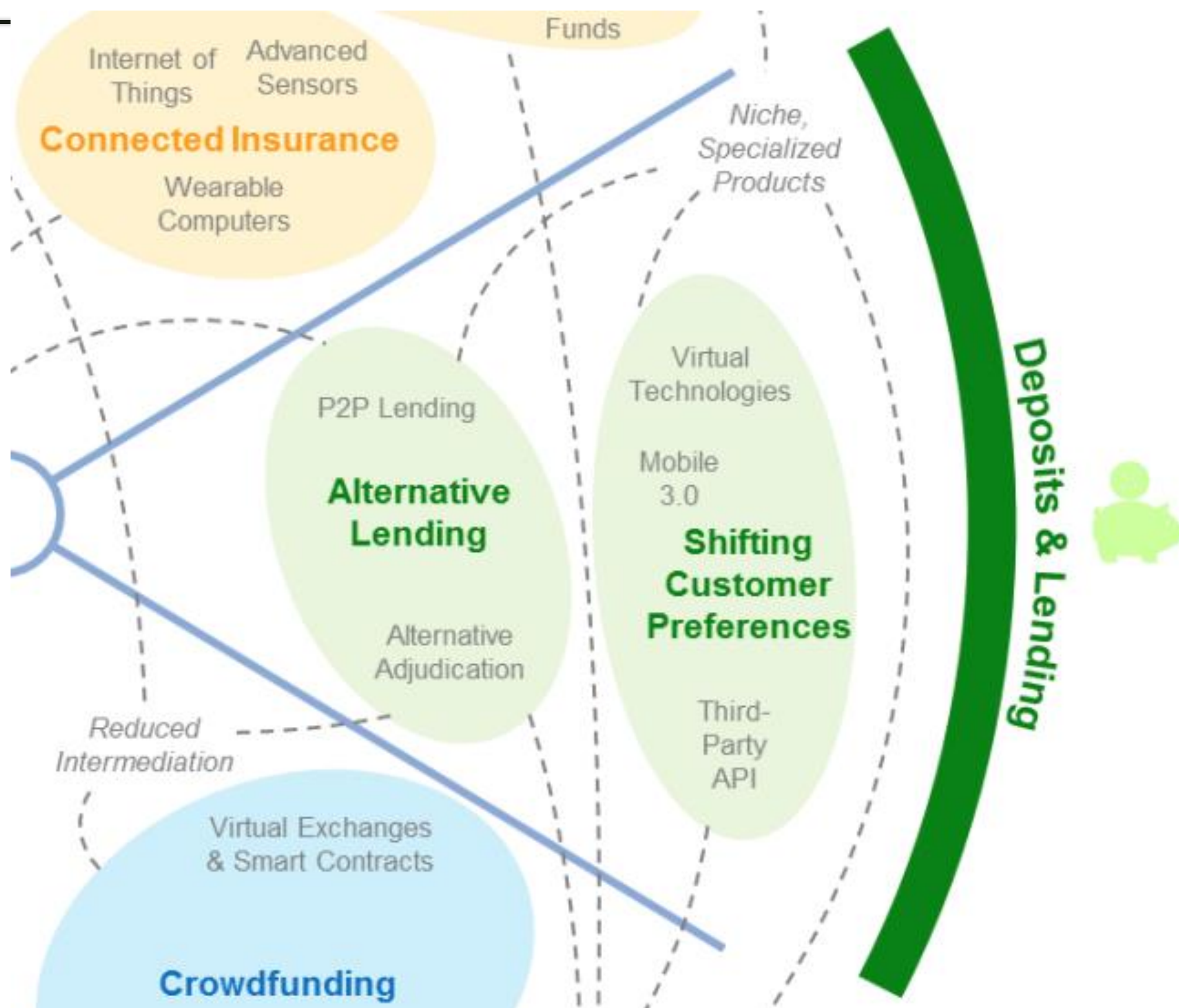
2

FinTech: Insurance



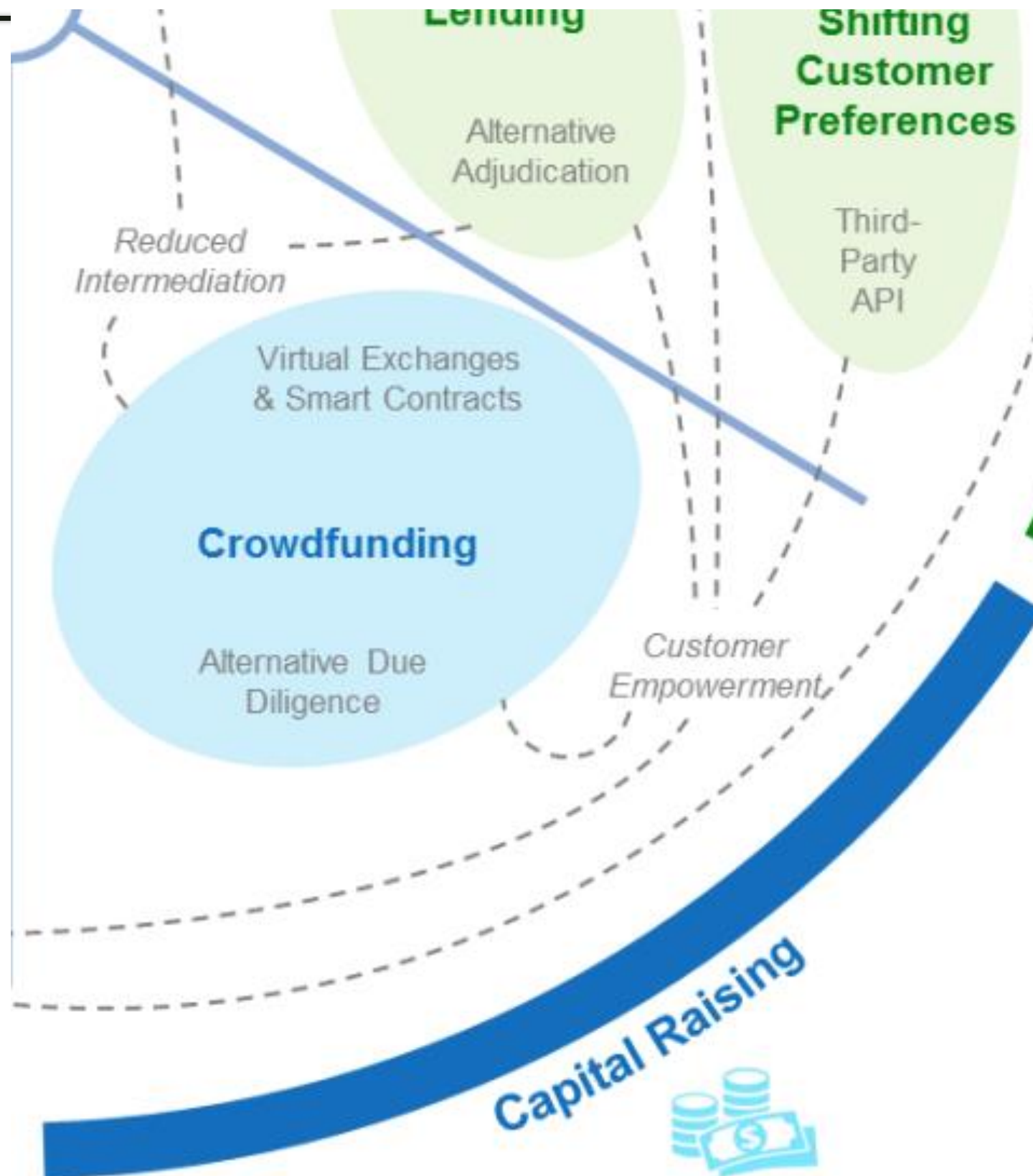
3

FinTech: Deposits & Lending



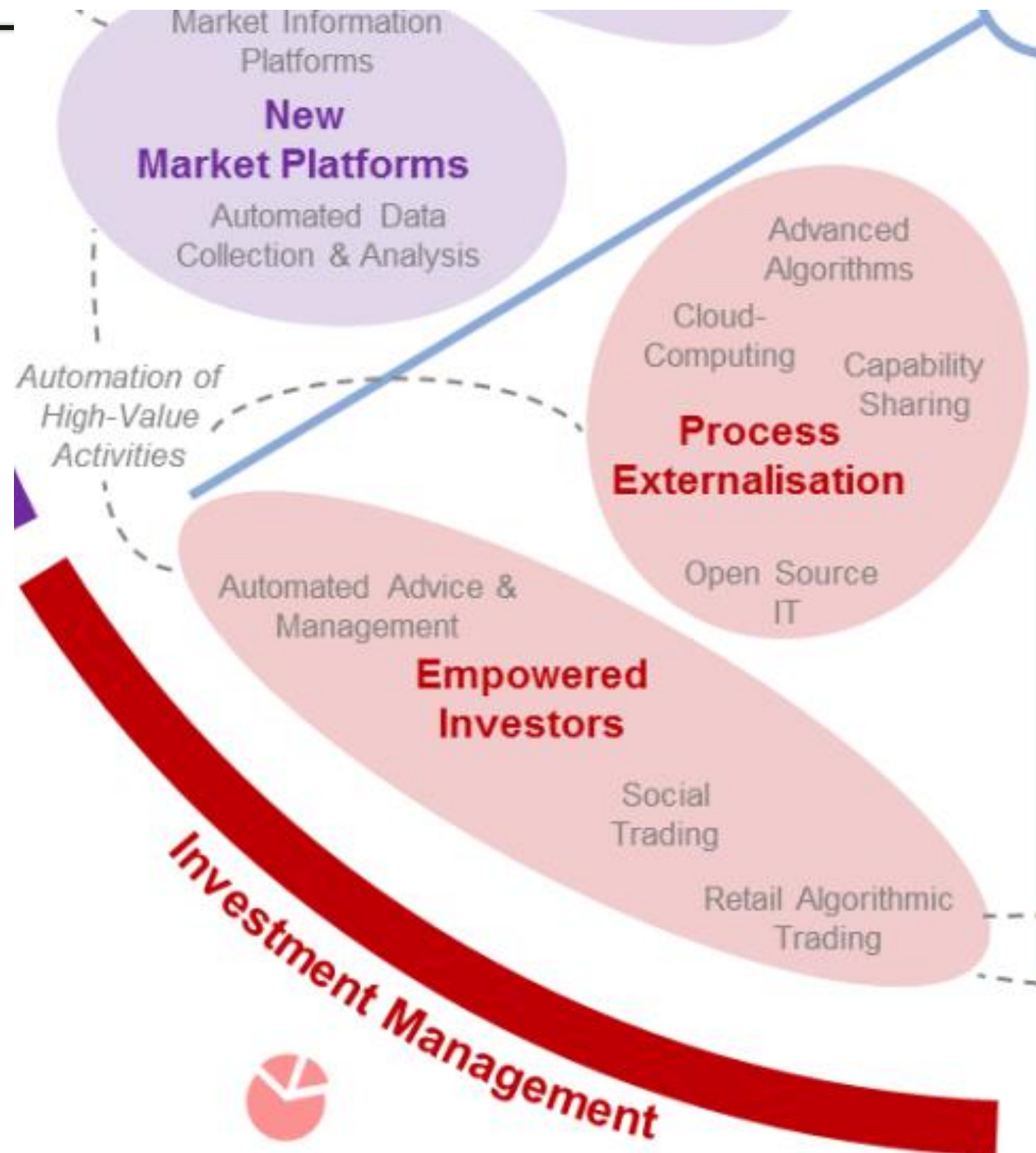
4

FinTech: Capital Raising



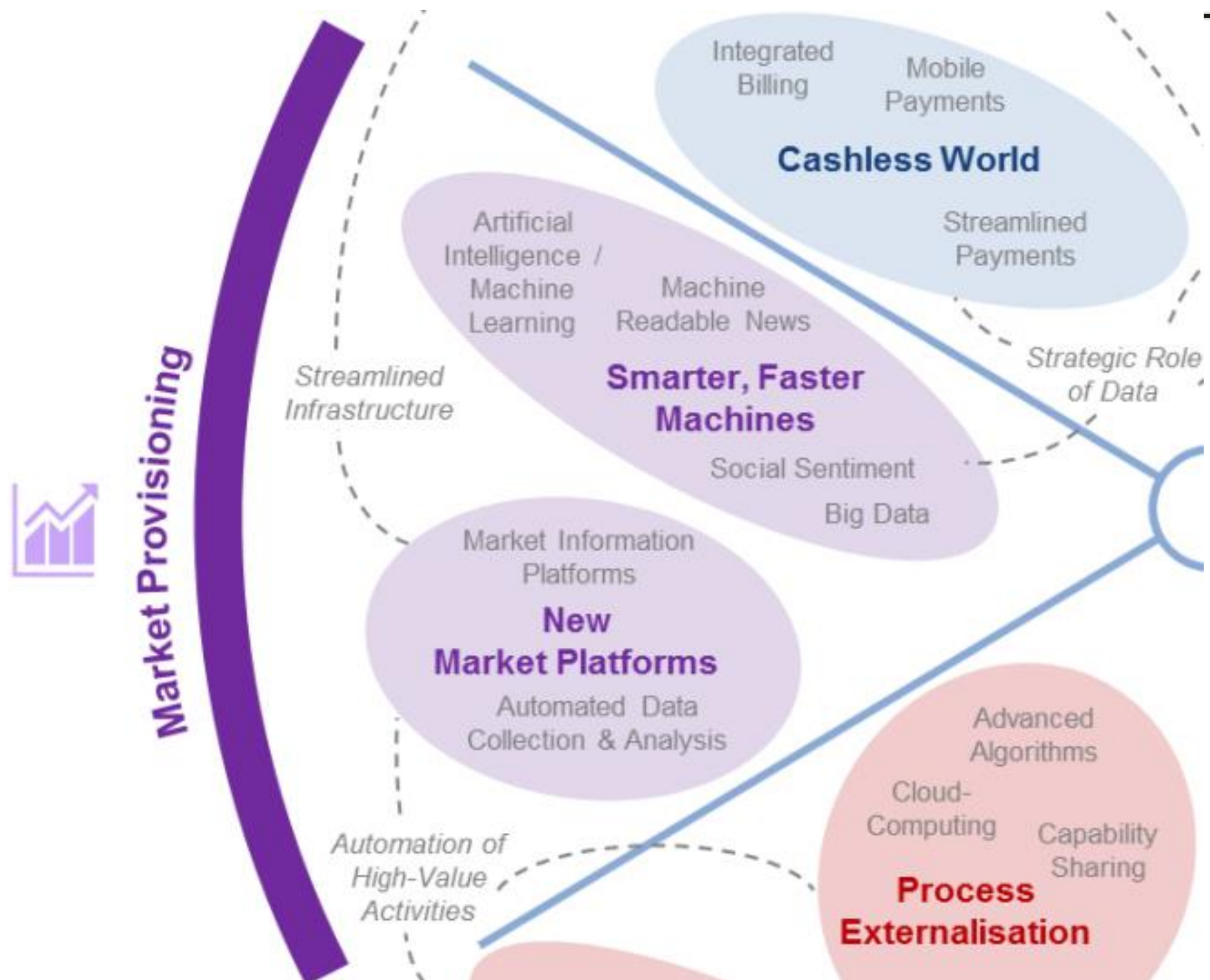
5

FinTech: Investment Management



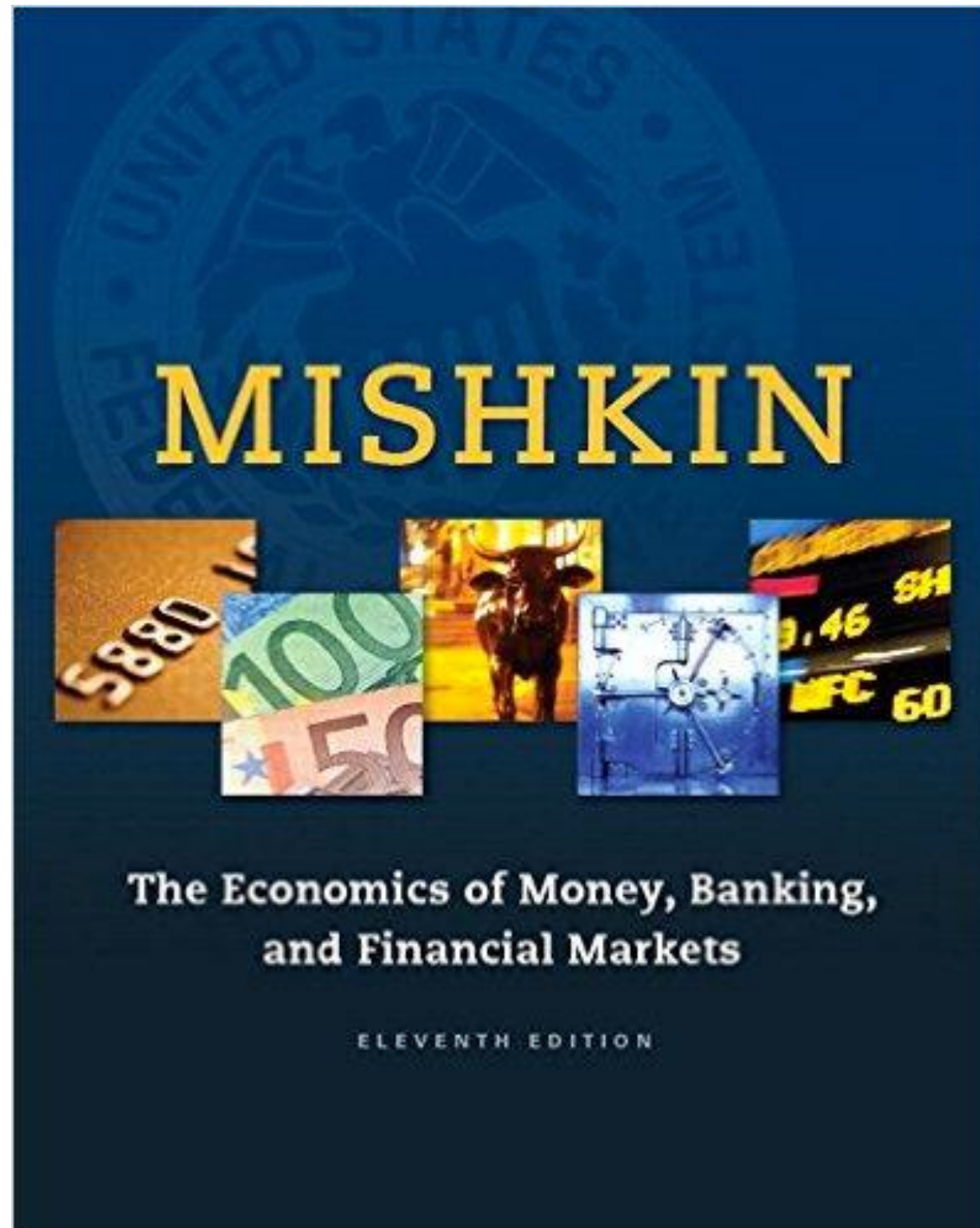
6

FinTech: Market Provisioning



The Economics of Money, Banking and Financial Markets

Frederic S. Mishkin (2015),
The Economics of Money, Banking and Financial Markets,
11th Edition, Pearson



Economics of Money, Banking and Financial Markets

1. **Money**, Banking, and Financial System
2. Financial Markets
3. Financial Institutions
4. Central Banking and the Conduct of Monetary Policy
5. International Finance and Monetary Policy
6. Monetary Theory
7. **Financial Services** Industry

INTRODUCTION

1. Why Study Money, Banking, and Financial Markets?
2. An Overview of the Financial System
3. What Is Money?

FINANCIAL MARKETS

- 4. Understanding Interest Rates
- 5. The Behavior of Interest Rates
- 6. The Risk and Term Structure of Interest Rates
- 7. The Stock Market, the Theory of Rational Expectations, and the Efficient Market Hypothesis

FINANCIAL INSTITUTIONS

- 8. An Economic Analysis of Financial Structure
- 9. Banking and the Management of Financial Institutions
- 10. Economic Analysis of Financial Regulation
- 11. Banking Industry: Structure and Competition
- 12. Financial Crises

CENTRAL BANKING AND THE CONDUCT OF MONETARY POLICY

101

- 13. Central Banks and the Federal Reserve System
- 14. The Money Supply Process
- 15. The Tools of Monetary Policy
- 16. The Conduct of Monetary Policy: Strategy and Tactics

MONETARY THEORY

- 19. Quantity Theory, Inflation, and the Demand for Money
- 20. The IS Curve
- 21. The Monetary Policy and Aggregate Demand Curves
- 22. Aggregate Demand and Supply Analysis
- 23. Monetary Policy Theory
- 24. The Role of Expectations in Monetary Policy
- 25. Transmission Mechanisms of Monetary Policy

Financial Services Industry

- 26. Financial Crises in Emerging Market Economies
- 27. The ISLM Model
- 28. Nonbank Finance
- 29. Financial Derivatives
- 30. Conflicts of Interest in the Financial Services Industry

Why Study Money, Banking, and Financial Markets?

Why Study Money, Banking, and Financial Markets?

- To examine how **financial markets** such as **bond**, **stock** and **foreign exchange** markets work
- To examine how **financial institutions** such as **banks** and **insurance companies** work
- To examine the **role of money** in the **economy**

Financial Markets

- Markets in which funds are transferred from people who have an excess of available funds to people who have a shortage of funds
 - Bond market
 - Stock market
 - Foreign exchange market

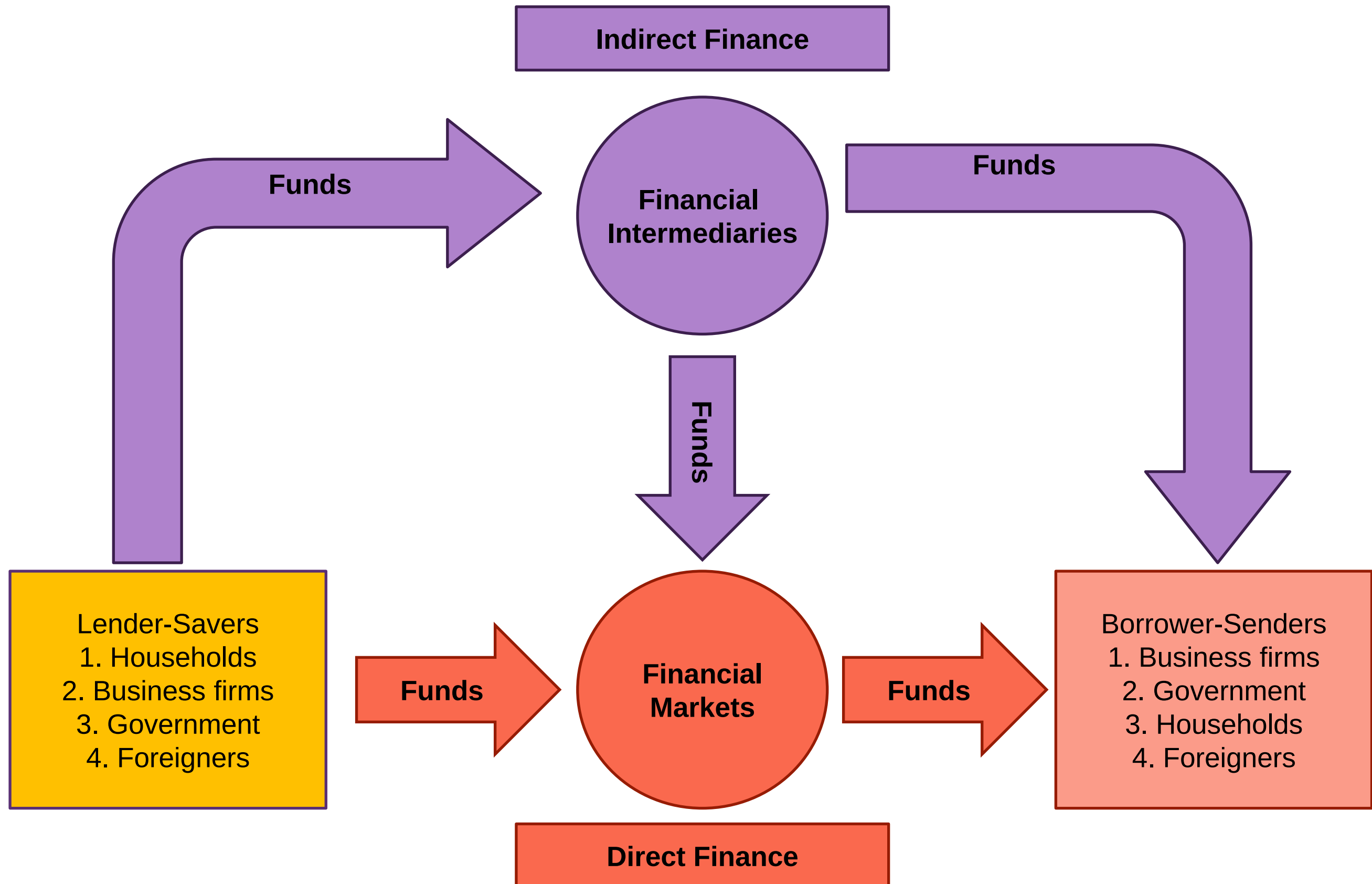
Financial Institutions

- Financial Intermediaries: institutions that borrow funds from people who have saved and make loans to other people:
 - **Banks**: accept deposits and make loans
 - Other Financial Institutions: **insurance companies, finance companies, pension funds, mutual funds and investment banks**
- **Financial Innovation**: the advent of the information age and e-finance

Money and Business Cycles

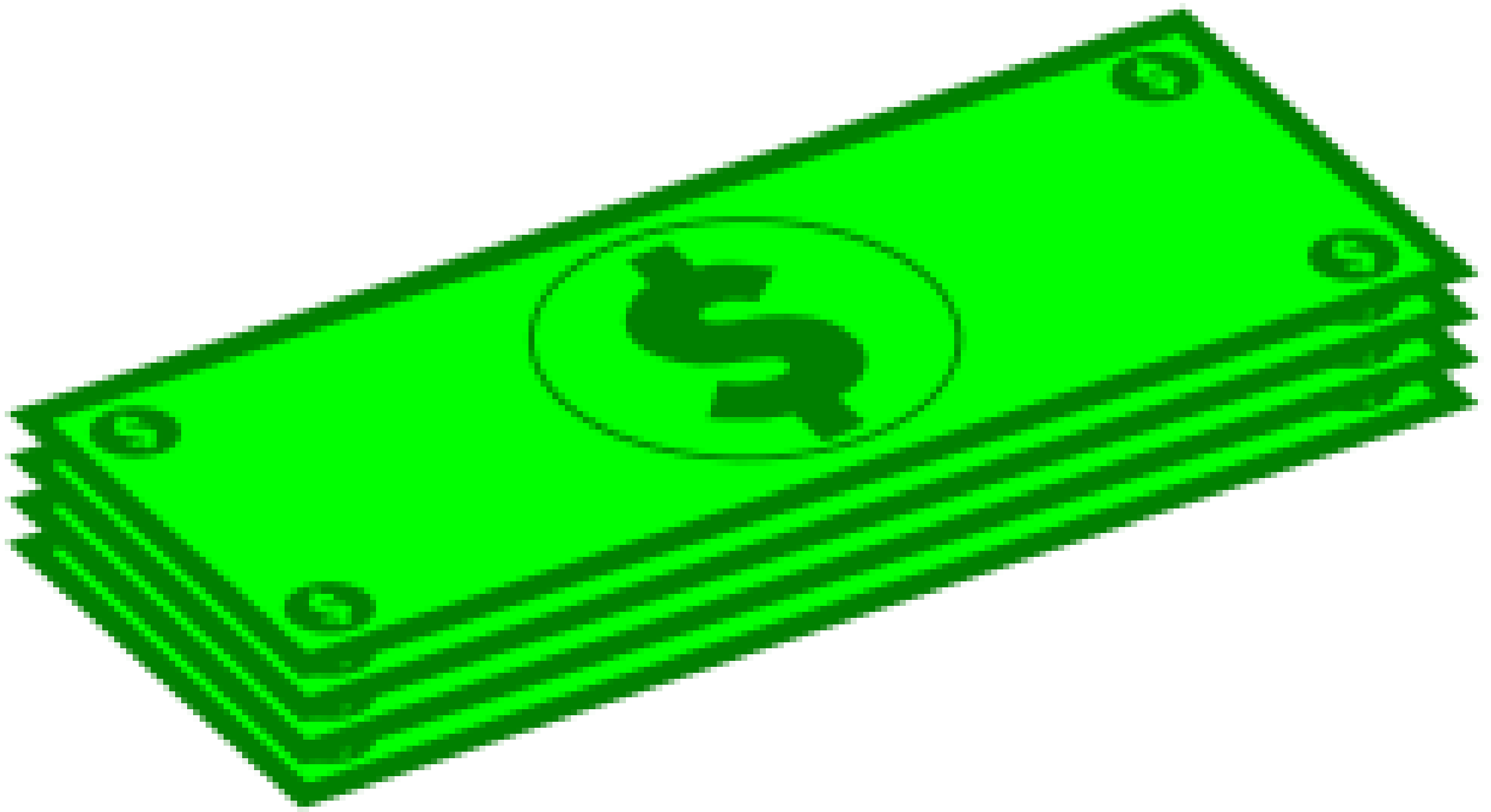
- Money plays an important role in generating business cycles
- Recessions (unemployment) and expansions affect all of us
- Monetary Theory ties changes in the money supply to changes in aggregate economic activity and the price level

Overview of the Financial System



What is Money?

Money



Bills



Meaning of Money

- **Money (=money supply)** any vehicle used as a means of **exchange** to pay for goods, services or debts.
- In today's society, any **asset** that can quickly be transferred into cash is considered money.
- The more **liquid** an asset is, the closer it is to money.
- In economics, **money** does not mean **wealth** nor does it mean **income**.

Functions of Money

- Medium of Exchange
- Unit of Account
- Store of Value

Medium of Exchange

- By eliminating barter, this function of money increases efficiency in a society.
- As human societies started to engage in exchange money had to be invented.
- Any technological change that reduces transaction costs increases the wealth of the society.
- Any technological change that allows people to specialize also increases wealth.

Unit of Account

- We use money to measure the value of goods and services.
- Suppose we had 4 goods and no money. How do we measure the price of each good?
 - A in terms of B
 - B in terms of C
 - C in terms of D
 - A in terms of C
 - A in terms of D
 - B in terms of D
- Money allows to quote prices in terms of currency only.

$$N!/2(N-2)!$$

Store of Value

- All **assets** are stored value.
- Money, although without any return, is still desirable to hold because it allows purchases immediately.
- Other assets take time (transaction costs) to use as a payment for purchases.
- The more liquid an asset is, the less transaction cost it carries.
- Inflation erodes the value of money.

Evolution of the Payments System

- Commodity Money:
 - valuable, easily standardized and divisible commodities (e.g. precious metals, cigarettes).
- Fiat Money:
 - paper money decreed by governments as legal tender.

Electronic Money

- Debit Cards
 - Instant transfer from your checking account to merchant's checking account.
- Stored Value Card
 - Gift cards.
- Electronic Cash
 - Account set up on a person's PC from her bank whereby she can buy products over the Internet.
- Electronic Checks
 - Checks written on PC and sent through the Internet.

Benefits of Paper Checks

- Cheaper than telecommunications network.
- Provide receipts.
- Allow float.
- May be more secure; avoid hacker problems.
- Do not leave a wealth of information trail.

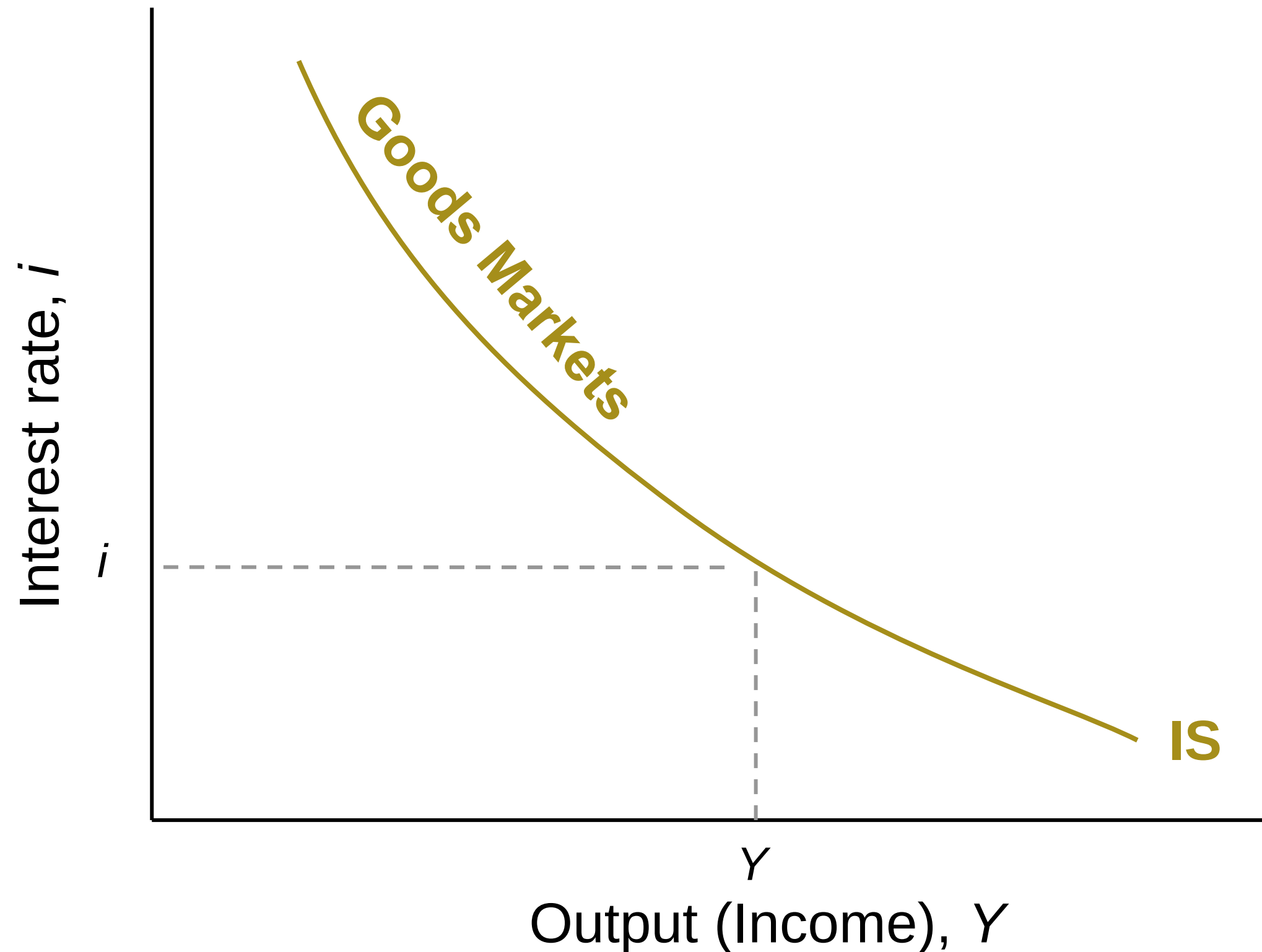
Measuring Money

- M1:
 - Currency, demand deposits, travelers checks.
- M2:
 - M1, saving deposits, small time deposits, retail MMMF.
- M3:
 - M2, large time deposits, repos, Eurodollar deposits, institutional MMMF.
- MZM:
 - M2, institutional MMMF minus small time deposits.
- Growth rates of these aggregates do not always go hand in hand, making monetary policy difficult since signals are conflicting.

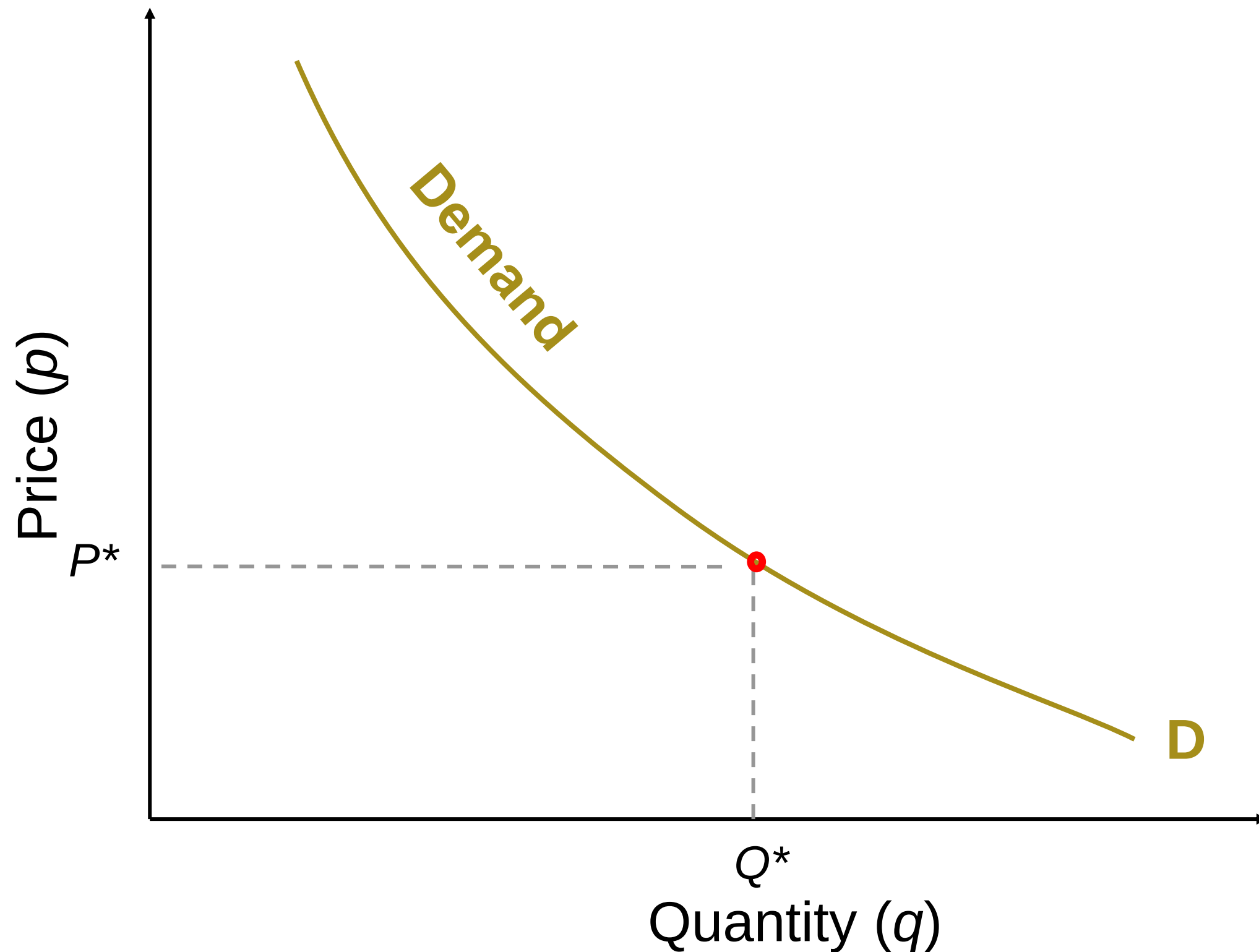
The IS Curve

The IS (Investment/Saving) Curve

The IS (Investment/Saving) Curve



Demand



The ISLM Model

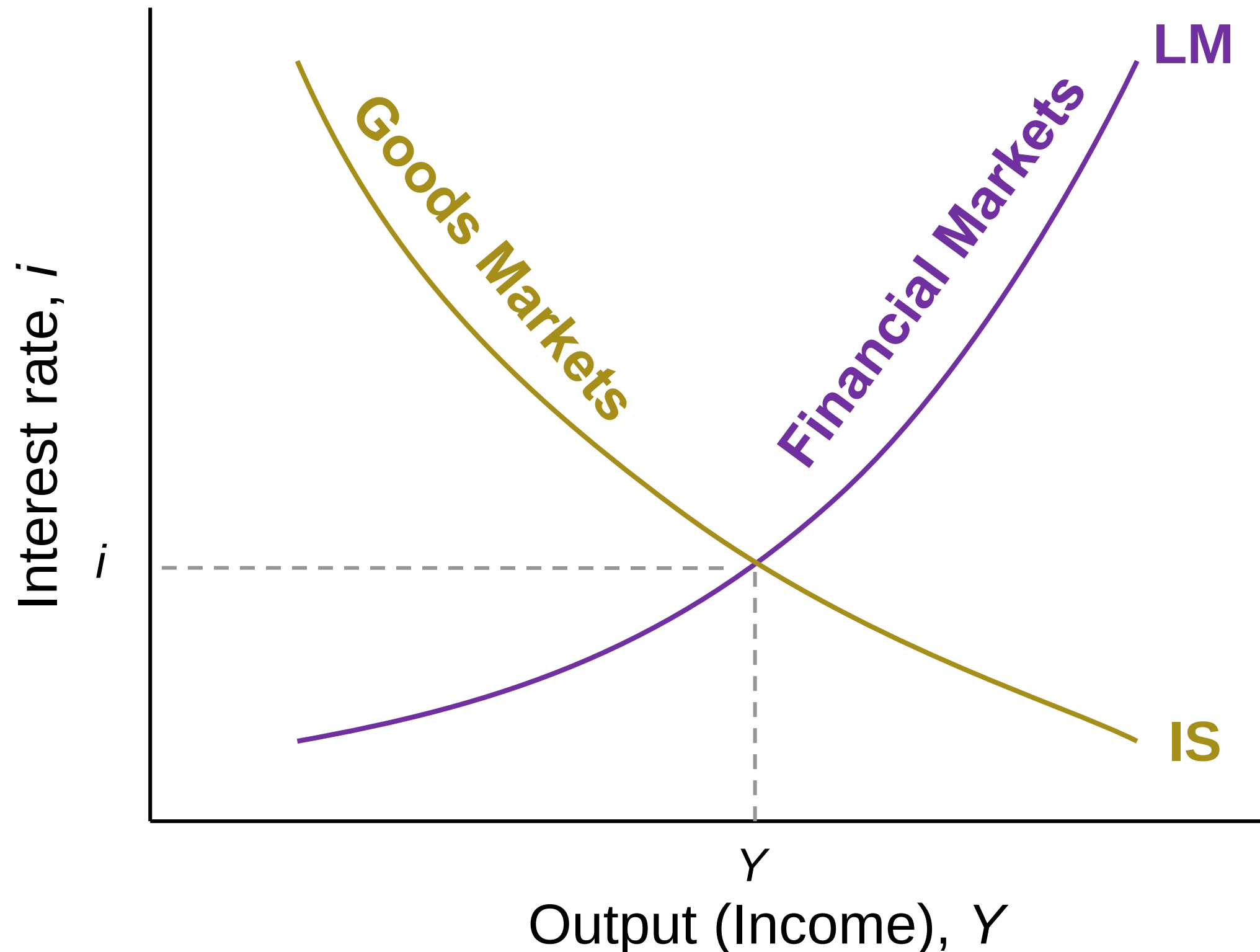
Goods and Financial Markets:

The ISLM Model

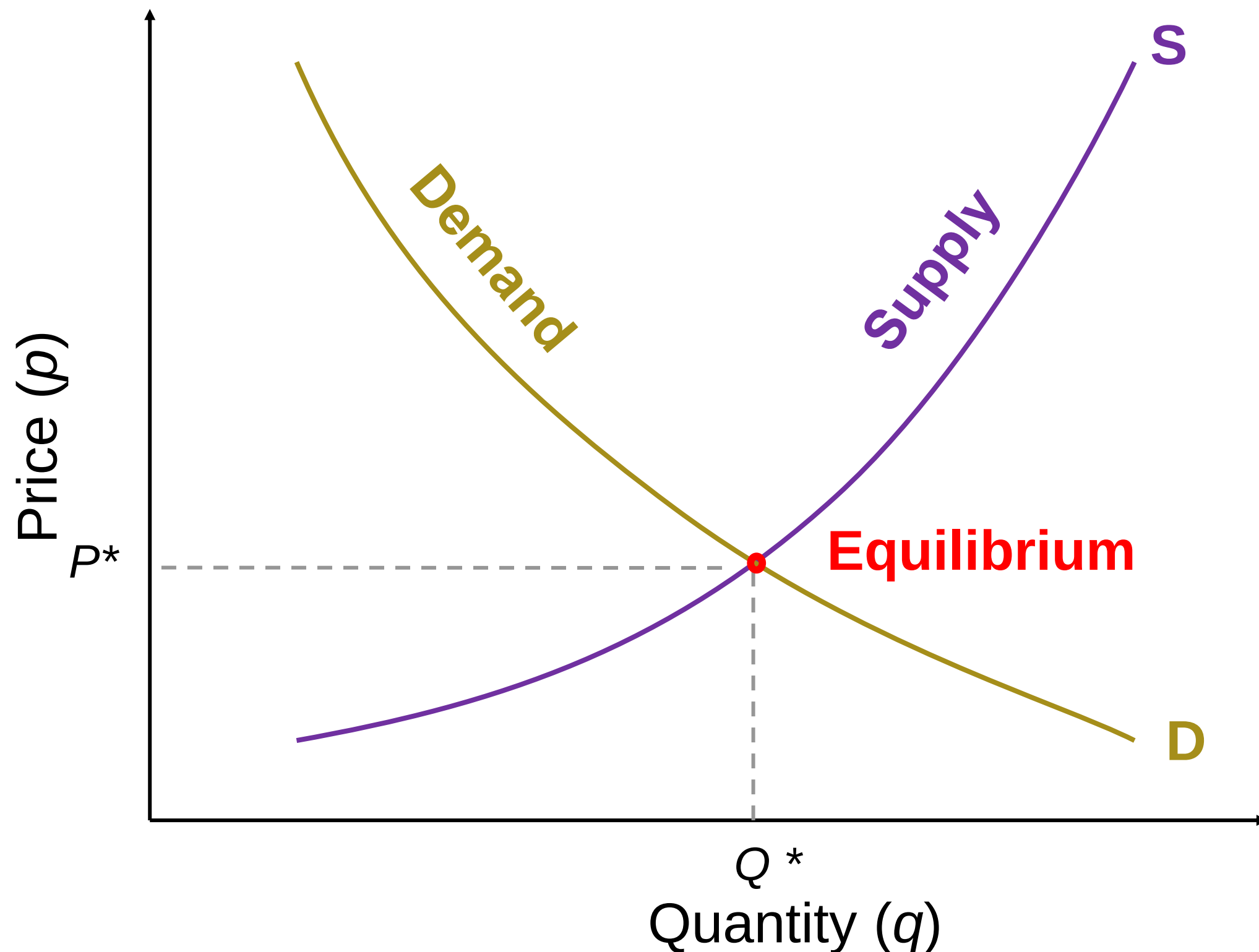
(Investment Saving –
Liquidity Preference Money Supply)
model

The ISLM Model

(Investment Saving –
Liquidity Preference Money Supply) model



Supply and Demand



Artificial Intelligence and Deep Learning for Fintech

From Algorithmic Trading to Personal Finance Bots: 41 Startups Bringing AI to Fintech

From Algorithmic Trading To Personal Finance Bots: 41

Startups Bringing AI To Fintech

AI in Fintech

41 Startups Bringing Artificial Intelligence To Fintech

General Purpose/ Predictive Analytics



Market Research & Sentiment Analysis



Search Engine



Quantitative Trading



Blockchain



Debt Collection



AI Assistants/Bots



Fraud Detection



Credit Scoring



Personal Banking



Artificial Intelligence (AI) in Fintech ¹³³

General Purpose/ Predictive Analytics



Market Research & Sentiment Analysis



Search Engine



Artificial Intelligence (AI) in Fintech

Quantitative Trading



CLONE ALGO



AI Assistants/Bots

KASIST



INSURIFY

SURE.



Credit Scoring

TypeScore aire



Wecash 闪银奇昇



Blockchain



Fraud Detection

feedzai

BIOCATCH
Less Friction. Less Fraud.

Personal Banking

personetics®
TAKING DIGITAL BANKING PERSONAL

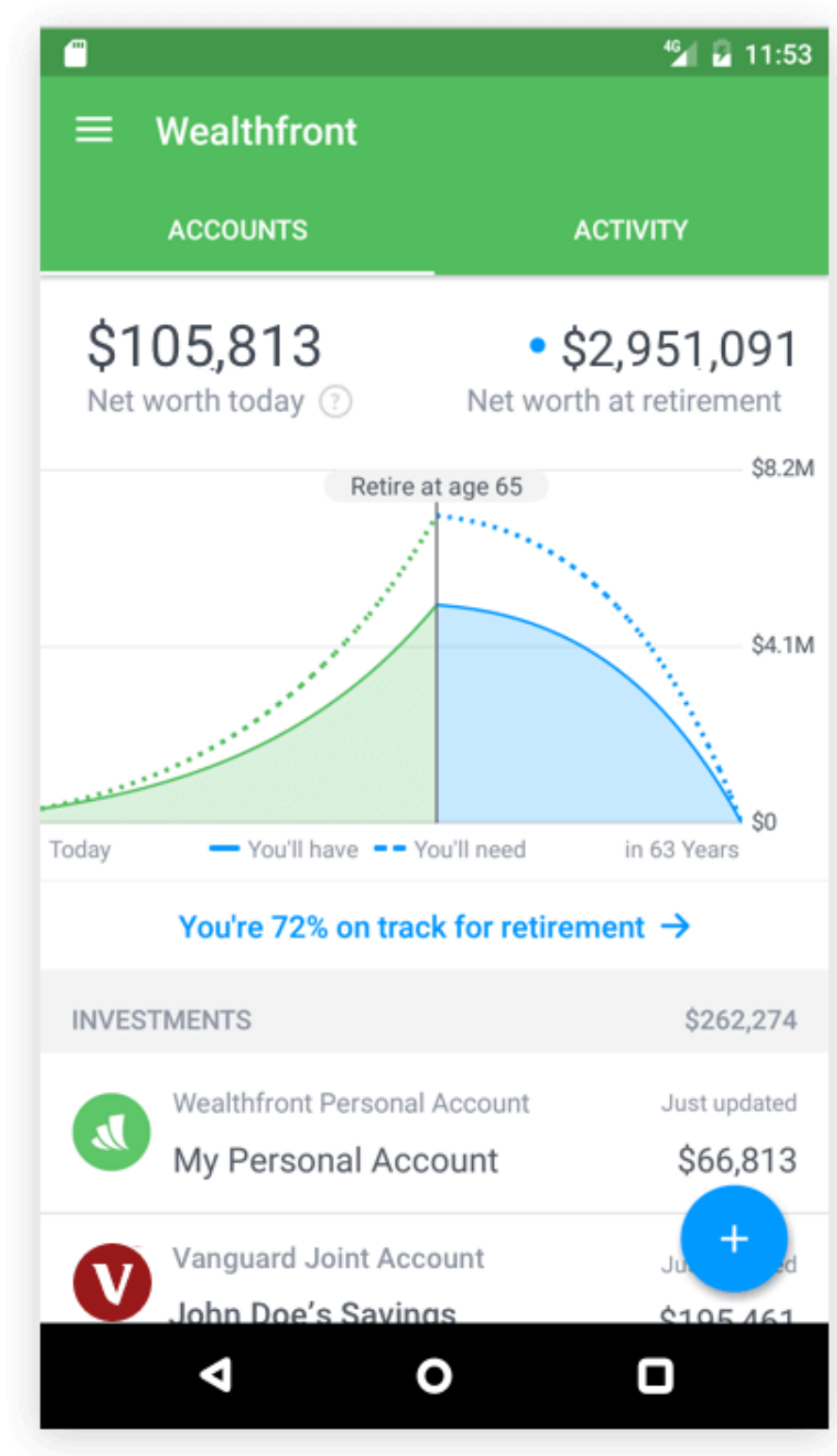
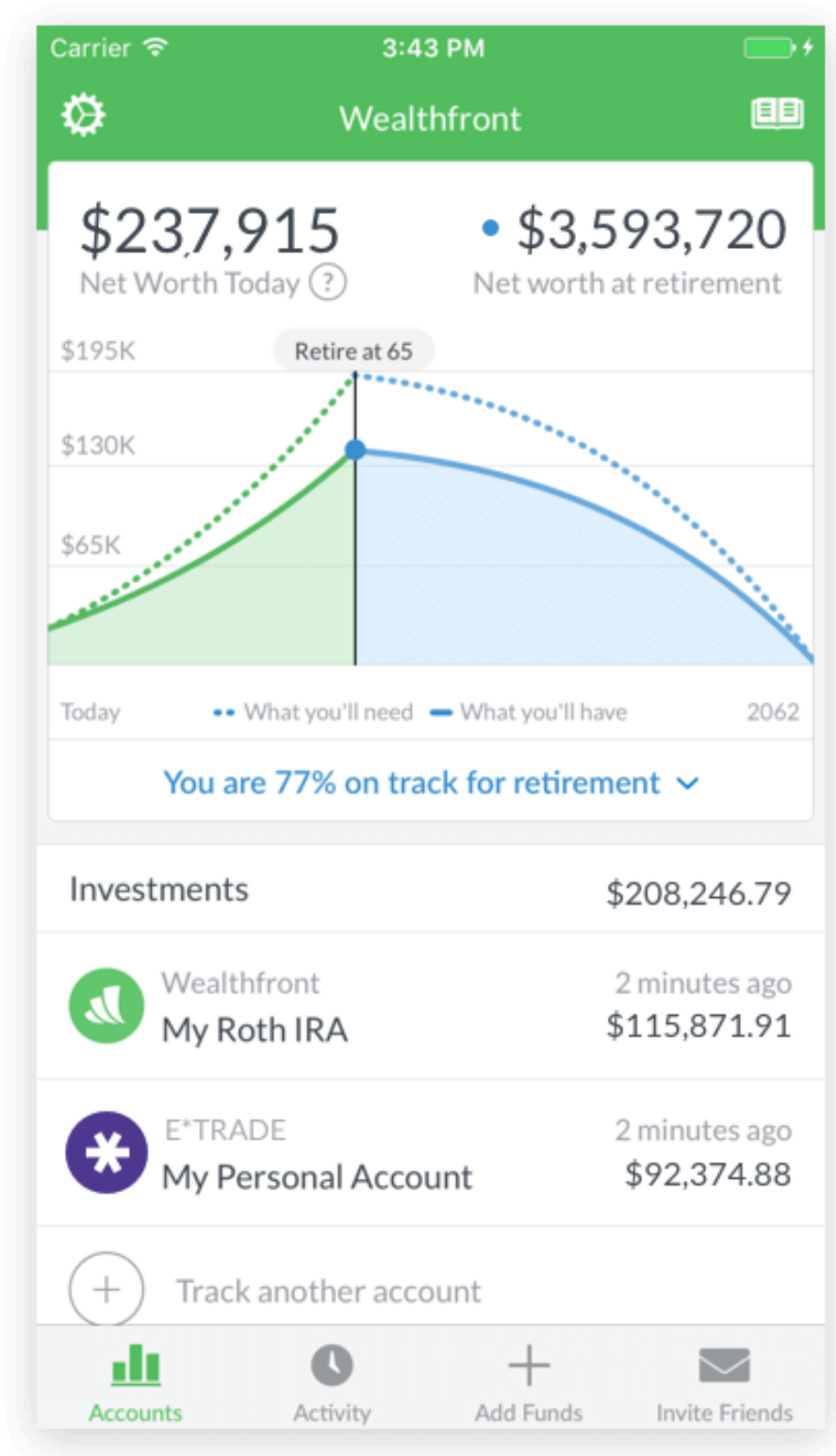


Debt Collection



TrueAccord

Wealthfront Robo Advisor



Financial Services

Technology Innovation

Innovation

Innovation:

a new idea,
method, or
device

Innovation:
something new

Novelty :

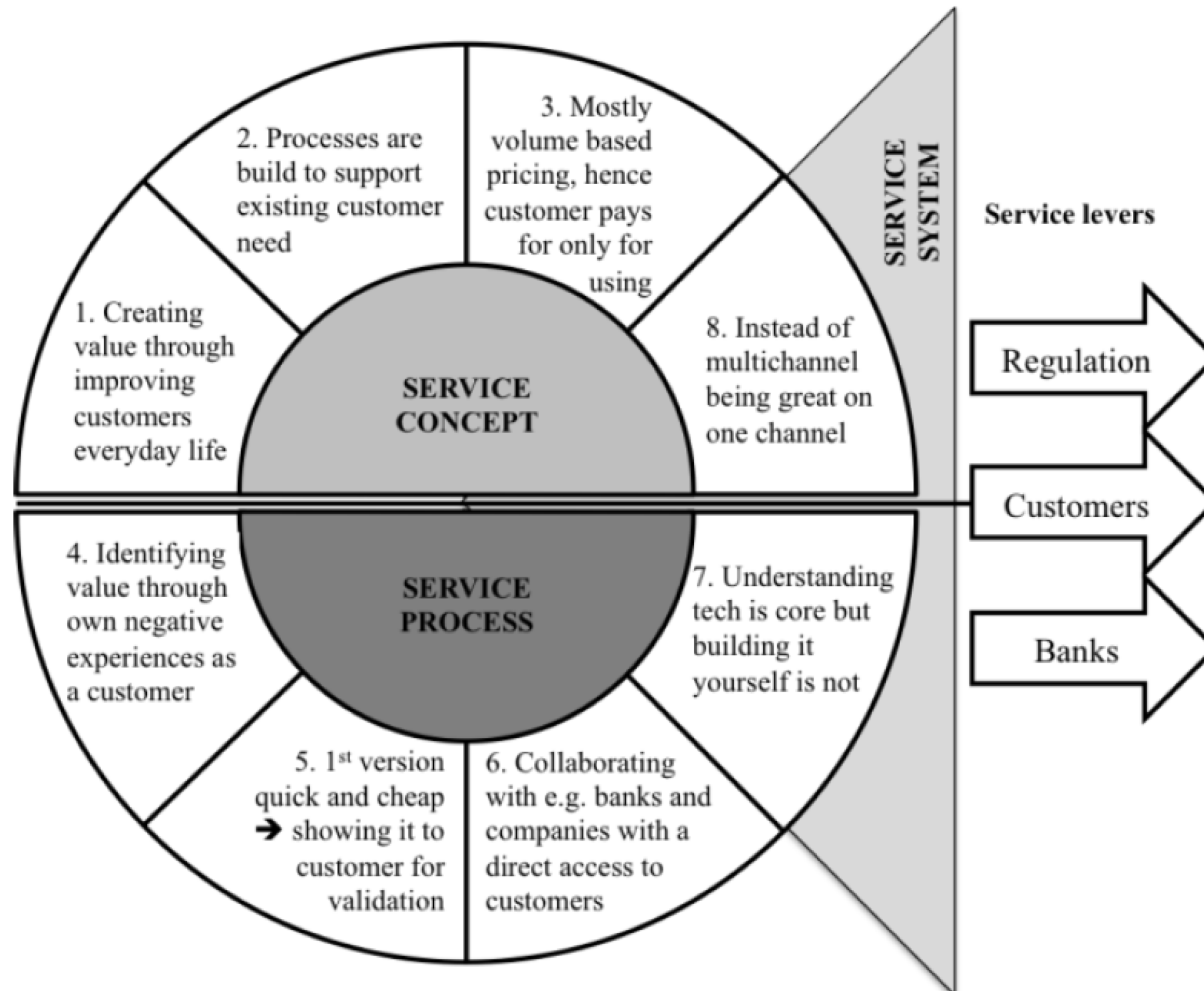
something new or unusual

the novelty of a self-driving car

Creativity is not a
new Idea.

Creativity is
an old belief
you leave behind

FinTechs as Service Innovators: Analysing Components of Innovation



Innovation

“a process of
searching and
recombining
existing knowledge
elements”

Search and recombination process to innovate:

A review of the empirical evidence and a research agenda



Source: Savino, Tommaso, Antonio Messeni Petruzzelli, and Vito Albino. "Search and recombination process to innovate: A review of the empirical evidence and a research agenda." *International Journal of Management Reviews* (2017).

Innovation Research in Economics, Sociology and Technology Management

Source: Gopalakrishnan, Shanti, and Fariborz Damanpour.

"A review of innovation research in economics, sociology and technology management." *Omega* 25, no. 1 (1997): 15-28.

Innovation Research in Economics, Sociology and Technology Management

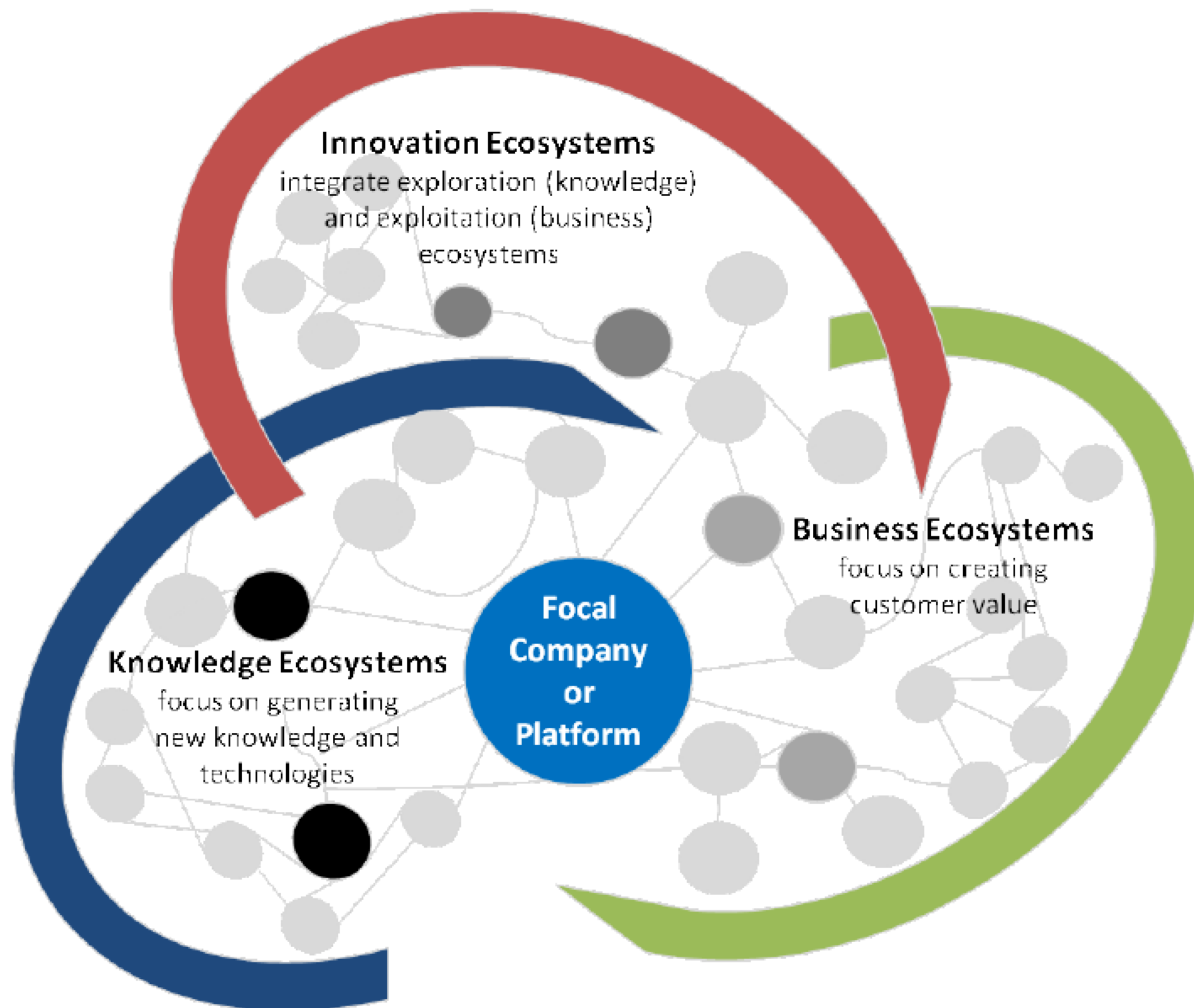
	Stage of process	Level of study	Type of innovation
<i>Economists</i>	Generation Idea generation Project definition	Industry	Product and process Only technical Only radical
<i>Technologists</i>			
Contextual technologists	Generation Commercialization and marketing Diffusion	Innovation (in the industry context)	Product and process Only technical Radical and incremental
Organizational technologists	Generation Idea generation Problem solving adoption Adoption Initiation	Organizational Sub-system	Product and process Only technical Radical and incremental
<i>Sociologists</i>			
Variance sociologists	Adoption Initiation Implementation	Organization	Product and process Technical and administrative Radical and incremental
Process sociologists	Adoption Initiation Implementation	Innovation (at the organizational level)	Product and process Technical and administrative Radical and incremental

Source: Gopalakrishnan, Shanti, and Fariborz Damanpour.

"A review of innovation research in economics, sociology and technology management." *Omega* 25, no. 1 (1997): 15-28.

Business, Innovation, and Knowledge Ecosystems

Business, Innovation, and Knowledge Ecosystems



Source: Valkokari, Katri. "Business, innovation, and knowledge ecosystems: how they differ and how to survive and thrive within them." *Technology Innovation Management Review* 5, no. 8 (2015).

Innovation Ecosystems

Characteristics

	Business Ecosystems	Innovation Ecosystems	Knowledge Ecosystems
Baseline of Ecosystem	Resource exploitation for customer value	Co-creation of innovation	Knowledge exploration
Relationships and Connectivity	Global business relationships both competitive and co-operative	Geographically clustered actors, different levels of collaboration and openness	Decentralized and disturbed knowledge nodes, synergies through knowledge exchange
Actors and Roles	Suppliers, customers, and focal companies as a core, other actors more loosely involved	Innovation policymakers, local intermediators, innovation brokers, and funding organizations	Research institutes, innovators, and technology entrepreneurs serve as knowledge nodes
Logic of Action	A main actor that operates as a platform sharing resources, assets, and benefits or aggregates other actors together in the networked business operations	Geographically proximate actors interacting around hubs facilitated by intermediating actors	A large number of actors that are grouped around knowledge exchange or a central non-proprietary resource for the benefit of all actors

Source: Valkokari, Katri. "Business, innovation, and knowledge ecosystems: how they differ and how to survive and thrive within them." *Technology Innovation Management Review* 5, no. 8 (2015).

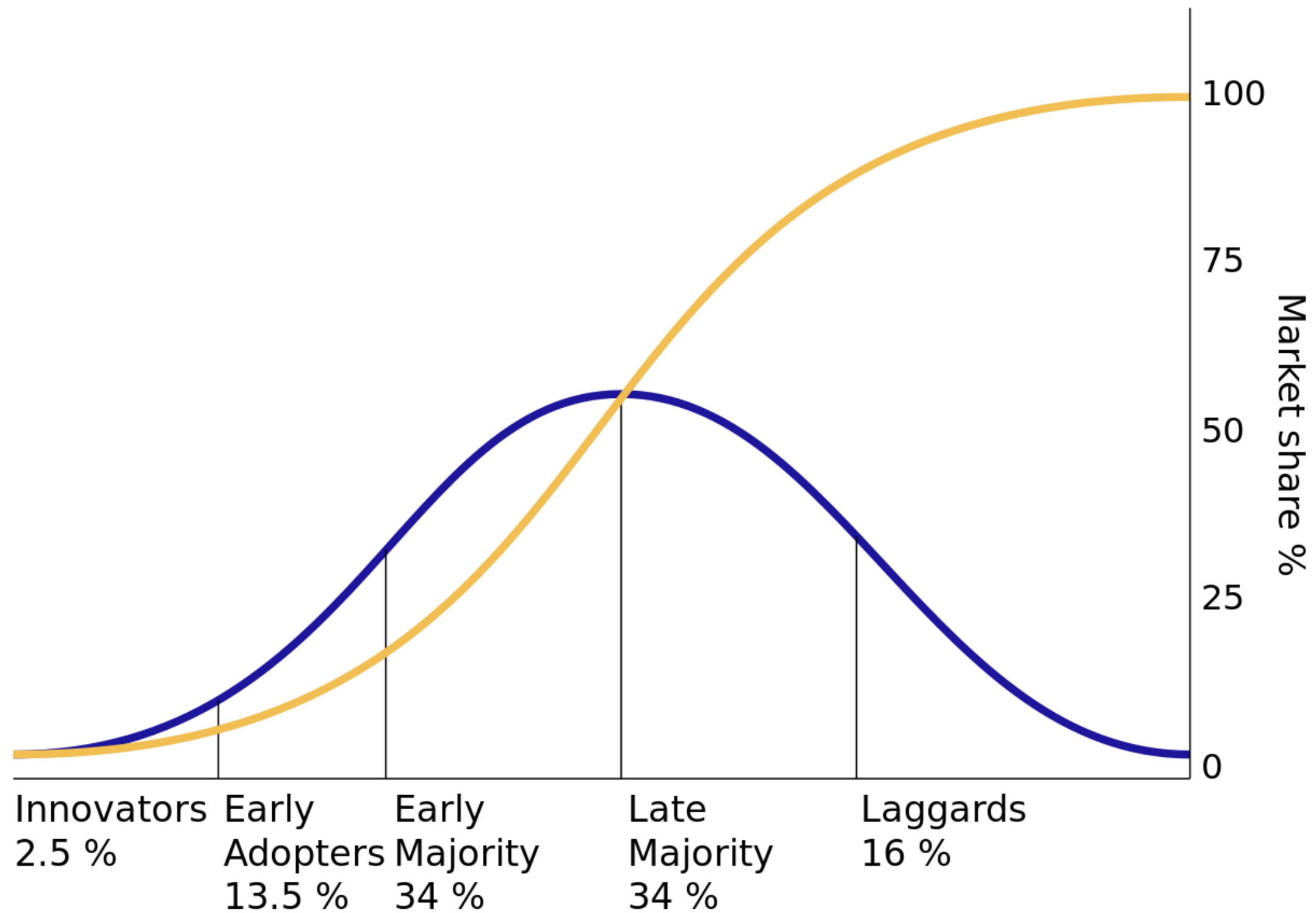
Diffusion of Innovation Theory (DOI)

Innovation

(Diffusion of Innovation)

1. Relative advantage
2. Compatibility
3. Complexity
4. Trialability
5. Observability

Diffusion of Innovation



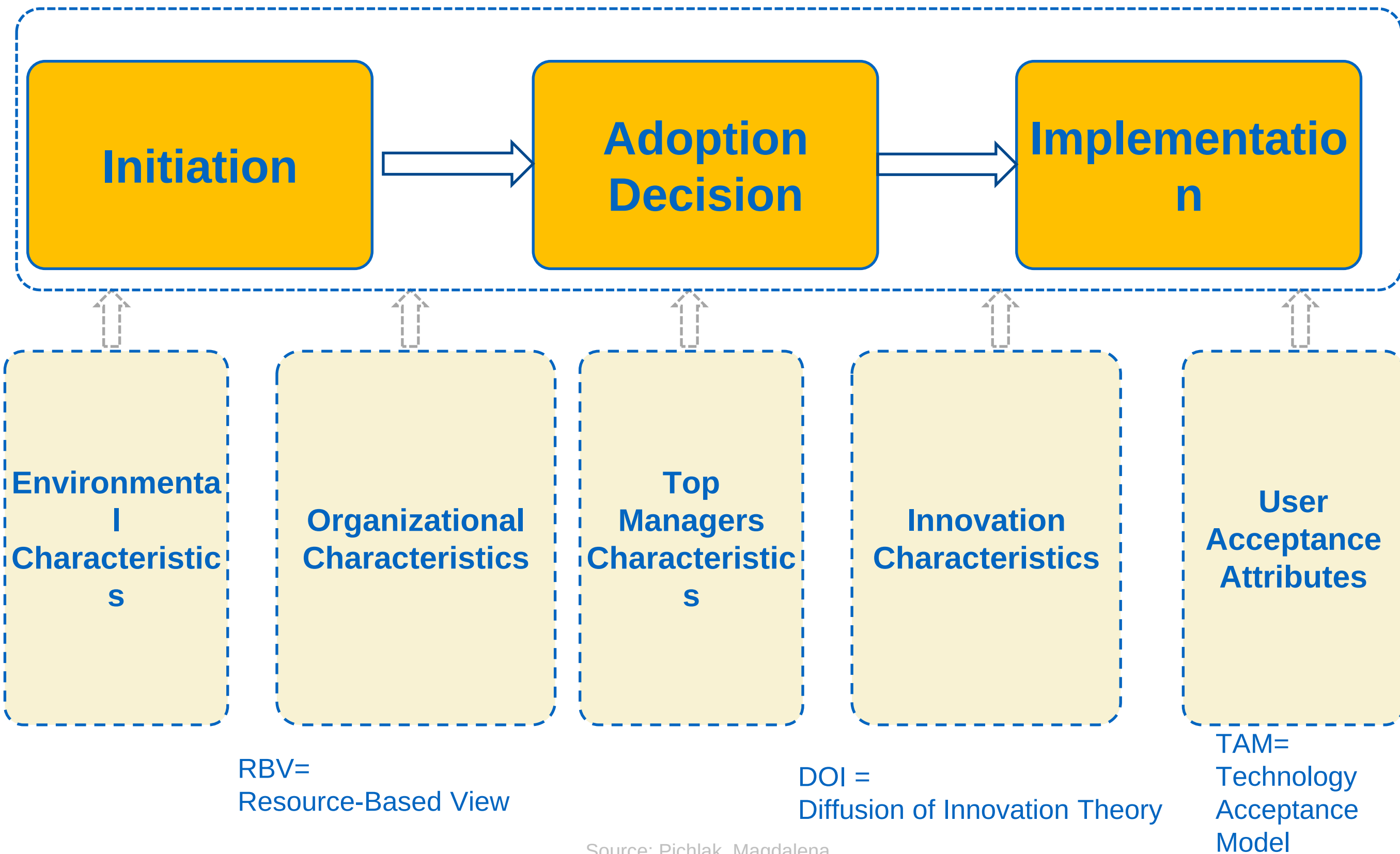
Innovation Adoption Process



Source: Pichlak, Magdalena.

"The innovation adoption process: A multidimensional approach." Journal of Management and Organization 22, no. 4 (2016): 476.

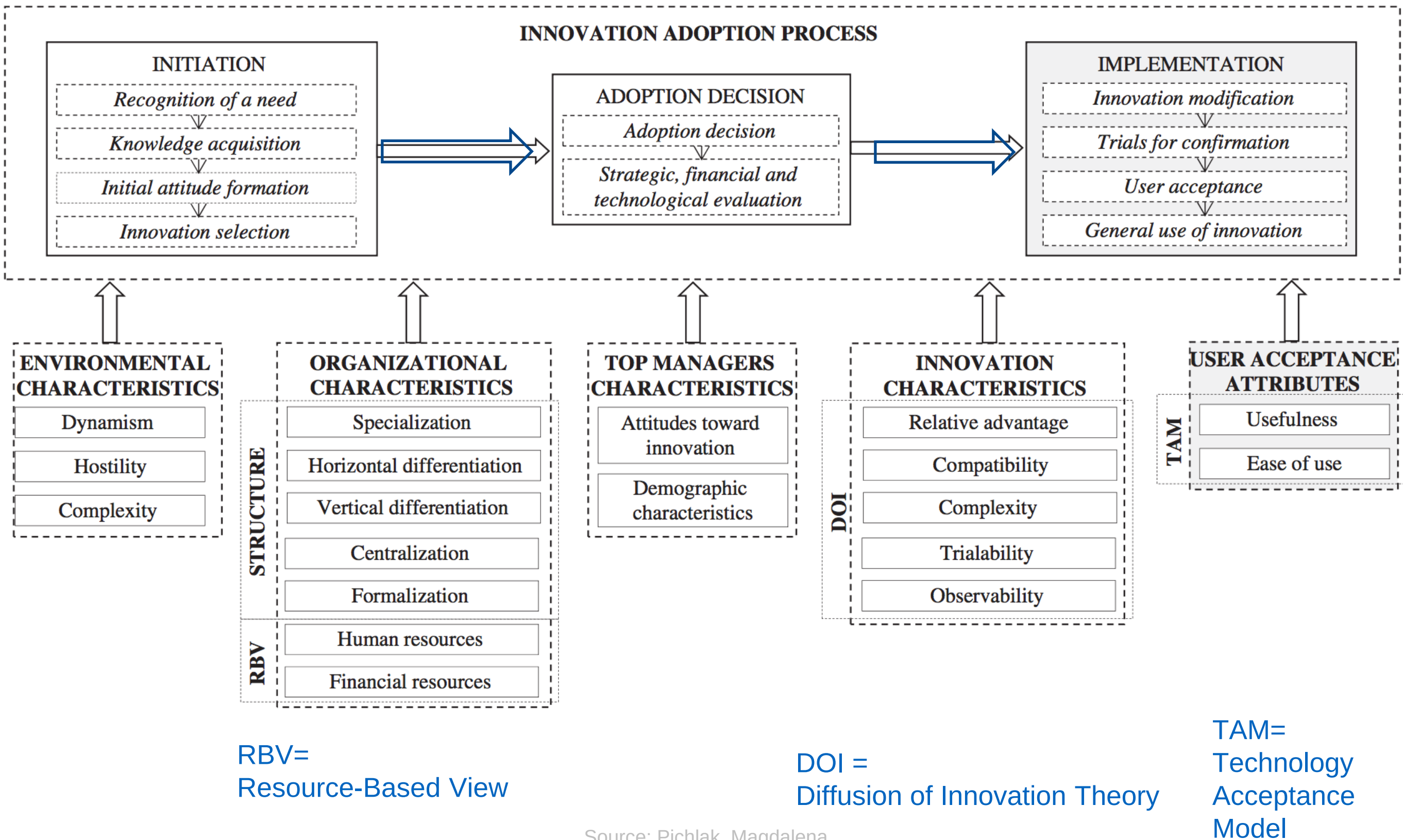
Innovation Adoption Process



Source: Pichlak, Magdalena.

"The innovation adoption process: A multidimensional approach." Journal of Management and Organization 22, no. 4 (2016): 476.

Innovation Adoption Process



Source: Pichlak, Magdalena.

"The innovation adoption process: A multidimensional approach." Journal of Management and Organization 22, no. 4 (2016): 476.

Innovation Adoption Process

Factors		Initiation					Adoption decision					Implementation				
		Mean	Me	Q3	Q1	QD	Mean	Me	Q3	Q1	QD	Mean	Me	Q3	Q1	QD
Environmental characteristics	Dynamism	3.4	3	4	2.75	0.625	3.6	4	4	3	0.5	4	4	5	4	0.5
	Hostility	3.3	3	4.25	3	0.625	3.9	4	4.25	3.75	0.25	3.7	4	4.5	3.5	0.5
Organizational characteristics	Complexity	4.5	5	5	4	0.5	3.2	3	4	2.75	0.625	3.3	3	4.25	3	0.625
	Specialization	3.8	4	4.25	3.75	0.25	2.9	3	4	2	1	2	2	3.25	2	0.625
	Horizontal differentiation	2.8	3	3.75	2.75	0.5	2.7	3	3.5	2	0.75	2	2	3.5	2	0.75
	Vertical differentiation	2.1	2	3.25	2	0.625	3.3	3	4	2.5	0.75	3.1	3	4	2.75	0.625
	Centralization	2	2	3.25	2	0.625	3.8	4	4.25	3.75	0.25	3.9	4	4.25	3.75	0.25
Top managers characteristics	Formalization	2.1	2	3	1.75	0.625	3	3	4.25	3	0.625	3.3	3	4	3	0.5
	Human resources	4.9	5	5	4.5	0.25	4	4	5	4	0.5	4.1	4	5	4	0.5
	Financial resources	3.2	3	4	2.5	0.75	4.1	4	4.25	3.75	0.25	4.8	5	5	4	0.5
	Top managers attitude towards innovation	4.1	4	4.5	4	0.25	3.9	4	4.25	3.75	0.25	4	4	4.5	3.5	0.5
	Top managers demographic characteristics	2.3	2	3.25	1.75	0.75	2	2.5	3	1	1	2.2	2	3	1.5	0.75
Innovation characteristics	Relative advantage	3	3	4	2.75	0.625	4.4	4.5	5	4	0.5	3.1	3	4	2.75	0.625
	Compatibility	2.8	3	3.5	2	0.75	3.9	4	4.25	3.75	0.25	3.9	4	4.25	3.75	0.25
	Complexity	3.6	4	4.25	3.75	0.25	3.8	4	4	3.75	0.125	3.9	4	4.25	3.75	0.25
	Trialability	3.2	3	4	2.75	0.625	3.1	3	4	2.5	0.75	4.1	4	5	4	0.5
	Observability	3.4	3.5	4.25	3	0.625	3.1	3.5	4	2	1	3.3	3	4.25	3	0.625
User acceptance attributes	Usefulness											3.2	3	4	2	1
	Ease of use											4	4	5	4	0.5

Note.
Me = median; Q = quartile; QD = quartile deviation.

Source: Pichlak, Magdalena.

"The innovation adoption process: A multidimensional approach." Journal of Management and Organization 22, no. 4 (2016): 476.

Innovation Adoption Process

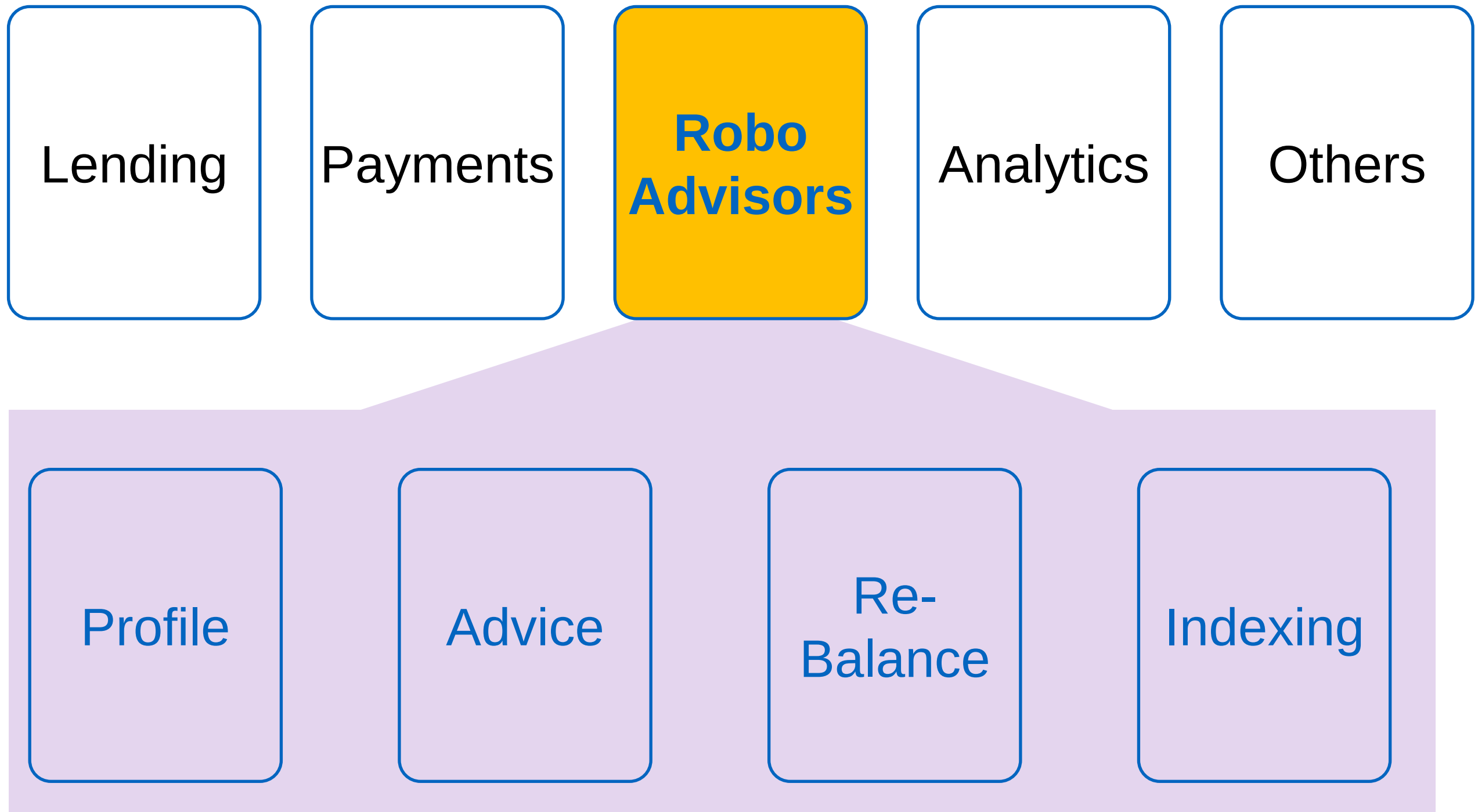
Initiation			Adoption decision			Implementation		
Factors	Round 1	Round 2	Factors	Round 1	Round 2	Factors	Round 1	Round 2
Complexity in the environment	4.5	4.2	Dynamism in the environment	3.6	3.4	Dynamism in the environment	4.0	3.8
Specialization	3.8	3.4	Hostility in the environment	3.9	4.0	Hostility in the environment	3.7	3.4
Horizontal differentiation	2.8	3.1	Centralization	3.8	3.8	Centralization	3.9	3.8
Human resources	4.9	5.0	Human resources	4.0	4.2	Formalization	3.3	3.2
Top managers attitude towards innovation	4.1	4.3	Financial resources	4.1	4.4	Human resources	4.1	4.4
Innovation complexity	3.6	3.3	Top managers attitude towards innovation	3.9	4.0	Financial resources	4.8	5.0
			Relative advantage	4.4	4.1	Top managers attitude towards innovation	4.0	4.4
			Innovation compatibility	3.9	3.6	Innovation compatibility	3.9	3.8
			Innovation complexity	3.8	3.8	Innovation complexity	3.9	3.9
						Innovation trialability	4.1	3.9
						Ease of use	4.0	4.2

Source: Pichlak, Magdalena.

"The innovation adoption process: A multidimensional approach." Journal of Management and Organization 22, no. 4 (2016): 476.

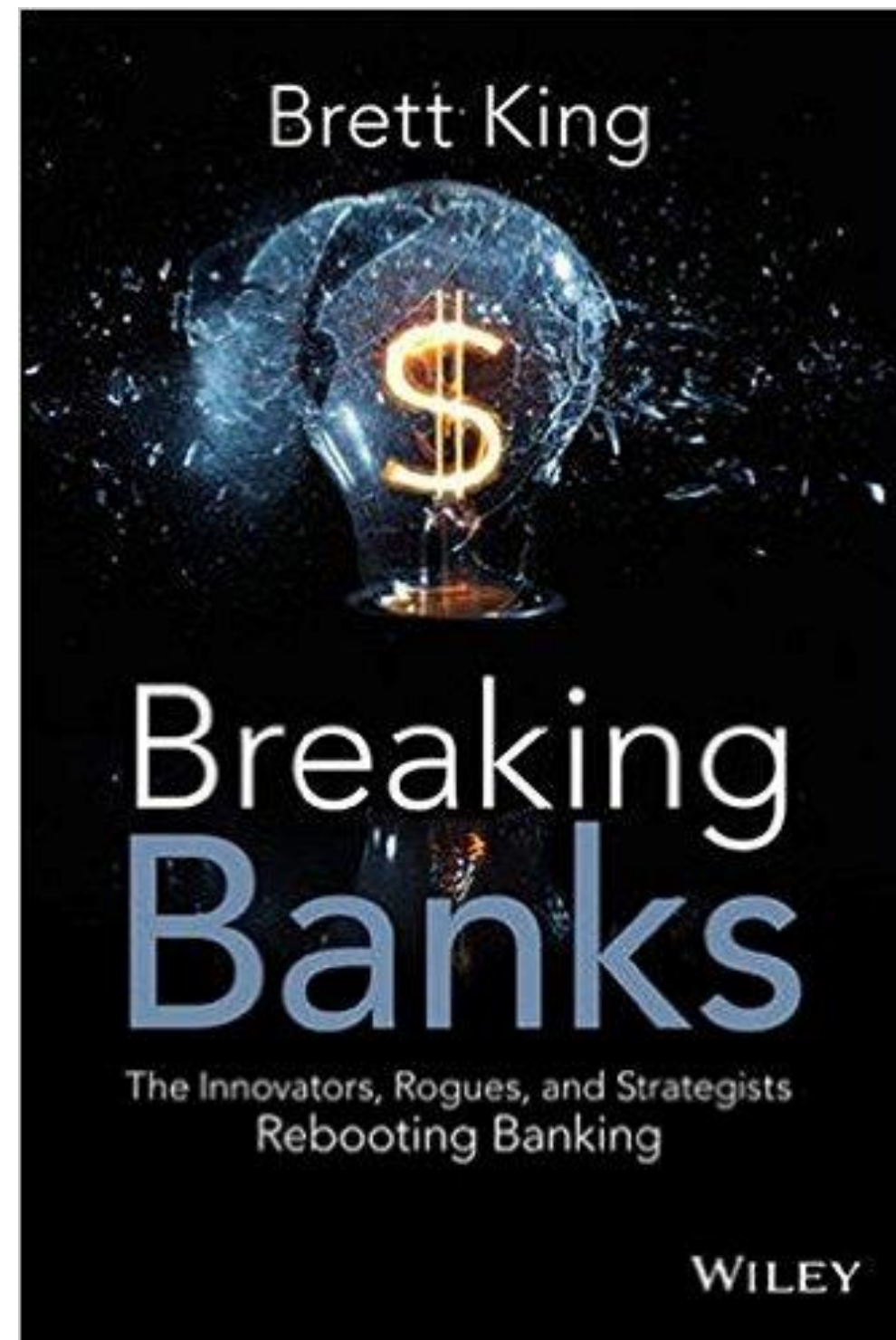
FinTech Innovation

FinTech high-level classification



Brett King (2014),
Breaking Banks:

The Innovators, Rogues, and Strategists Rebooting Banking
Wiley



“In the next 10 years,
we'll see more
disruption and changes
to the banking and
financial industry
than we've seen in the
preceding 100 years.”

(Brett King, 2014)

Disrupting Banking: The Fintech Startups That Are Unbundling Wells Fargo, Citi and Bank of America

Fintech: Unbundling the Bank

Wealth Management: Wealthfront

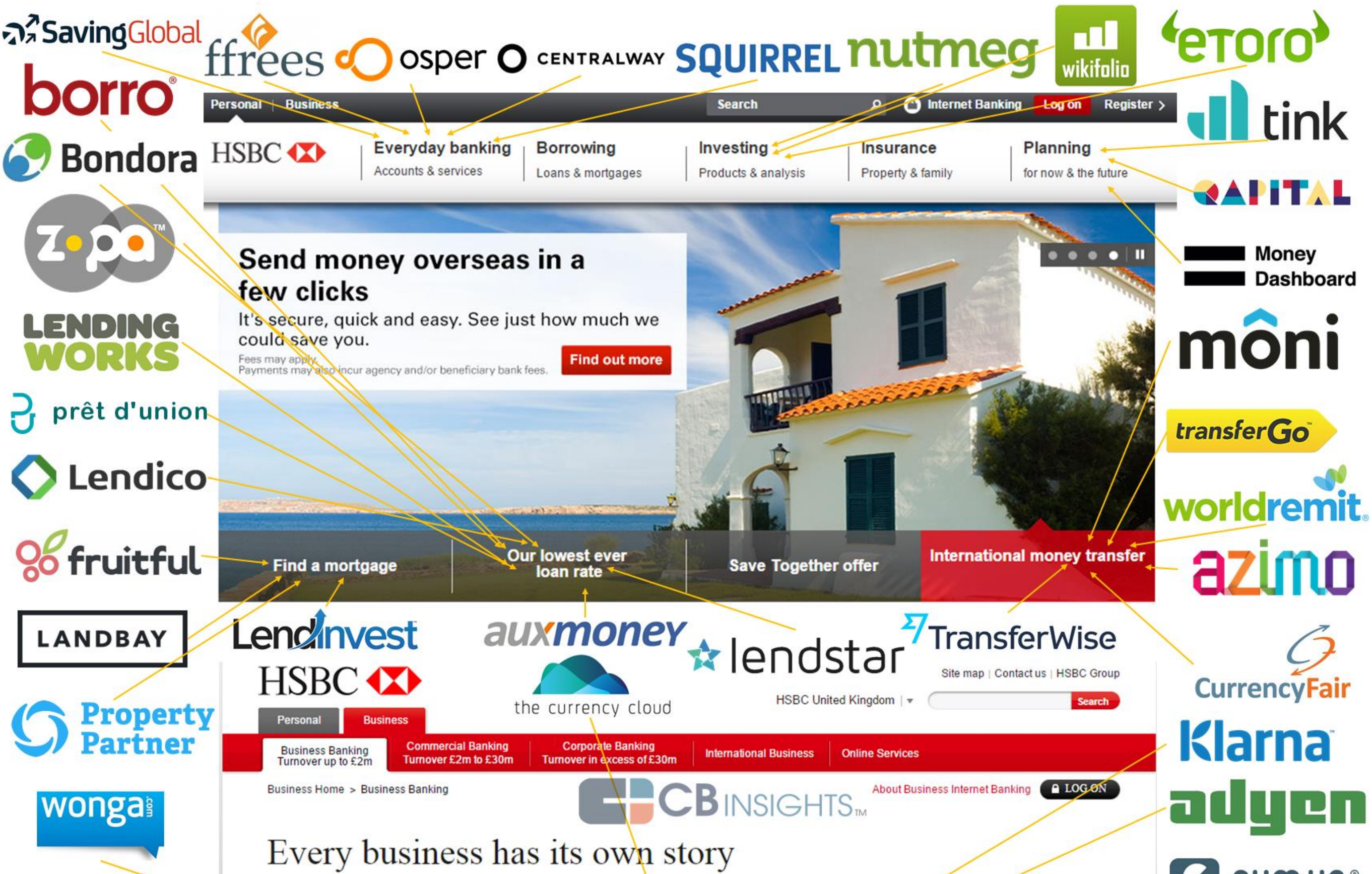


Disrupting European Banking: The FinTech Startups That Are Unbundling HSBC, Santander, and BNP

Unbundling of a European Bank



Unbundling of a European Bank



Financial Technology (Fintech) Categories

1. Banking Infrastructure
2. Business Lending
3. Consumer and Commercial Banking
4. Consumer Lending
5. Consumer Payments
6. Crowdfunding
7. Equity Financing
8. Financial Research and Data
9. Financial Transaction Security
10. Institutional Investing
11. International Money Transfer
12. Payments Backend and Infrastructure
13. Personal Finance
14. Point of Sale Payments
15. Retail Investing
16. Small and Medium Business Tools

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